ANSI/ASNT CP-105-2024
American National Standard

ASNT STANDARD

TOPICAL OUTLINES FOR QUALIFICATION OF NONDESTRUCTIVE TESTING PERSONNEL

2024
EDITION

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ASNT Mission Statement:
ASNT's mission is to advance the field of nondestructive testing.

ASNT Code of Ethics:
The ASNT Code of Ethics was developed to provide members of the Society with broad ethical statements to guide their professional lives. In spirit and in word, each ASNT member is responsible for knowing and adhering to the values and standards set forth in the Society's Code. More information, as well as the complete version of the Code of Ethics, can be found on ASNT's website, asnt.org.

ANSI/ASNT CP-105-2024
American National Standard
ASNT Standard Topical Outlines for
Qualification of Nondestructive Testing Personnel

The American Society for Nondestructive Testing Inc.
Approved 5 April 2024
American National Standards Institute

Abstract

This standard applies to personnel whose specific tasks or jobs require appropriate knowledge of the technical principles underlying nondestructive testing (NDT) methods for which they have responsibilities within the scope of their employment. These specific tasks or jobs include, but are not limited to, performing, specifying, reviewing, monitoring, supervising, and evaluating NDT work.

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Employers or other persons utilizing NDT services are cautioned that they retain full responsibility for the ultimate determination of the qualifications of NDT personnel and for the certification process. The process of personnel qualification and certification as detailed in the standard does not relieve the employer of the ultimate legal responsibility to ensure that the NDT personnel are fully qualified for the tasks being undertaken.

This standard is subject to revision or withdrawal at any time by ASNT.

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American National Standard Approval of an American National Standard requires verification by ANSI that the requirements for due process, consensus, and other criteria for approval have been met by the standards developer.

Consensus is established when, in the judgment of the ANSI Board of Standards Review, substantial agreement has been reached by directly and materially affected interests. Substantial agreement means much more than a simple majority, but not necessarily unanimity. Consensus requires that all views and objections be considered, and that a concerted effort be made toward their resolution.

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FOREWORD

(Not a part of American National Standard CP-105-2024)

An essential element in the effectiveness of nondestructive testing (NDT) is the qualification of the personnel who are responsible for and who perform NDT. Formal training and actual experience are important and necessary elements in acquiring the skills necessary to effectively perform nondestructive tests.

The American Society for Nondestructive Testing Inc. (ASNT) has, therefore, undertaken the preparation and publication of this standard, which specifies the procedures, essential factors, and minimum requirements for qualifying and certifying NDT personnel.

In 2018 ASNT accepted the ASTM E1316 definitions of calibration and standardization for use in its publications. These terms are used as follows:

- *Calibration* is the comparison (which may include adjustment) of a test instrument to a known reference that is normally traceable to some recognized authority (e.g., NIST). Calibration is typically performed by an organization considered qualified to do so (e.g., an accredited laboratory, or in some cases, the instrument manufacturer) at a determined, periodic interval. Calibration of electronic instrumentation typically involves verification of the linearity of the instrument's response over its usable range.
- *Standardization* is typically completed prior to performing an NDT test, and may also be performed at times during the performance of the test and at the completion of the test as a validation of proper instrument operation. It is the adjustment of an NDT instrument using a reference standard (that contains a known condition) to obtain or establish a known and reproducible response.

ASNT CP-105: *ASNT Standard Topical Outlines for Qualification of Nondestructive Testing Personnel* was initially processed and approved for submittal to the American National Standards Institute (ANSI) by the ASNT Standards Writing Committee. This revision was processed by the ASNT Standards Council. Council approval of the standard does not necessarily imply that all committee members voted for its approval. At the time it approved this standard, the Standards Council and the CP-105 Committee had the following members:

Ethics Statement

The NDT profession plays a critical role in ensuring public safety by providing information on the condition and integrity of critical infrastructure and products through its various NDT methods and techniques. The NDT profession has a responsibility to act ethically in all activities including the proper collection, interpretation, and reporting of NDT data. To this end, ASNT highly encourages all individuals and companies using this standard/document to incorporate the topic of ethics and ethical behavior of its employees and contractors into their training programs and examinations.

The outlines contained in this American National Standard were approved by the ASNT Technical and Education (T&E) Council through its method committees. At the time this standard was approved, the T&E Council Methods Division had the following members:

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SUMMARY OF CHANGES

CP-105 2024 Edition

1. Where applicable for accuracy, the words “technique” and “method” were changed.
2. Throughout, the word “manmade” was changed to “manufactured”.
3. Where applicable, the words “radiographic” and “radiography” were changed.
4. References were updated as needed and are not marked.

Acoustic Emission Testing

Acoustic Emission Testing Level II Topical Outline

Page 5 Section 3.0 through paragraph 3.3.5 were inserted.

Acoustic Emission Testing Level II Topical Outline

Page 8 Section 4.0 through paragraph 4.11 were inserted.

Laser Methods Testing

Page 35 deleted “Laser Method Testing Profilometry, Level II Topical Outline, and Level III Topical Outline References”.

Leak Testing

Level I Topical Outline

FUNDAMENTALS IN LEAK TESTING COURSE

Page 37 2.4.1 was changed.

Page 37 2.4.5–2.4.7 were inserted.

Page 37 2.6.4 was changed.

Page 37 2.7–2.7.4 were inserted.

Page 37 deleted 2.8.2.1.2.

Page 38 2.9.2.1 and 2.9.2.2.1.6–2.9.2.2.1.10 were inserted.

SAFETY IN LEAK TESTING

Page 38 Section 8.0 was inserted.

Leak Testing Methods Course

Page 39 1.1.6 was changed.

Page 39 1.1.6.3, 1.1.9.4, 1.1.9.5, 1.1.10.7, and 1.1.10.8 were inserted.

Page 39 1.1.11 and 1.1.11.1 were changed.

Page 39 1.1.11.3, 1.1.13, and 1.1.14 were inserted.

Page 39 2.4–2.4.4, and 2.6.7–2.6.10.5 were inserted.

Leak Testing Level II Topical Outline

PRINCIPLES OF LEAK TESTING COURSE

Page 39 1.1.1.1 was inserted.

Page 40 1.2.7, 1.2.8, 1.5.7, 2.5–2.5.4, 3.1.3, and 3.1.4 were inserted.

PRESSURE AND VACUUM TECHNOLOGY COURSE

Page 41 2.2.3, 2.2.4, and 2.3.2.2 were changed.

Page 41 2.2.3.4 and 2.2.4.2 were inserted.

Page 41 2.3.6–2.3.6.3.2 were inserted.

Page 42 2.4.5.4, 2.4.8, 2.6.1.3, and 2.6.3.4 were inserted.

Page 42 Section 3.0 was inserted.

Leak Testing Level III Topical Outline

Page 42 the “Leak Test Selection Course” was moved under the “Leak Testing Level III Topical Outline”.

Page 43 2.2.5 was changed.

Page 43 3.1.2 was inserted.

Page 43 3.3.1.1, 3.3.1.3–3.3.2, 3.4.1–3.4.3, 3.4.7–3.4.9.4 were inserted.

Page 44 3.8, 3.9, and 3.10.1–3.11.3 were inserted.

Page 44 3.12 and 4.3.4 were changed.

Page 45 4.4.4.6–4.4.4.8, 4.4.5.6–4.4.5.8, 5.2.2–5.2.2.2, 6.1.7–6.1.10, and 6.2.6–6.2.8 were inserted.

Liquid Penetrant Testing

Liquid Penetrant Testing Level II Topical Outline

Page 47 3.6.1, 3.6.2, and 4.2 were inserted.

Page 47 deleted Section 5.0.

Liquid Penetrant Testing Level III Topical Outline

Page 48 3.1.1, 3.1.2, and 4.1–4.8 were inserted.

Magnetic Particle Testing Topical Outlines

Magnetic Particle Testing Level I Topical Outline

Page 51 Section 1.0 through paragraph 1.7 were inserted.

Page 51 2.1.2 was inserted.

Page 51 2.2.1–2.2.8 replaced paragraphs 1.2.1–1.3.

Page 51 Section 3.0 through paragraph 6.2.3 replaced previous sections.

Page 51 7.3 was inserted.

Page 51 8.2, 8.5, and 8.7 were changed.

Page 51 Section 8.0 was moved to Section 9.0.

Page 51 8.2, 8.3, and 8.5–8.5.4 were deleted.

Page 52 the original Section 9.0 was moved to paragraph 11.5.

Page 52 9.3–9.10.5 were inserted.

Page 52 Section 11.0 through paragraph 11.4 were inserted.

Page 52 Section 12.0 through paragraph 12.10 were changed and inserted.

Magnetic Particle Testing Level II Topical Outline

Page 52 Section 1.0 through paragraph 1.7 were inserted.

Page 52 Section 3.0 was moved to paragraph 2.4.

Page 52 2.4.3 was inserted.

Page 53 Section 4.0 was moved to 2.5.

Page 53 2.5.2.5–2.5.2.6 were inserted.

Page 53 Section 5.0 was moved to paragraph 2.6.

Page 53 Section 6.0 through paragraph 6.4 were deleted.

Page 53 Section 3.0 was inserted.

Page 53 4.1, 4.2–4.2.3, 4.4.2, 4.5.7 and 4.6–4.6.5 were inserted.

Page 53 7.1–7.2.4, 7.6–7.6.4., and 7.7–7.7.2 were inserted.

Page 53 Section 9.0 was deleted.

Pages 53–54 Sections 6.0–8.0 were inserted.

Magnetic Particle Testing Level III Topical Outline

Page 54 insertions and deletions were made in Sections 4.0–6.0.

Microwave Technology Testing Level I Topical Outline

Page 55 4.1 was changed and 4.2–4.3.1 were inserted.

Microwave Technology Testing Level II Topical Outline

THEORY COURSE

Page 56 8.2 and 8.3 were inserted.

TECHNIQUE COURSE

Page 56 2.1 was changed and 2.2–2.3.1 were inserted.

Microwave Technology Testing Level III Topical Outline

Page 57 Section 7.0 through paragraph 7.3 were deleted.

Page 57 Section 10.0 was changed.

Radiographic Testing Topical Outlines

Computed Radiography Level II Topical Outline

ADVANCED COMPUTED RADIOGRAPHY COURSE

Page 69 Section 8.0 and paragraph 8.2 were changed.

Computed Tomography Level II Topical Outline

RADIOGRAPHIC INTERPRETATION AND EVALUATION COURSE

Page 71 3.2 was changed.

Computed Tomography (CT)

Page 76 Sections 1.0–3.0 were deleted and replaced, and Sections 4.0–6.0 were inserted.

Pages 78–79 under “Digital Radiography and Computed Radiography Technique Course,” Sections 1.0–5.0 were inserted.

Page 79 under “Digital Radiography and Computed Radiography Interpretation and Evaluation Course,” Sections 1.0–5.0 were inserted.

Pages 79–80 under “Computed Tomography Technique Course,” Sections 1.0–4.0 were inserted.

Page 80 Sections 1.0–4.0 were inserted.

Thermal/Infrared Testing Topical Outlines

Basic Thermal/Infrared Physics Course

BASIC THERMAL/INFRARED OPERATING COURSE

Page 81 3.2 was changed and paragraphs 3.2.1–3.2.5 were deleted.

Thermal/Infrared Testing Level III Topical Outline

Page 86 2.1.18.1–2.1.18.4 were deleted.

Page 88 6.2.1–6.2.7 were deleted.

Vibration Analysis Topical Outlines

Vibration Analysis Level I Topical Outline

BASIC VIBRATION ANALYSIS PHYSICS COURSE

Page 101 1.2.1 and 1.2.2 were inserted.

Page 101 changes and insertions were made to paragraphs 2.1–2.4.

BASIC VIBRATION ANALYSIS OPERATING COURSE

Page 101 changes and insertions were made to paragraphs 1.1.1–1.1.3.

Pages 101–102 changes and insertions were made to Sections 2.0–6.0.

Vibration Analysis Level II Topical Outline

INTERMEDIATE VIBRATION ANALYSIS PHYSICS COURSE

Page 102 changes and insertions were made to paragraphs 1.2–3.12.

INTERMEDIATE VIBRATION ANALYSIS TECHNIQUES COURSE

Pages 102–103 changes and insertions were made to Section 1.0 through paragraph 2.7.2.

Page 103 3.4–3.7 and 4.3.1–4.3.2 were inserted.

Page 103 under Section 6.0 inserted a sentence.

Page 103 6.3 was inserted.

Vibration Analysis Level III Topical Outline

Page 103 1.2.1–1.2.3 were changed.

Page 104 under Section 3.0 deleted a sentence.

Page 104 changes and insertions were made to paragraphs 4.4–5.2.

Page 104 6.3 was inserted.

Visual Testing Level II Topical Outline

Page 108 11.1 and 11.2 were inserted.

SCOPE

ASNT Standard Topical Outlines for Qualification of Nondestructive Testing Personnel

1.0 Scope

1.1 This standard establishes the minimum topical outline and the minimum training course content requirements for the qualification of nondestructive testing (NDT) personnel.

1.2 The amount of time spent on each topic in each method should be determined by the NDT Level III and the applicable certification document.

1.3 These topical outlines are progressive; i.e., consideration as Level I is based on satisfactory completion of the Level I training course; consideration as Level II is based on satisfactory completion of both Level I and Level II training courses.

1.4 Topics in the outlines may be deleted or expanded to meet the employer's specific applications or for limited certification, unless stated otherwise by the applicable certification procedure or written practice.

AE

ACOUSTIC EMISSION TESTING TOPICAL OUTLINES

Acoustic Emission Testing Level I Topical Outlines

Basic Acoustic Emission Physics Course

1.0 Principles of Acoustic Emission Testing (AE)

1.1 Characteristics of acoustic emission
- 1.1.1 Continuous emission
- 1.1.2 Burst emission
- 1.1.3 Emission/signal levels and frequencies

1.2 Sources of acoustic emission
- 1.2.1 Sources in crystalline materials – introduction
- 1.2.2 Sources in nonmetals – introduction
- 1.2.3 Sources in composites – introduction
- 1.2.4 Other sources

1.3 Wave propagation – introduction
- 1.3.1 Wave velocity in materials
- 1.3.2 Attenuation
- 1.3.3 Reflections, multiple paths
- 1.3.4 Source input versus signal output

1.4 Repeated loadings: Kaiser and Felicity effects and Felicity ratio
- 1.4.1 In metals
- 1.4.2 In composites

1.5 Terminology (refer to AE Glossary, ASTM E1316)

2.0 Sensing the AE Wave

2.1 Sensors
- 2.1.1 Principles of operation
- 2.1.2 Construction
- 2.1.3 Frequency

2.2 Sensor attachment
- 2.2.1 Coupling materials
- 2.2.2 Attachment devices

Basic Acoustic Emission Technique Course

1.0 Instrumentation and Signal Processing

1.1 Cables
- 1.1.1 Coaxial cable
- 1.1.2 Twisted pair cable
- 1.1.3 Noise problems in cables
- 1.1.4 Connectors

1.2 Signal conditioning
- 1.2.1 Preamplifiers
- 1.2.2 Amplifiers
- 1.2.3 Filters
- 1.2.4 Units of gain measurement

1.3 Signal detection
- 1.3.1 Threshold comparator
- 1.3.2 Units of threshold measurement
- 1.3.3 Sensitivity determined by gain and/or threshold

1.4 Signal processing
- 1.4.1 Waveform characteristics
- 1.4.2 Discrimination techniques
- 1.4.3 Distribution techniques

1.5 Source location techniques
- 1.5.1 Single-channel location
- 1.5.2 Linear location
- 1.5.3 Planar location
- 1.5.4 Other location techniques

1.6 Acoustic emission test systems
- 1.6.1 Single-channel systems
- 1.6.2 Multichannel systems
- 1.6.3 Dedicated industrial systems

1.7 Accessory techniques
- 1.7.1 Audio indicators
- 1.7.2 X-Y and strip-chart recording
- 1.7.3 Oscilloscopes
- 1.7.4 Others

2.0 Acoustic Emission Test Techniques

2.1 Equipment standardization and setup for test
- 2.1.1 Standardization signal generation techniques
- 2.1.2 Standardization procedures
- 2.1.3 Sensor placement
- 2.1.4 Adjustment of equipment controls
- 2.1.5 Discrimination technique adjustments

2.2 Loading procedures
- 2.2.1 Type of loading
- 2.2.2 Maximum test load
- 2.2.3 Load holds
- 2.2.4 Repeated and programmed loadings
- 2.2.5 Rate of loading

- 2.3 Data display
  - 2.3.1 Selection of display mode
  - 2.3.2 Use and reading of different kinds of display
- 2.4 Noise sources and pre-test identification techniques
  - 2.4.1 Electromagnetic noise
  - 2.4.2 Mechanical noise
- 2.5 Precautions against noise
  - 2.5.1 Electrical shielding
  - 2.5.2 Electronic techniques
  - 2.5.3 Prevention of movement
  - 2.5.4 Attenuating materials and applications
- 2.6 Data interpretation and evaluation – introduction
  - 2.6.1 Separating relevant acoustic emission indications from noise
  - 2.6.2 Accept/reject techniques and evaluation criteria
- 2.7 Reports
  - 2.7.1 Purpose
  - 2.7.2 Content and structure

3.0 Codes, Standards, and Procedures

- 3.1 Guide-type standards (glossaries, user manuals, etc.)
- 3.2 Standardized/codified acoustic emission test procedures
- 3.3 User-developed test procedures

4.0 Applications of AE (course should include at least three categories from 4.1 and at least four categories from 4.2)

- 4.1 Laboratory studies (material characterization)
  - 4.1.1 Crack growth and fracture mechanics
  - 4.1.2 Environmentally assisted cracking
  - 4.1.3 Dislocation movement (metals)
  - 4.1.4 Clarifying deformation mechanisms (composites)
  - 4.1.5 Phase transformation and phase stability
  - 4.1.6 Creep
  - 4.1.7 Residual stress
  - 4.1.8 Corrosion
  - 4.1.9 Fatigue
  - 4.1.10 Rupture
  - 4.1.11 Ductile/brittle transition
  - 4.1.12 Other material characterization applications
- 4.2 Structural applications
  - 4.2.1 Pressure vessels (metal)
  - 4.2.2 Storage tanks (metal)
  - 4.2.3 Pressure vessels/storage tanks (composite)
  - 4.2.4 Piping and pipelines
  - 4.2.5 Bucket trucks
  - 4.2.6 Aircraft
  - 4.2.7 Bridges
  - 4.2.8 Mines
  - 4.2.9 Dams, earthen slopes
  - 4.2.10 Pumps, valves, etc.
  - 4.2.11 Rotating plant
  - 4.2.12 In-process weld monitoring
  - 4.2.13 Leak detection and monitoring
  - 4.2.14 Other structural applications

Acoustic Emission Testing Level II Topical Outline

Acoustic Emission Physics Course

1.0 Principles of AE

- 1.1 Characteristics of AE
  - 1.1.1 Introductory concepts of source, propagation, measurement, display, and evaluation
  - 1.1.2 Relationships between AE and other NDT methods
  - 1.1.3 Significance of applied load in AE
  - 1.1.4 Basic math review (exponents, graphing, metric units)
- 1.2 Materials and deformation
  - 1.2.1 Constitution of crystalline and noncrystalline materials
  - 1.2.2 Stress and strain
  - 1.2.3 Elastic and plastic deformation, crack growth
- 1.3 Sources of acoustic emission
  - 1.3.1 Burst emission, continuous emission
  - 1.3.2 Emission/signal levels, units of amplitude measurement
  - 1.3.3 Sources in crystalline materials
    - 1.3.3.1 Dislocations – plastic deformation
    - 1.3.3.2 Phase transformations
    - 1.3.3.3 Deformation twinning
    - 1.3.3.4 Nonmetallic inclusions
    - 1.3.3.5 Subcritical crack growth
      - 1.3.3.5.1 Subcritical crack growth under increasing load
      - 1.3.3.5.2 Ductile tearing under increasing load
      - 1.3.3.5.3 Fatigue crack initiation and growth
      - 1.3.3.5.4 Hydrogen embrittlement cracking
      - 1.3.3.5.5 Stress corrosion cracking
  - 1.3.4 Sources in nonmetals
    - 1.3.4.1 Microcracking
    - 1.3.4.2 Gross cracking
    - 1.3.4.3 Crazing
    - 1.3.4.4 Other sources in nonmetals
  - 1.3.5 Sources in composites
    - 1.3.5.1 Fiber breakage
    - 1.3.5.2 Matrix cracking
    - 1.3.5.3 Fiber-matrix debonding
    - 1.3.5.4 Delamination
    - 1.3.5.5 Fiber pull-out, relaxation
    - 1.3.5.6 Friction
  - 1.3.6 Other sources
    - 1.3.6.1 Pressure leaks
    - 1.3.6.2 Oxide and scale cracking
    - 1.3.6.3 Slag cracking
    - 1.3.6.4 Frictional sources
    - 1.3.6.5 Liquefaction and solidification

1.3.6.6 Loose parts, intermittent contact
1.3.6.7 Fluids and nonsolids
1.3.6.8 Crack closure

1.4 Wave propagation
1.4.1 Near-field impulse response
1.4.2 Modes of propagation
1.4.3 Mode conversion, reflection, and refraction
1.4.4 Wave velocity in material
1.4.5 Anisotropic propagation in composites
1.4.6 Specimen geometry effects

1.5 Attenuation
1.5.1 Geometric attenuation
1.5.2 Dispersion
1.5.3 Scattering, diffraction
1.5.4 Attenuation due to energy loss mechanisms
1.5.5 Attenuation versus frequency

1.6 Kaiser and Felicity effects, and Felicity ratio
1.6.1 In metals
1.6.2 In composites
1.6.3 In other materials

1.7 Terminology (refer to AE Glossary, ASTM E1316)

2.0 Sensing the AE Wave

2.1 Transducing processes (piezoelectricity, etc.)

2.2 Sensors
2.2.1 Construction
2.2.2 Conversion efficiencies
2.2.3 Standardization (sensitivity curve)

2.3 Sensor attachment
2.3.1 Coupling materials
2.3.2 Attachment devices
2.3.3 Waveguides

2.4 Sensor utilization
2.4.1 Flat response sensors
2.4.2 Resonant response sensors
2.4.3 Integral-electronics sensors
2.4.4 Special sensors (directional, mode responsive)
2.4.5 Sensor selection

3.0 Machine Learning

3.1 Introduction to machine learning and pattern recognition
3.1.1 Supervised and unsupervised learning
3.1.2 Regression and classification problems
3.1.3 Data presentation in machine learning
3.1.3.1 Vectors
3.1.3.2 Matrices
3.1.3.3 Categorical

3.2 Mathematical functions and operations in machine learning
3.2.1 Linear and nonlinear
3.2.2 Univariate and multivariate
3.2.3 Inner products
3.2.4 Norms/distances
3.2.5 Vector-matrix multiplications
3.2.6 Correlations

3.3 Basic concepts in statistics and probability
3.3.1 Introduction
3.3.2 Mean, median, and mode
3.3.3 Variance and standard deviation
3.3.4 Rules of probability – addition and multiplication
3.3.5 Conditional probability

Acoustic Emission Technique Course

1.0 Instrumentation and Signal Processing

1.1 Cables
1.1.1 Coaxial cable
1.1.2 Twisted pair cable
1.1.3 Optical fiber cable
1.1.4 Noise problems in cables
1.1.5 Impedance matching
1.1.6 Connectors

1.2 Signal conditioning
1.2.1 Preamplifiers
1.2.2 Amplifiers
1.2.3 Filters
1.2.4 Units of gain measurement

1.3 Signal detection
1.3.1 Threshold comparator
1.3.2 Units of threshold measurement
1.3.3 Sensitivity determined by gain and/or threshold

1.4 Signal processing
1.4.1 Waveform characteristics
1.4.1.1 Amplitude analysis
1.4.1.2 Pulse duration analysis
1.4.1.3 Rise time analysis
1.4.1.4 Event and event rate processing
1.4.1.5 MARSE (measured area of the rectified signal envelope)
1.4.2 Discrimination techniques
1.4.3 Distribution techniques

1.5 Source location techniques
1.5.1 Single-channel location
1.5.2 Linear location
1.5.3 Planar location
1.5.4 Other location techniques

1.6 Acoustic emission test systems
1.6.1 Single-channel systems
1.6.2 Multichannel systems
1.6.3 Dedicated industrial systems

1.7 Accessory techniques
1.7.1 Audio indicators
1.7.2 X-Y and strip-chart recording
1.7.3 Oscilloscopes
1.7.4 Magnetic recorders
1.7.5 Others

1.8 Advanced signal processing techniques
1.8.1 Signal definition
1.8.2 Signal capture
1.8.3 Frequency analysis
1.8.4 Pattern recognition

2.0 Acoustic Emission Test Techniques

- 2.1 Factors affecting test equipment selection
  - 2.1.1 Material being monitored
  - 2.1.2 Location and nature of emission
  - 2.1.3 Type of information desired
  - 2.1.4 Size and shape of test part
- 2.2 Equipment standardization and setup for test
  - 2.2.1 Standardization signal generation techniques
  - 2.2.2 Standardization procedures
  - 2.2.3 Sensor selection and placement
  - 2.2.4 Adjustment of equipment controls
  - 2.2.5 Discrimination technique adjustments
- 2.3 Loading procedures
  - 2.3.1 Type of loading
  - 2.3.2 Maximum test load
  - 2.3.3 Load holds
  - 2.3.4 Repeated and programmed loadings
  - 2.3.5 Rate of loading
- 2.4 Special test procedures
  - 2.4.1 High-temperature/low-temperature tests
  - 2.4.2 Interrupted tests (including cyclic fatigue)
  - 2.4.3 Long-term tests
  - 2.4.4 Tests in high-noise environments
- 2.5 Data display
  - 2.5.1 Selection of display mode
  - 2.5.2 Use and reading of different kinds of display
- 2.6 Noise sources and pretest identification techniques
  - 2.6.1 Electromagnetic noise
  - 2.6.2 Mechanical noise
- 2.7 Precautions against noise
  - 2.7.1 Electrical shielding
  - 2.7.2 Electronic techniques
  - 2.7.3 Prevention of movement
  - 2.7.4 Attenuating materials and applications
- 2.8 Data interpretation
  - 2.8.1 Recognizing noise in the recorded data
  - 2.8.2 Noise elimination by data-filtering techniques
  - 2.8.3 Relevant and nonrelevant acoustic emission response
- 2.9 Data evaluation
  - 2.9.1 Methods for ranking, grading, accepting/rejecting
  - 2.9.2 Comparison with standardization signals
  - 2.9.3 Source evaluation by complementary NDT methods
- 2.10 Reports
  - 2.10.1 Purpose
  - 2.10.2 Content and structure

3.0 Codes, Standards, Procedures, and Societies

- 3.1 Guide-type standards (glossaries, user manuals, etc.)
- 3.2 Standardized/codified AE procedures
- 3.3 User-developed test procedures
- 3.4 Societies active in AE

4.0 Applications of AE (course should include at least three categories from 4.1 and at least four categories from 4.2)

- 4.1 Laboratory studies (material characterization)
  - 4.1.1 Crack growth and fracture mechanics
  - 4.1.2 Environmentally assisted cracking
  - 4.1.3 Dislocation movement (metals)
  - 4.1.4 Clarifying deformation mechanisms (composites)
  - 4.1.5 Phase transformation and phase stability
  - 4.1.6 Creep
  - 4.1.7 Residual stress
  - 4.1.8 Corrosion
  - 4.1.9 Fatigue
  - 4.1.10 Rupture
  - 4.1.11 Ductile/brittle transition
  - 4.1.12 Other material characterization applications
- 4.2 Structural applications
  - 4.2.1 Pressure vessels (metal)
  - 4.2.2 Storage tanks (metal)
  - 4.2.3 Pressure vessels/storage tanks (composite)
  - 4.2.4 Piping and pipelines
  - 4.2.5 Bucket trucks
  - 4.2.6 Aircraft
  - 4.2.7 Bridges
  - 4.2.8 Mines
  - 4.2.9 Dams, earthen slopes
  - 4.2.10 Pumps, valves, etc.
  - 4.2.11 Rotating plant
  - 4.2.12 In-process weld monitoring
  - 4.2.13 Leak detection and monitoring
  - 4.2.14 Other structural applications

Acoustic Emission Testing Level III Topical Outline

1.0 Principles and Theory

- 1.1 Characteristics of AE
  - 1.1.1 Concepts of source, propagation, loading, measurement, display, and evaluation
  - 1.1.2 Proper selection of acoustic emission as method of choice
    - 1.1.2.1 Differences between AE and other techniques
    - 1.1.2.2 Complementary roles of AE and other methods
    - 1.1.2.3 Potential or conflicting results between methods
    - 1.1.2.4 Factors that qualify/disqualify the use of AE
  - 1.1.3 Math review (exponents, logarithms, metric units, and conversions)
- 1.2 Materials and deformation
  - 1.2.1 Materials constitution
    - 1.2.1.1 Crystalline/noncrystalline
    - 1.2.1.2 Metals/composites/other
  - 1.2.2 Stress and strain (including triaxial, residual, and thermal)
  - 1.2.3 Elastic and plastic deformation, crack growth
  - 1.2.4 Materials properties (strength, toughness, etc.)

- 1.3 Sources of acoustic emission
  - 1.3.1 Broadband nature of source spectra
  - 1.3.2 Emission/signal levels, units of amplitude measurement
  - 1.3.3 Sources in crystalline materials
    - 1.3.3.1 Dislocations – plastic deformation
    - 1.3.3.2 Phase transformations
    - 1.3.3.3 Deformation twinning
    - 1.3.3.4 Nonmetallic inclusions
    - 1.3.3.5 Subcritical crack growth
      - 1.3.3.5.1 Subcritical crack growth under increasing load
      - 1.3.3.5.2 Ductile tearing under increasing load
      - 1.3.3.5.3 Fatigue crack initiation and growth
      - 1.3.3.5.4 Hydrogen embrittlement cracking
      - 1.3.3.5.5 Stress corrosion cracking
  - 1.3.4 Sources in nonmetals
    - 1.3.4.1 Microcracking
    - 1.3.4.2 Gross cracking
    - 1.3.4.3 Crazing
    - 1.3.4.4 Other sources in nonmetals
  - 1.3.5 Sources in composites
    - 1.3.5.1 Fiber breakage
    - 1.3.5.2 Matrix cracking
    - 1.3.5.3 Fiber-matrix debonding
    - 1.3.5.4 Delamination
    - 1.3.5.5 Fiber pull-out, relaxation
    - 1.3.5.6 Friction
  - 1.3.6 Other sources
    - 1.3.6.1 Pressure leaks, cavitation
    - 1.3.6.2 Oxide and scale cracking
    - 1.3.6.3 Slag cracking
    - 1.3.6.4 Frictional sources
    - 1.3.6.5 Liquefaction and solidification
    - 1.3.6.6 Loose parts, intermittent contact
    - 1.3.6.7 Fluids and nonsolids
    - 1.3.6.8 Crack closure
    - 1.3.6.9 Corrosion
- 1.4 Wave propagation
  - 1.4.1 Near-field impulse response
  - 1.4.2 Modes of propagation (including lamb waves)
  - 1.4.3 Mode conversion, reflection, and refraction
  - 1.4.4 Wave velocity in material (including velocity dispersion)
  - 1.4.5 Anisotropic propagation in composites
  - 1.4.6 Specimen geometry effects
- 1.5 Attenuation
  - 1.5.1 Geometric attenuation
  - 1.5.2 Dispersion
  - 1.5.3 Scattering, diffraction
  - 1.5.4 Effects of contained fluids
  - 1.5.5 Attenuation due to energy loss mechanisms
  - 1.5.6 Attenuation versus frequency
- 1.6 Kaiser and Felicity effects, and Felicity ratio
  - 1.6.1 In metals
  - 1.6.2 In composites
  - 1.6.3 Emission during load holds
- 1.7 Terminology (refer to AE Glossary, ASTM E1316)

2.0 Equipment and Materials

- 2.1 Transducing processes (piezoelectricity, etc.)
- 2.2 Sensors
  - 2.2.1 Construction
    - 2.2.1.1 Single-ended
    - 2.2.1.2 Differential
    - 2.2.1.3 Test environment considerations
    - 2.2.1.4 Wave mode sensitivity
  - 2.2.2 Conversion efficiencies, temperature effects
  - 2.2.3 Calibration
    - 2.2.3.1 Methods and significance
    - 2.2.3.2 Calculations from absolute sensitivity curve
  - 2.2.4 Reciprocity
- 2.3 Sensor attachment
  - 2.3.1 Coupling materials – selection and effective use
  - 2.3.2 Attachment devices
  - 2.3.3 Waveguides – design considerations, effect on signal
- 2.4 Sensor utilization
  - 2.4.1 Flat response sensors
  - 2.4.2 Resonant response sensors
  - 2.4.3 Integral-electronics sensors
  - 2.4.4 Special sensors (directional, mode responsive, accelerometers)
  - 2.4.5 Sensor selection
- 2.5 Simulated AE sources
  - 2.5.1 Hsu-Nielsen source (pencil lead break)
  - 2.5.2 Piezoelectric transducers and associated electronics
  - 2.5.3 Gas jet
  - 2.5.4 Other devices
- 2.6 Cables
  - 2.6.1 Cable types
    - 2.6.1.1 Coaxial
    - 2.6.1.2 Twisted pair
    - 2.6.1.3 Multiscreened
    - 2.6.1.4 Optical
    - 2.6.1.5 Others
  - 2.6.2 Shielding and other factors governing cable selection
  - 2.6.3 Cable length effects
  - 2.6.4 Noise problems in cables
  - 2.6.5 Cables as transmission lines
  - 2.6.6 Impedance matching
  - 2.6.7 Connectors

2.7 Signal conditioning
- 2.7.1 Preamplifiers (dynamic range, cable drive capability, etc.)
- 2.7.2 Amplifiers
- 2.7.3 Filters – selection, roll-off rates
- 2.7.4 Units of gain measurement
- 2.7.5 Electronic noise

2.8 Signal detection
- 2.8.1 Threshold comparator
- 2.8.2 Units of threshold measurement
- 2.8.3 Sensitivity determined by gain and/or threshold
- 2.8.4 Use of floating threshold
- 2.8.5 Dead time

2.9 Signal processing
- 2.9.1 Waveform characteristics
  - 2.9.1.1 Amplitude
  - 2.9.1.2 Pulse duration
  - 2.9.1.3 Rise time
  - 2.9.1.4 Signal strength (MARSE)
  - 2.9.1.5 Threshold crossing counts
  - 2.9.1.6 Hit versus event processing

2.10 Source location
- 2.10.1 Single-channel location
- 2.10.2 Linear location
- 2.10.3 Hit-sequence zonal location
- 2.10.4 Other location methods
- 2.10.5 Guard channels and spatial filtering

2.11 Advanced signal processing
- 2.11.1 Data filtering
- 2.11.2 Signal definition
- 2.11.3 Signal capture
- 2.11.4 Frequency analysis (Fourier theorem, theory of spectrum)
- 2.11.5 Pattern recognition
- 2.11.6 Source function determination by deconvolution/ Green's function

2.12 AE test systems
- 2.12.1 Single-channel systems
- 2.12.2 Multichannel systems
- 2.12.3 Dedicated industrial systems
- 2.12.4 Interpreting and writing system specifications

2.13 Accessory materials
- 2.13.1 Audio indicators
- 2.13.2 X-Y and strip-chart recording
- 2.13.3 Oscilloscopes
- 2.13.4 Magnetic recorders
- 2.13.5 Computers and their use
  - 2.13.5.1 Operating systems
  - 2.13.5.2 Data storage and transfer
  - 2.13.5.3 Data output
- 2.13.6 Others

2.14 Factors affecting test equipment selection
- 2.14.1 Material being monitored
- 2.14.2 Location and nature of emission
- 2.14.3 Type of information desired
- 2.14.4 Size and shape of test part

3.0 Techniques

3.1 Equipment standardization and setup for test
- 3.1.1 Standardization signal generation techniques
- 3.1.2 Standardization procedures
- 3.1.3 Sensor selection and placement
- 3.1.4 Adjustment of equipment controls
- 3.1.5 Discrimination technique adjustments

3.2 Establishing loading procedures
- 3.2.1 Type of loading
- 3.2.2 Maximum test load
- 3.2.3 Load holds
- 3.2.4 Repeated and programmed loadings
- 3.2.5 Rate of loading

3.3 Precautions against noise
- 3.3.1 Noise identification
  - 3.3.1.1 Electromagnetic noise
  - 3.3.1.2 Mechanical noise
- 3.3.2 Noise elimination/discrimination before test
  - 3.3.2.1 Electrical shielding
  - 3.3.2.2 Grounding
  - 3.3.2.3 Frequency filtering
  - 3.3.2.4 Gain and/or threshold adjustment
  - 3.3.2.5 Floating threshold
  - 3.3.2.6 Attenuating materials and applications
  - 3.3.2.7 Prevention of movement, friction
  - 3.3.2.8 Guard channels, spatial filtering
  - 3.3.2.9 Time-based and load-based gating
  - 3.3.2.10 Discrimination based on waveform characteristics

3.4 Special test procedures
- 3.4.1 High-temperature/low-temperature tests
- 3.4.2 Interrupted tests (including cyclic fatigue)
- 3.4.3 Long-term tests, permanent/continuous monitoring
- 3.4.4 Tests in high-noise environments

3.5 Data displays
- 3.5.1 Purpose and value of different displays
  - 3.5.1.1 Time-based and load-based plots
  - 3.5.1.2 Location displays
  - 3.5.1.3 Distribution functions
  - 3.5.1.4 Crossplots
  - 3.5.1.5 Other displays
- 3.5.2 Selection of displays

4.0 Machine Learning

4.1 Supervised and unsupervised learning
- 4.1.1 Reinforcement learning
- 4.1.2 Regression and classification problems
- 4.1.3 Clustering

4.2 Data representation in machine learning
- 4.2.1 Vectors
- 4.2.2 Matrices
- 4.2.3 Categorical
- 4.2.4 Time series

4.3 Mathematical functions and operations in machine learning
 4.3.1 Inner products
 4.3.2 Norms/distances
 4.3.3 Vector-matrix multiplications
 4.3.4 Correlation and its connection to inner product
 4.3.5 Discrete Fourier transformation
 4.3.6 Convolution/correlation
4.4 Basic concepts in statistics and probability
 4.4.1 Introduction
 4.4.2 Multivariate normal distribution
 4.4.3 Conditional probability
 4.4.4 Bayes' theorem and Bayesian inference
4.5 Optimization
 4.5.1 Introduction to objective functions
 4.5.2 Gradient descent algorithm
 4.5.3 Optimizing a linear regression model
4.6 Machine learning terminology
 4.6.1 Training and test setting
 4.6.2 Validation (set)
 4.6.3 Feature scaling
 4.6.4 Input and output size
 4.6.5 Weights and hyper-parameters
 4.6.6 Loss functions
 4.6.7 Mean squared error (MSE) and mean absolute error (MAE)
 4.6.8 True and false positives
 4.6.9 Precision and recall
 4.6.10 Confusion matrix
4.7 Overfitting and underfitting
4.8 Reproducibility and interpretability
4.9 Deep learning and transfer learning
4.10 Machine learning algorithms
 4.10.1 Linear regression
 4.10.2 K-means clustering
 4.10.3 Logistic regression
 4.10.4 Neural networks
 4.10.5 Decision trees
4.11 Application of machine learning to AE

5.0 Interpretation and Evaluation

5.1 Data interpretation
 5.1.1 Relevant and nonrelevant acoustic emission response
 5.1.2 Recognizing noise versus true acoustic emission in the recorded data
 5.1.3 Distribution function analysis
 5.1.4 Crossplot analysis
 5.1.5 Noise elimination – data-filtering techniques
  5.1.5.1 Spatial filtering
  5.1.5.2 Filtering on waveform characteristics
  5.1.5.3 Time-based and parametric-based filtering
5.2 Data evaluation
 5.2.1 Methods for ranking, grading, accepting/rejecting
 5.2.2 Comparison with standardization signals
 5.2.3 Source evaluation by complementary NDT methods
5.3 Reports
 5.3.1 Purpose
 5.3.2 Content and structure
 5.3.3 Developing a standard report format

6.0 Procedures

6.1 Guide-type standards (glossaries, user manuals, etc.)
6.2 Standardized/codified AE test procedures
6.3 User-developed test procedures
6.4 Societies active in AE
6.5 Interpretation of codes, standards, and procedures
6.6 Developing and writing AE test procedures
6.7 Training and examining Level I and II NDT personnel

7.0 Safety and Health

7.1 Hazards associated with structural failure during test
7.2 Other hazards associated with AE
7.3 Importance of local regulations

8.0 Applications

8.1 Laboratory studies (material characterization)
 8.1.1 Crack growth and fracture mechanics
 8.1.2 Environmentally assisted cracking
 8.1.3 Dislocation movement (metals) (strain rate and volume effects)
 8.1.4 Clarifying deformation mechanisms (composites)
 8.1.5 Phase transformation and phase stability
 8.1.6 Creep
 8.1.7 Residual stress
 8.1.8 Corrosion
 8.1.9 Fatigue
 8.1.10 Rupture
 8.1.11 Ductile/brittle transition
 8.1.12 Other material characterization applications
8.2 Structural applications
 8.2.1 Pressure vessels (metal)
 8.2.2 Storage tanks (metal)
 8.2.3 Pressure vessels/storage tanks (composite)
 8.2.4 Piping and pipelines
 8.2.5 Bucket trucks
 8.2.6 Aircraft
 8.2.7 Bridges
 8.2.8 Mines
 8.2.9 Dams, earthen slopes
 8.2.10 Pumps, valves, etc.
 8.2.11 Rotating plant
 8.2.12 In-process weld monitoring
 8.2.13 Leak detection and monitoring
 8.2.14 Other structural applications

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ET
ELECTROMAGNETIC TESTING TOPICAL OUTLINES

Alternating Current Field Measurement Testing Level I Topical Outline

Theory Course

1.0 Introduction to Electromagnetic Testing (ET)
- 1.1 Brief history of testing
- 1.2 Basic principles of NDT

2.0 Electromagnetic Theory
- 2.1 Eddy current theory
  - 2.1.1 Generation of eddy currents by means of an alternating current (AC) field
  - 2.1.2 Effects of fields created by eddy currents
  - 2.1.3 Properties of eddy currents
    - 2.1.3.1 Travel in circular direction
    - 2.1.3.2 Eddy current distribution
    - 2.1.3.3 Effects of liftoff and geometry
    - 2.1.3.4 Relationship of magnetic field in relation to a current in a coil
    - 2.1.3.5 Effects of permeability variations in magnetic materials
    - 2.1.3.6 Effect of discontinuities
    - 2.1.3.7 Relationship between frequency and depth of penetration
    - 2.1.3.8 Standard depths of penetration
- 2.2 Flux leakage theory
  - 2.2.1 Terminology and units
  - 2.2.2 Principles of magnetization
    - 2.2.2.1 *B-H* curve
    - 2.2.2.2 Magnetic properties
    - 2.2.2.3 Magnetic fields
    - 2.2.2.4 Magnetic permeability
    - 2.2.2.5 Factors affecting magnetic permeability
- 2.3 Basic electrical theory
  - 2.3.1 Basic units of electrical measurement
  - 2.3.2 Direct current circuits
  - 2.3.3 Ohm's law
  - 2.3.4 Faraday's law
  - 2.3.5 Resistance
  - 2.3.6 Inductance
  - 2.3.7 Magnetic effect of electrical currents

Technique Course

1.0 Alternating Current Field Measurement (ACFM) Theory
- 1.1 Production of uniform fields
- 1.2 Current flow, B_x, B_z, and B_y relationships
- 1.3 Relationship of the B_x, B_z, and butterfly plots
- 1.4 Other sources that influence the signals

2.0 Types of Probes
- 2.1 Coil arrangements
  - 2.1.1 Primary induction coil
  - 2.1.2 B_x and B_y sensor coils
- 2.2 Coil factors (liftoff)
- 2.3 Theory of operation
- 2.4 Applications
- 2.5 Limitations
- 2.6 Probe markings

3.0 Probe Software
- 3.1 Probe software versions and compatibility
- 3.2 Manufacturers' sensitivity settings
  - 3.2.1 Gain
  - 3.2.2 Scalings
  - 3.2.3 Relationship between gain and current settings
- 3.3 Sensitivity checks

4.0 Factors Affecting the Choice of Probes
- 4.1 Type of part to be inspected
- 4.2 Type of discontinuity to be inspected
- 4.3 Speed of testing required
- 4.4 Probable location of discontinuity

5.0 Types of Hardware and Operating Software Applications
- 5.1 Choice of systems for specific applications
- 5.2 Choice of software for specific applications
  - 5.2.1 Depth and length sizing capabilities
  - 5.2.2 Probe resolution
  - 5.2.3 Coating thickness

6.0 Scanning for Detection
- 6.1 Initial setup
- 6.2 Setting position indicators
- 6.3 Probe orientation
- 6.4 Scanning speed
- 6.5 Scanning pattern for tubulars and pipes
- 6.6 Scanning pattern for linear sections
- 6.7 Scanning for transverse cracks

7.0 Signal Interpretation
- 7.1 Review of display format
- 7.2 Detection and examination procedure
- 7.3 Crack signals – linear cracks, angled cracks, line contacts and multiple cracks, transverse cracks
- 7.4 Other signal sources – liftoff, geometry, materials, magnetism, edges, and corners

Alternating Current Field Measurement Testing Level II Topical Outline

Principles Course

1.0 Review of Electromagnetic Theory
- 1.1 Eddy current theory
- 1.2 ACFM theory
- 1.3 Types of ACFM sensing probes

2.0 Factors that Affect Depth of Penetration
- 2.1 Conductivity
- 2.2 Permeability
- 2.3 Frequency
- 2.4 Coil size

3.0 Factors that Affect ACFM
- 3.1 Residual fields
- 3.2 Defect geometry
- 3.3 Defect location – scanning pattern for attachments, corners, and ratholes
- 3.4 Defect orientation
- 3.5 Distance between adjacent defects

Techniques and Applications Course

1.0 Software Commands
- 1.1 Probe file production
  - 1.1.1 Selection of gain and frequency settings for specific applications
  - 1.1.2 Selection of current for specific applications
  - 1.1.3 Selections of sensitivity settings and scalings for specific applications
- 1.2 Standardization settings
  - 1.2.1 Alarm settings
  - 1.2.2 Butterfly plot scalings
- 1.3 Adjustment of communication rates

2.0 User Standards and Operating Procedures
- 2.1 Explanation of standards applicable to ACFM
- 2.2 Explanation of operating procedures applicable to ACFM

Eddy Current Testing Level I Topical Outline

Theory Course

1.0 Introduction to Eddy Current Testing
- 1.1 Historical and developmental process
  - 1.1.1 Founding fathers: Arago, Lenz, Faraday, and Maxwell
  - 1.1.2 Advances in electronics
- 1.2 Basic physics and controlling principles
  - 1.2.1 Varying magnetic fields
  - 1.2.2 Electromagnetic induction
  - 1.2.3 Primary and secondary force relationships

2.0 Electromagnetic Theory
- 2.1 Eddy current theory
  - 2.1.1 Generation of eddy currents by means of an AC field
  - 2.1.2 Effect of fields created by eddy currents (impedance changes)
  - 2.1.3 Effect of change of impedance on instrumentation
  - 2.1.4 Properties of eddy current
    - 2.1.4.1 Travel in circular direction
    - 2.1.4.2 Strongest on surface of test material
    - 2.1.4.3 Zero value at center of solid conductor placed in an alternating magnetic field
    - 2.1.4.4 Strength, time relationship, and orientation as functions of test-system parameters and test-part characteristics
    - 2.1.4.5 Small magnitude of current flow
    - 2.1.4.6 Relationships of frequency and phase
    - 2.1.4.7 Electrical effects, conductivity of materials
    - 2.1.4.8 Magnetic effects, permeability of materials
    - 2.1.4.9 Geometrical effects

3.0 Lab Demonstration
- 3.1 Generation of Z-curves with conductivity samples
- 3.2 Generation of liftoff curves

Basic Technique Course

1.0 Types of Eddy Current Sensing Elements
- 1.1 Probes
  - 1.1.1 Types of arrangements
    - 1.1.1.1 Probe coils
    - 1.1.1.2 Encircling coils
    - 1.1.1.3 Inside coils
  - 1.1.2 Modes of operation
    - 1.1.2.1 Absolute
    - 1.1.2.2 Differential
    - 1.1.2.3 Hybrids
  - 1.1.3 Theory of operation
  - 1.1.4 Hall effect sensors
    - 1.1.4.1 Theory of operation
    - 1.1.4.2 Differences between coil and Hall element systems

1.1.5 Applications
1.1.5.1 Measurement of material properties
1.1.5.2 Flaw detection
1.1.5.3 Geometrical features
1.1.6 Advantages
1.1.7 Limitations
1.2 Factors affecting choice of sensing elements
1.2.1 Type of part to be inspected
1.2.2 Type of discontinuity to be detected
1.2.3 Speed of testing required
1.2.4 Amount of testing (percentage) required
1.2.5 Probable location of discontinuity

2.0 Selection of Inspection Parameters
2.1 Frequency
2.2 Coil drive – current/voltage
2.3 Hall effect element drive – current/voltage
2.4 Channel gain
2.5 Display sensitivity selections
2.6 Standardization
2.7 Filtering
2.8 Thresholds

3.0 Readout Mechanisms
3.1 Calibrated or uncalibrated meters
3.2 Impedance plane displays
3.2.1 Analog
3.2.2 Digital
3.3 Data recording systems
3.4 Alarms, lights, etc.
3.5 Numerical readouts
3.6 Marking systems
3.7 Sorting gates and tables
3.8 Cutoff saw or shears
3.9 Automation and feedback

4.0 Lab Demonstration
4.1 Demo filter effects on rotating reference standards
4.2 Demo liftoff effects
4.3 Demo frequency effects
4.4 Demo rotational and forward speed effects
4.5 Generate a Z-curve with conductivity standards

Eddy Current Testing Level II Topical Outline

Principles Course

1.0 Review of Electromagnetic Theory
1.1 Eddy current theory
1.2 Types of eddy current sensing probes

2.0 Factors that Affect Coil Impedance
2.1 Test part
2.1.1 Conductivity
2.1.2 Permeability
2.1.3 Mass
2.1.4 Homogeneity
2.2 Test system
2.2.1 Frequency
2.2.2 Coupling
2.2.3 Field strength
2.2.4 Test coil and shape
2.2.5 Hall effect elements

3.0 Signal-to-Noise Ratio
3.1 Definition
3.2 Relationship to eddy current testing
3.3 Methods of improving signal-to-noise ratio (SNR)

4.0 Selection of Test Frequency
4.1 Relationship of frequency to type of test
4.2 Considerations affecting choice of test
4.2.1 SNR
4.2.2 Causes of noise
4.2.3 Methods to reduce noise
4.2.3.1 DC saturation
4.2.3.2 Shielding
4.2.3.3 Grounding
4.2.4 Phase discrimination
4.2.5 Response speed
4.2.6 Skin effect

5.0 Coupling
5.1 Fill factor
5.2 Liftoff

6.0 Field Strength and its Selection
6.1 Permeability changes
6.2 Saturation
6.3 Effect of AC field strength on eddy current testing

7.0 Instrument Design Considerations
7.1 Amplification
7.2 Phase detection
7.3 Differentiation of filtering

Techniques and Applications Course

1.0 User Standards and Operating Procedures
1.1 Explanation of standards and specifications used in eddy current testing

2.0 Inspection System Output
2.1 Accept/reject criteria
2.1.1 Sorting, go/no-go
2.2 Signal classification processes
2.2.1 Discontinuity
2.2.2 Flaw
2.3 Detection of signals of interest
2.3.1 Near surface
2.3.2 Far surface
2.4 Flaw-sizing techniques
2.4.1 Phase to depth
2.4.2 Volts to depth
2.5 Calculation of flaw frequency

2.6 Sorting for properties related to conductivity
2.7 Thickness evaluation
2.8 Measurement of ferromagnetic properties
2.8.1 Comparative circuits

Remote Field Testing Level I Topical Outline

Theory Course

1.0 Introduction to Remote Field Testing (RFT)
1.1 Historical and developmental process
1.1.1 Founding fathers: McLean, Schmidt, Atherton, and Lord
1.1.2 The computer age and its effect on the advancement of RFT
1.2 Basic physics and controlling principles
1.2.1 Varying magnetic fields
1.2.2 Electromagnetic induction
1.2.3 Primary and secondary field relationships

2.0 Electromagnetic Theory
2.1 Generation of eddy currents in conductors
2.2 Eddy current propagation and decay, standard depth of penetration
2.3 Near field, transition, and remote field zones
2.4 Properties of remote field eddy currents
2.4.1 Through-transmission nature
2.4.2 Magnetic flux is predominant energy
2.4.3 The ferrous tube as a waveguide
2.4.4 Strength of field in three zones
2.4.5 External field is source of energy in remote field
2.4.6 Factors affecting phase lag and amplitude
2.4.7 Geometric factors – fill factor, external support plates, tube sheets
2.4.8 Speed of test, relationship to thickness, frequency, conductivity, and permeability
2.4.9 Effect of deposits – magnetite, copper, calcium
2.4.10 RFT in nonferrous tubes

Basic Technique Course

1.0 Types of Remote Field Sensing Elements
1.1 Probes
1.1.1 Types of arrangements
1.1.1.1 Absolute bobbin coils
1.1.1.2 Differential bobbin coils
1.1.1.3 Arrays
1.1.2 Modes of operation
1.1.2.1 RFT voltage plane and reference curve
1.1.2.2 X-Y voltage plane
1.1.2.3 Chart recordings
1.1.3 Theory of operation
1.1.4 Applications
1.1.4.1 Heat exchanger and boiler tubes
1.1.4.2 Pipes and pipelines
1.1.4.3 External and through-transmission probes
1.1.5 Advantages
1.1.5.1 Equal sensitivity to internal and external flaws
1.1.5.2 Easy to understand – increasing depth of flaw signals rotate CCW
1.1.6 Limitations
1.1.6.1 Speed
1.1.6.2 Difficult to differentiate internal versus external flaws
1.1.6.3 Small signals from small-volume flaws
1.1.6.4 Finned tubes
1.2 Factors affecting choice of probe type
1.2.1 Differential for small-volume flaws (e.g., pits)
1.2.2 Absolute for large-area defects (e.g., steam erosion, fretting)
1.2.3 Test (probe travel) speed
1.2.4 Single versus dual exciters and areas of reduced sensitivity
1.2.5 Bobbin coils and solid-state sensors
1.2.6 Finned tubes

2.0 Selection of Inspection Parameters
2.1 Frequency
2.2 Coil drive – current/voltage
2.3 Pre-amp gain
2.4 Display gain
2.5 Standardization

3.0 Readout Mechanisms
3.1 Display types
3.1.1 RFT voltage plane displays
3.1.2 Voltage vector displays
3.2 RFT reference curve
3.3 Chart recordings
3.4 Odometers
3.5 Storing and recalling data on computers

Principles Course

1.0 Review of Electromagnetic Theory
1.1 RFT theory
1.2 Types of RFT sensing probes

2.0 Factors that Affect Coil Impedance
2.1 Test part
2.1.1 Conductivity
2.1.2 Permeability
2.1.3 Mass
2.1.4 Homogeneity
2.2 Test system
2.2.1 Frequency
2.2.2 Coupling (fill factor)
2.2.3 Field strength (drive volts, frequency)
2.2.4 Coil shapes

3.0 Signal-to-Noise Ratio
- 3.1 Definition
- 3.2 Relationship to RFT
- 3.3 Methods of improving SNR
  - 3.3.1 Speed
  - 3.3.2 Fill factor
  - 3.3.3 Frequency
  - 3.3.4 Filters
  - 3.3.5 Drive
  - 3.3.6 Shielding
  - 3.3.7 Grounding (3.3.6 and 3.3.7 also apply to other methods)

4.0 Selection of Test Frequency
- 4.1 Relationship of frequency to depth of penetration
- 4.2 Relationship of frequency to resolution
- 4.3 Dual-frequency operation
- 4.4 Beat frequencies
- 4.5 Optimum frequency

5.0 Coupling
- 5.1 Fill factor
- 5.2 Importance of centralizing the probe

6.0 Field Strength
- 6.1 Probe drive and penetration
- 6.2 Effect of increasing thickness, conductivity, or permeability
- 6.3 Position of receive coils versus field strength

7.0 Instrument Design Considerations
- 7.1 Amplification
- 7.2 Phase and amplitude detection (lock-in amplifier)
- 7.3 Differentiation and filtering

Remote Field Testing Level II Topical Outline

Techniques and Applications Course

1.0 Equipment
- 1.1 Probes
  - 1.1.1 Absolute bobbin coils
  - 1.1.2 Differential bobbin coils
  - 1.1.3 Arrays
  - 1.1.4 Dual exciter or dual detector probes
  - 1.1.5 Solid-state sensors
  - 1.1.6 External probes
  - 1.1.7 Effect of fill factor
  - 1.1.8 Centralizing the probe
  - 1.1.9 Quality of the "ride"
  - 1.1.10 Cable length considerations
  - 1.1.11 Preamplifiers – internal and external
- 1.2 Instruments
  - 1.2.1 Measuring phase and amplitude
  - 1.2.2 Displays – RFT, voltage plane, impedance plane differences
  - 1.2.3 Chart recordings
  - 1.2.4 Storing, retrieving, archiving data
  - 1.2.5 Standardization frequency
- 1.3 Reference standards
  - 1.3.1 Material
  - 1.3.2 Thickness
  - 1.3.3 Size
  - 1.3.4 Heat treatment
  - 1.3.5 Simulated defects
  - 1.3.6 ASTM E2096
  - 1.3.7 How often to standardize

2.0 Techniques
- 2.1 Factors affecting signals
  - 2.1.1 Probe speed/smoothness of travel
  - 2.1.2 Depth, width, and length of flaw versus probe footprint
  - 2.1.3 Probe drive, pre-amp gain, view gain, filters
  - 2.1.4 Position of flaw versus other objects (e.g., support plates)
  - 2.1.5 Fill factor
  - 2.1.6 SNR
  - 2.1.7 Thickness, conductivity, and permeability of the tube
  - 2.1.8 Correct display of the signal
- 2.2 Selection of test frequencies
  - 2.2.1 Single or dual or multifrequency
  - 2.2.2 Sharing the time slice
  - 2.2.3 Number of readings per cycle
  - 2.2.4 Beat frequencies, harmonics, and base frequencies
  - 2.2.5 Optimum frequency
  - 2.2.6 Saturating the input amplifier (large-volume defects)
  - 2.2.7 Small-volume defects – optimizing the settings to detect

3.0 Applications
- 3.1 Tubulars using internal probes
  - 3.1.1 Heat exchanger tubes
  - 3.1.2 Boiler tubes
  - 3.1.3 Pipes
  - 3.1.4 Pipelines
  - 3.1.5 Furnace tubes
- 3.2 Tubulars using external probes
  - 3.2.1 Boiler tubes
  - 3.2.2 Process pipes
  - 3.2.3 Pipelines
  - 3.2.4 Structural pipes
- 3.3 Other applications
  - 3.3.1 Flat plate
  - 3.3.2 Finned tubes
  - 3.3.3 Hydrogen furnace tubes
  - 3.3.4 Nonferrous tubes and pipes
  - 3.3.5 Cast-iron water mains
  - 3.3.6 Oil well casings

4.0 Inspection System Output
- 4.1 Accept/reject criteria
  - 4.1.1 Customer specified or code specified
- 4.2 Signal classification processes
  - 4.2.1 Discontinuity
  - 4.2.2 Flaw
- 4.3 Detection of signals of interest
  - 4.3.1 Near/under support plates and tubesheets
  - 4.3.2 Flaws in the free span
  - 4.3.3 Internal and external flaws
  - 4.3.4 Recognition of signals from nonflaws
- 4.4 Signal recognition, data analysis, and flaw-sizing techniques
  - 4.4.1 Understanding the RFT reference curve and using it for flaw sizing
  - 4.4.2 Using phase angle to calculate flaw depth on the X-Y display
  - 4.4.3 Coil footprint considerations

Electromagnetic Testing Level III Topical Outline

Eddy Current Testing

1.0 Principles/Theory
- 1.1 Eddy current theory
  - 1.1.1 Generation of eddy currents
  - 1.1.2 Effect of fields created by eddy currents (impedance changes)
  - 1.1.3 Properties of eddy currents
    - 1.1.3.1 Travel mode
    - 1.1.3.2 Depth of penetration
    - 1.1.3.3 Effects of test-part characteristics – conductivity and permeability
    - 1.1.3.4 Current flow
    - 1.1.3.5 Frequency and phase
    - 1.1.3.6 Effects of permeability variations – noise
    - 1.1.3.7 Effects of discontinuity orientation

2.0 Equipment/Materials
- 2.1 Probes – general
  - 2.1.1 Advantages/limitations
- 2.2 Through, encircling, or annular coils and Hall effect elements
  - 2.2.1 Advantages/limitations/differences
- 2.3 Factors affecting choice of sensing elements
  - 2.3.1 Type of part to be inspected
  - 2.3.2 Type of discontinuity to be detected
  - 2.3.3 Speed of testing required
  - 2.3.4 Amount of testing required
  - 2.3.5 Probable location of discontinuity
  - 2.3.6 Applications other than discontinuity detection
- 2.4 Readout selection
  - 2.4.1 Meter
  - 2.4.2 Oscilloscope, X-Y, and other displays
  - 2.4.3 Alarm, lights, etc.
  - 2.4.4 Strip-chart recorder
- 2.5 Instrument design considerations
  - 2.5.1 Amplification
  - 2.5.2 Phase detection
  - 2.5.3 Differentiation or filtering
  - 2.5.4 Thresholds, box gates, etc.

3.0 Techniques/Standardization
- 3.1 Factors that affect coil impedance
  - 3.1.1 Test part
  - 3.1.2 Test system
- 3.2 Selection of test frequency
  - 3.2.1 Relation of frequency to type of test
  - 3.2.2 Consideration affecting choice of test
    - 3.2.2.1 SNR
    - 3.2.2.2 Phase discrimination
    - 3.2.2.3 Response speed
    - 3.2.2.4 Skin effect
- 3.3 Coupling
  - 3.3.1 Fill factor
  - 3.3.2 Liftoff
- 3.4 Field strength
  - 3.4.1 Permeability changes
  - 3.4.2 Saturation
  - 3.4.3 Effect of AC field strength on eddy current testing
- 3.5 Comparison of techniques
- 3.6 Standardization
  - 3.6.1 Techniques
  - 3.6.2 Reference standards
- 3.7 Techniques – general
  - 3.7.1 Thickness
  - 3.7.2 Sorting
  - 3.7.3 Conductivity
  - 3.7.4 Surface or subsurface flaw detection
  - 3.7.5 Tubing

4.0 Interpretation/Evaluation
- 4.1 Flaw detection
- 4.2 Sorting for properties
- 4.3 Thickness gauging
- 4.4 Process control
- 4.5 General interpretations

5.0 Procedures

Remote Field Testing

1.0 RFT Principles and Theories
- 1.1 Three zones in RFT
  - 1.1.1 Near field (direct field)
  - 1.1.2 Transition zone
  - 1.1.3 Remote field zone
- 1.2 Through-transmission nature of RFT
- 1.3 Standard depth of penetration factors
  - 1.3.1 Thickness
  - 1.3.2 Permeability
  - 1.3.3 Conductivity
  - 1.3.4 Frequency
  - 1.3.5 Geometry

1.4 Signal analysis
1.5 Display options
   1.5.1 Voltage plane (polar coordinates)
   1.5.2 X-Y display (rectilinear coordinates)
   1.5.3 Chart recordings – phase, log-amplitude, magnitude, X-Y
1.6 Advanced applications
   1.6.1 Array probes
   1.6.2 Large pipes
   1.6.3 Flat plates
   1.6.4 Nonferrous applications
   1.6.5 Effects of tilt and shields
   1.6.6 Effects of cores and magnets

2.0 Codes and Practices
2.1 Writing procedures
2.2 ASTM E2096
2.3 SNT-TC-1A
   2.3.1 Responsibility of Level III
2.4 Supervision and training
2.5 Administering exams
2.6 Ethics
2.7 Reports – essential elements, legal responsibility

Alternating Current Field Measurement Testing

1.0 Principles and Theory
1.1 Generation of eddy currents
1.2 Effect of fields created by eddy currents
1.3 Properties of eddy currents
   1.3.1 Depth of penetration
   1.3.2 Effects of test-part characteristics
   1.3.3 Current flow
   1.3.4 Frequency
   1.3.5 Effects of permeability variations
   1.3.6 Effects of discontinuity orientation

2.0 Equipment and Materials
2.1 Alternating current measurement probes, general
   2.1.1 Advantages and limitations
2.2 Factors affecting choice of probes
   2.2.1 Type of part to be inspected
   2.2.2 Type of discontinuity to be inspected
   2.2.3 Speed of testing required
   2.2.4 Amount of testing required
   2.2.5 Probable location of discontinuity
   2.2.6 Applications other than discontinuity detection
2.3 Techniques/equipment sensitivity
   2.3.1 Selection of test frequency
   2.3.2 Selection of correct probe scalings in relation to the test
   2.3.3 Selection of correct communication rates

3.0 Interpretation and Evaluation of Signals
3.1 Flaw detection

4.0 Procedures

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GPR

GROUND PENETRATING RADAR TOPICAL OUTLINES

Ground Penetrating Radar Level I Topical Outline

Theory Course

1.0 Introduction to Ground Penetrating Radar (GPR)
- 1.1 Radar
  - 1.1.1 Reflection
  - 1.1.2 Radar equation
  - 1.1.3 Polarization
  - 1.1.4 Interference
- 1.2 History of GPR

2.0 Electromagnetic Theory
- 2.1 Electromagnetic wave propagation
- 2.2 Velocity
- 2.3 Wavelength
- 2.4 Interference
  - 2.4.1 Attenuation
  - 2.4.2 Dispersion
  - 2.4.3 Noise
  - 2.4.4 Clutter
- 2.5 Electrical properties
  - 2.5.1 Relative dielectric permittivity
  - 2.5.2 Electrical conductivity
  - 2.5.3 Dielectric materials
    - 2.5.3.1 Conductors
    - 2.5.3.2 Insulators
    - 2.5.3.3 Semiconductors
  - 2.5.4 Types of materials
    - 2.5.4.1 Soil
    - 2.5.4.2 Concrete
    - 2.5.4.3 Rocks
    - 2.5.4.4 Water – salt/fresh
    - 2.5.4.5 Ice
    - 2.5.4.6 Others
- 2.6 Magnetic properties in materials
  - 2.6.1 Ferromagnetic
  - 2.6.2 Paramagnetic
  - 2.6.3 Super paramagnetic

3.0 GPR Equipment
- 3.1 Antennas
  - 3.1.1 Polarization
  - 3.1.2 Fresnel reflection
  - 3.1.3 Snell angle
  - 3.1.4 Near field/far field
  - 3.1.5 Frequencies
- 3.2 Coupling
  - 3.2.1 Impedance matching
  - 3.2.2 Unloading
  - 3.2.3 Ringing
- 3.3 Waveguides

Basic Techniques Course

1.0 Surveys
- 1.1 Defining the objectives
- 1.2 Antenna selection and orientation
  - 1.2.1 Speed
  - 1.2.2 Frequency
  - 1.2.3 Application
  - 1.2.4 Materials
  - 1.2.5 Targets
- 1.3 Depth of penetration
  - 1.3.1 Gain controls
  - 1.3.2 Sensitivity controls
  - 1.3.3 Standardization
- 1.4 Range settings
- 1.5 Filter settings
- 1.6 Scanning parameters
  - 1.6.1 Mapping
  - 1.6.2 Grid layout
  - 1.6.3 Spacing

2.0 Applications
- 2.1 Test methods
- 2.2 Advantages
- 2.3 Limitations

3.0 Data Display and Interpretation
- 3.1 Material properties
- 3.2 Layer reflection
  - 3.2.1 Trench effect
- 3.3 Target reflection
  - 3.3.1 Point targets
- 3.4 Detection accuracy
- 3.5 Horizontal accuracy and resolution
- 3.6 Depth accuracy and resolution
- 3.7 Measurement techniques

Ground Penetrating Radar Level II Topical Outline

Principles and Applications Course

1.0 Review of Electromagnetic Theory
- 1.1 Radar equation
- 1.2 Stokes vector
- 1.3 Mueller matrix
- 1.4 Poincaré sphere

2.0 Types of Tests

3.0 Factors Affecting Tests
- 3.1 Depth of investigation
- 3.2 Orientation
- 3.3 Noise
  - 3.3.1 Signal-to-noise ratios
  - 3.3.2 Causes
  - 3.3.3 Filters
- 3.4 Interferences
- 3.5 Noninvasive surfaces

4.0 Field Strength
- 4.1 Antenna drive
- 4.2 Effects of conductivity
- 4.3 Permeability effects
- 4.4 Ground truth
- 4.5 Hyperbolic shape analysis

5.0 Instrument Design Considerations
- 5.1 Waveguides
- 5.2 Multipathing
- 5.3 Near-field and far-field factors
- 5.4 Resonance
- 5.5 Resolution

6.0 Data
- 6.1 Data acquisition
- 6.2 Data processing
  - 6.2.1 Displays
- 6.3 Modeling
- 6.4 Interpretation
  - 6.4.1 Uncertainty

Ground Penetrating Radar Level III Topical Outline

Theory Course

1.0 Introduction
- 1.1 History
- 1.2 Applications

2.0 Systems Design
- 2.1 Range
  - 2.1.1 Antenna loss
  - 2.1.2 Transmission loss
  - 2.1.3 Coupling loss
  - 2.1.4 Mismatch
  - 2.1.5 Target scatter
  - 2.1.6 Material attenuation
- 2.2 Velocity propagation
- 2.3 Clutter
- 2.4 Resolution
  - 2.4.1 Depth
  - 2.4.2 Plan

3.0 Modeling
- 3.1 Time domain models
- 3.2 Antenna radiation
- 3.3 Numerical modeling
  - 3.3.1 Schemes
  - 3.3.2 Applications
  - 3.3.3 Absorbing boundary conditions properties
- 3.4 Target shape effects

4.0 Material Properties
- 4.1 Dielectric materials
- 4.2 Ice, water
- 4.3 Soils, rocks
- 4.4 Suitability
- 4.5 Manufactured materials
- 4.6 Techniques
- 4.7 Measurement techniques

5.0 Antennas
- 5.1 Types
  - 5.1.1 Element
  - 5.1.2 Traveling wave
  - 5.1.3 Impulse radiating
  - 5.1.4 Frequency dependent
  - 5.1.5 Horn
  - 5.1.6 Dielectric antennas
- 5.2 Arrays
- 5.3 Polarization

6.0 Signal Modulation
- 6.1 Ultra-wideband signal resolution
  - 6.1.1 Waveform characteristics
  - 6.1.2 Signals
    - 6.1.2.1 Time domain
    - 6.1.2.2 Noise
    - 6.1.2.3 Comparisons
    - 6.1.2.4 Spectra comparisons
- 6.2 Amplitude modulation
- 6.3 Continuous wave frequency modulation
- 6.4 Polarization modulation

7.0 Signal processing
- 7.1 A-scan processing
  - 7.1.1 Zero offset
  - 7.1.2 Noise
  - 7.1.3 Clutter
  - 7.1.4 Gain
  - 7.1.5 Filtering
  - 7.1.6 Target resonances
- 7.2 B-scan processing
- 7.3 C-scan processing
- 7.4 Migration
- 7.5 Image processing
- 7.6 Deconvolution
- 7.7 Data acquisition
- 7.8 Data processing
- 7.9 Microwave
- 7.10 Clutter
- 7.11 Anomalies

Applications Course

1.0 Archeology
- 1.1 Application
- 1.2 Identification

2.0 Civil Engineering
- 2.1 Roads and pavement
- 2.2 Conduits
- 2.3 Techniques
- 2.4 Concrete
  - 2.4.1 Walls
  - 2.4.2 Reinforcement
  - 2.4.3 Dowels and anchors
- 2.5 Buildings
  - 2.5.1 Masonry
  - 2.5.2 Floors
  - 2.5.3 Walls
  - 2.5.4 Joints
- 2.6 Tunnels

3.0 Forensic Applications
- 3.1 Principles of search
- 3.2 Methods
- 3.3 Graves
- 3.4 Remains
- 3.5 Excavation
- 3.6 Examples

4.0 Geophysical Applications
- 4.1 Radar systems
- 4.2 Frozen materials
  - 4.2.1 Short pulse and frequency-modulated continuous-wave radar
  - 4.2.2 Firm layering and isochrones
  - 4.2.3 Crevasses detection
  - 4.2.4 Hydraulic pathways
  - 4.2.5 Topography
  - 4.2.6 Lake ice
- 4.3 Polythermal glaciers
  - 4.3.1 Radar system
  - 4.3.2 Bottom topography
  - 4.3.3 Internal structure
  - 4.3.4 Snow cover
- 4.4 Oil spills
- 4.5 Contaminations
- 4.6 Soil erosion
- 4.7 Coal and salt
- 4.8 Rocks

5.0 Borehole Radar (Long Distance)
- 5.1 Radar design
- 5.2 Data
- 5.3 Subsurface fracture characterization
- 5.4 Radar polarimetry
- 5.5 Specifications
- 5.6 Data acquisition
- 5.7 Problems
- 5.8 Signal/image processing
- 5.9 Electromagnetic modeling

6.0 Mine Detection
- 6.1 Humanitarian and military programs
  - 6.1.1 United States
  - 6.1.2 Europe
  - 6.1.3 Others
- 6.2 Performance assessment
- 6.3 Mine detection
  - 6.3.1 Handheld
  - 6.3.2 Vehicle mounted
  - 6.3.3 Airborne
  - 6.3.4 Data processing
  - 6.3.5 Clutter characteristics and removal

7.0 Utilities
- 7.1 Technology
- 7.2 Pipes and cables
- 7.3 Pipe hawk
- 7.4 Mapping
- 7.5 Drainage
- 7.6 Inspection of pipe

8.0 Remote Sensing
- 8.1 Airborne systems
- 8.2 Satellite systems
- 8.3 Planetary exploration
  - 8.3.1 Interplanetary body measurement
  - 8.3.2 Scientific objectives
  - 8.3.3 Models
  - 8.3.4 Performance

9.0 Equipment
- 9.1 Survey methods
- 9.2 Site characteristics
- 9.3 Surface characteristics
- 9.4 Material characteristics
- 9.5 Target characteristics

10.0 Regulations, Radiological Aspects, and Electromagnetic Compatibility (EMC)
- 10.1 US regulations
- 10.2 European regulations
- 10.3 Radiological aspects
- 10.4 EMC

GROUND PENETRATING RADAR METHOD LEVEL I, II, AND III TRAINING REFERENCES

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* Available from The American Society for Nondestructive Testing Inc., Columbus, OH.

GW

GUIDED WAVE TESTING TOPICAL OUTLINES

Guided Wave Testing Level I Topical Outline

Note: It is recommended that the trainee receive instruction in this course prior to performing work in guided wave testing (GW).

GW, or long-range ultrasonic testing, is uniquely different and specialized such that it should be considered a separate method for Level I and II personnel. The intent of this document is to provide the basic body of knowledge requirements for GW of piping that is consistent with other methods and not intended to replace specific training or schemes identified by the various equipment manufacturers.

1.0 Introduction
- 1.1 Basic understanding of nondestructive testing
- 1.2 Responsibilities of levels of certification
- 1.3 Terminology of GW
- 1.4 History of GW
- 1.5 Product technology
  - 1.5.1 Pipe designs and manufacturing processes
  - 1.5.2 Process versus service-induced defects

2.0 Basic Principles of GW
- 2.1 Utility of GW
- 2.2 Characteristics of guided wave propagation
- 2.3 Basic concepts of GW of piping
- 2.4 Various types of guided waves

3.0 Equipment
- 3.1 Transmission and reception of guided waves
- 3.2 Piezoelectric effect
- 3.3 Magnetostriction effect
- 3.4 Transduction
  - 3.4.1 Pulser-receiver unit(s)
  - 3.4.2 Sensor rings
  - 3.4.3 Types of sensors
  - 3.4.4 Array arrangement
  - 3.4.5 Directionality
  - 3.4.6 Frequency limits
  - 3.4.7 Dead zone and near field
  - 3.4.8 Influence of transduction and frequency on inspection
  - 3.4.9 Guided wave focusing
  - 3.4.10 Factors influencing selection of test conditions
  - 3.4.11 Influence of pipe geometry and pipe configuration
- 3.5 Cables
- 3.6 Computer interface
- 3.7 Electromagnetic acoustic transducers (EMATs)

4.0 Testing Techniques
- 4.1 Pulse-echo
- 4.2 Through-transmission

5.0 Standardization
- 5.1 Distance standardization
- 5.2 Amplitude standardization distance amplitude correction (DAC) and time corrected gain (TCG)

6.0 Procedures
- 6.1 Specific applications
  - 6.1.1 Weld/feature location
  - 6.1.2 Corrosion and wall loss evaluation
    - 6.1.2.1 Aboveground piping – uninsulated
    - 6.1.2.2 Aboveground piping – insulated
- 6.2 Basic data collection
  - 6.2.1 Choosing a test location
  - 6.2.2 Selection of wave modes
  - 6.2.3 Selection of frequency
- 6.3 Basic data evaluation
  - 6.3.1 Data quality
  - 6.3.2 Influence of geometry and structure
  - 6.3.3 Recognition of symmetrical features
  - 6.3.4 Distance-amplitude correction
  - 6.3.5 Recognition of nonsymmetrical features
  - 6.3.6 Influence of coatings, linings, and pipe condition of pulse-echo analysis
  - 6.3.7 Recognizing false indications
- 6.4 Basic reporting
  - 6.4.1 A-scan presentations
  - 6.4.2 Feature list

Guided Wave Testing Level II Topical Outline

1.0 Introduction
- 1.1 Review of Level I knowledge
- 1.2 Terminology of GW
- 1.3 Level II responsibilities
- 1.4 Product technology
  - 1.4.1 Various types of pipeline supports
  - 1.4.2 Advanced information on girth welds and other accessories welded to pipe (including typical defects)
  - 1.4.3 Structural integrity of pipelines – codes and standards for the specific sector

2.0 Principles of GW
- 2.1 Various types of guided wave modes (deeper knowledge)
- 2.2 Factors influencing selection of test parameters
- 2.3 Sensitivity to cross-sectional changes
- 2.4 Effect of feature geometry
- 2.5 Transducer configuration

3.0 Equipment
- 3.1 Advanced transduction configuration
- 3.2 Hardware and software requirements for optimization of test parameters
- 3.3 Data presentation (deeper knowledge)

4.0 Procedures
- 4.1 Contents and requirements of instructions, procedures, and standards
- 4.2 Preparation of written instructions
- 4.3 Cased piping (road crossings)
- 4.4 Buried piping
- 4.5 Coated piping

5.0 Advanced GW Data Analysis

6.0 Evaluation
- 6.1 Comparison procedures
  - 6.1.1 Standards and references
  - 6.1.2 Application of results of other NDT methods
- 6.2 Object appraisal
  - 6.2.1 History of pipe
  - 6.2.2 Intended use of pipe
  - 6.2.3 Existing and applicable code interpretation
  - 6.2.4 Type of discontinuity and location
- 6.3 Follow-up

7.0 Reporting
- 7.1 Influence of frequency
- 7.2 Influence of focusing
- 7.3 Estimation of indication severity

Guided Wave Testing Level III Topical Outline

1.0 Introduction
- 1.1 Terminology of GW
- 1.2 Level III responsibilities
  - 1.2.1 Preparation of inspection procedures
  - 1.2.2 Preparation of training materials for Level I and Level II
  - 1.2.3 Developing material for Level I and Level II tests including written and practical tests

2.0 Principles of GW
- 2.1 Review of mathematical basics
  - 2.1.1 Advanced GW propagation theory
  - 2.1.2 Dispersion effect and compensation factors
  - 2.1.3 Effect of material properties
  - 2.1.4 Bi-layer systems
  - 2.1.5 Attenuation due to viscoelastic coatings and embedded medium (parameters affecting and mathematical prediction)
  - 2.1.6 Sensitivity to stiffness changes
  - 2.1.7 Properties of guided waves in pipes and plates
- 2.2 Various types of GW modes
  - 2.2.1 Torsional, longitudinal, and flexural
  - 2.2.2 Modes in bends

3.0 Nonconventional Applications
- 3.1 Advanced array configuration
- 3.2 Transduction selection parameters
- 3.3 Advanced standardization systems
- 3.4 Underwater inspection
- 3.5 Guided wave monitoring
- 3.6 Guided wave focusing
- 3.7 Advanced transduction systems

4.0 Interpretations/Evaluations
- 4.1 Identification of discontinuities in various industrial environments
- 4.2 Variables affecting test results
  - 4.2.1 Transducer performance
  - 4.2.2 Instrument performance
  - 4.2.3 Effect of testing environment
  - 4.2.4 Pipe specifications (diameter, thickness, manufacturing method, tolerances) and condition (temperature, roughness, stress)
- 4.3 Range and sensitivity
- 4.4 Signal-to-noise ratio
- 4.5 Detailed knowledge on how to classify and assess observations and identification of best NDT method for sizing (UT, RT, etc.) or monitoring defect growth (GW, UT, etc.)

5.0 Writing Procedures for Specific Applications
- 5.1 General and bare or painted piping
- 5.2 Insulated piping
- 5.3 Inspection under supports (simple, welded, clamped)
- 5.4 Road crossings
- 5.5 Buried piping
- 5.6 Plate
- 5.7 Steel cable or wire rope
- 5.8 Rods or rail stock
- 5.9 Tubes

6.0 Understanding of Codes, Standards, and Specifications

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LM

LASER METHODS TESTING TOPICAL OUTLINES

Laser Methods Testing – Holography/Shearography Level I Topical Outline

Basic Holography/Shearography Principles Course

1.0 Introduction

- 1.1 Nondestructive testing (NDT)
- 1.2 Overview of shearography and holography NDT (basic premise)
  - 1.2.1 Relationship between stress and strain for a material or structure
  - 1.2.2 Inherent material or structural stiffness (e.g., Young's modulus)
  - 1.2.3 Looking for subsurface defects by observing the surface of the test article as it is acted upon by an applied stress
- 1.3 History of holography/shearography
- 1.4 Definition of speckle interferometry
- 1.5 Application of shearography (SNDT) and holography (HNDT)
- 1.6 Overview of international NDT certification
- 1.7 Responsibilities of levels of certification

2.0 Basic Principles of Light and Shearography

- 2.1 Wave nature of light
- 2.2 Wave particle duality
- 2.3 Visible light versus monochromatic light
- 2.4 Definition of coherence
- 2.5 Interference
- 2.6 Interferometry
  - 2.6.1 What is an interferometer?
    - 2.6.1.1 Michelson interferometer example
  - 2.6.2 Wavelength of light used as a measuring stick
  - 2.6.3 Shearography and holography cameras are interferometers

3.0 Lasers

- 3.1 Introduction to lasers
- 3.2 Properties of laser light
  - 3.2.1 Practical sources of monochromatic-coherent light
  - 3.2.2 High-power densities
  - 3.2.3 Polarized outputs
- 3.3 Interference and the formation of laser speckle
  - 3.3.1 Laser speckle and interferometry provide the basis for holography and shearography
- 3.4 Commonly used lasers based on medium (basic properties)
  - 3.4.1 Solid-state lasers
  - 3.4.2 Gas lasers
  - 3.4.3 Ion lasers
  - 3.4.4 Diode lasers
  - 3.4.5 Diode pumped solid-state lasers (DPSS)
- 3.5 Use of multiple laser sources
  - 3.5.1 Multiple beams from single lasers (beam splitting)
  - 3.5.2 Independent laser sources (e.g., laser diodes)
    - 3.5.2.1 Coherence and phase limitations
- 3.6 Use of fiber-optic delivery systems

4.0 Laser Safety

- 4.1 Introduction
- 4.2 Potential dangers
  - 4.2.1 Eye exposure
    - 4.2.1.1 Dangers of intra-beam (collimated) viewing
    - 4.2.1.2 Dangers of concentrated (magnified) viewing
  - 4.2.2 Skin exposure
  - 4.2.3 Potential ignitions source
  - 4.2.4 Hazardous material exposure from laser mediums
  - 4.2.5 Hazardous byproducts (e.g., ozone production)
- 4.3 Levels of laser classification 1–4 based on increasing level of potential danger
  - 4.3.1 Classification limits and safety requirements
- 4.4 Expanded beams – inverse square law
- 4.5 Laser system classification
  - 4.5.1 Potential exposure during normal operation and maintenance versus service
  - 4.5.2 Safety during service operations
  - 4.5.3 Measurement locations
  - 4.5.4 Aversion response time (blink response)
  - 4.5.5 Enclosures and interlocks
- 4.6 Rules for the safe use of lasers
- 4.7 Keeping laser systems safe
- 4.8 Safety requirements for the production and field applications
- 4.9 Laser safety officers
- 4.10 Laser safety references

5.0 Basic Holography/Shearography Systems

- 5.1 Laser illumination (use of monochromatic-coherent light)
- 5.2 Differences between holography camera and shearography camera
  - 5.2.1 Independent reference beam versus sheared images
  - 5.2.2 Shear vector (definition)
    - 5.2.2.1 Magnitude and sensitivity
    - 5.2.2.2 Direction (orientation) and sensitivity
  - 5.2.3 Beam ratio (definition) – holography
  - 5.2.4 Holography sensitive to absolute displacement
    - 5.2.4.1 Single-lobed indication similar to a topographical map
    - 5.2.4.2 Increased stability requirements
  - 5.2.5 Shearography sensitive to relative displacements.
    - 5.2.5.1 Double-lobed indications showing first derivative of displacement
- 5.3 Basic holography/shearography
  - 5.3.1 Basic premise
  - 5.3.2 Image capture
  - 5.3.3 Application of stress
  - 5.3.4 Observation of surface deformation
- 5.4 Basic image formation process (general overview)
  - 5.4.1 Subtraction
  - 5.4.2 Phase stepping
    - 5.4.2.1 Wrapped versus unwrapped images
  - 5.4.3 Continuous phase stepping with variable reference
  - 5.4.4 Hybrid techniques (e.g., phase reversal, additive subtractive phase modulation, etc.)

Basic Operating Course

1.0 Holography/Shearography System Setup

- 1.1 Camera and test article stability
  - 1.1.1 Three-point mounting
  - 1.1.2 Additional requirements for holography
  - 1.1.3 Benefits of common camera/test article platform
- 1.2 Camera settings
  - 1.2.1 Focus/clarity
    - 1.2.1.1 Relationship between iris and focus setting
    - 1.2.1.2 Proper procedure for setting focus at minimum depth of field
  - 1.2.2 Iris/aperture
    - 1.2.2.1 Selecting the proper iris setting
    - 1.2.2.2 Saturation
    - 1.2.2.3 Relationship between iris and speckle size
  - 1.2.3 Shear vector
    - 1.2.3.1 Relationship between shear distance and camera shear angle
    - 1.2.3.2 Shear distance and system sensitivity
    - 1.2.3.3 Common shearing convention and its importance
    - 1.2.3.4 Nominal shear magnitudes
    - 1.2.3.5 Shear orientation and direction of maximum sensitivity
      - 1.2.3.5.1 Minimizing effect of part or camera motion
    - 1.2.3.6 Effect of shear vector orientation on indication appearance
  - 1.2.4 Beam ratio (holography only)
    - 1.2.4.1 Optimum 1:1 ratio for digital holography
  - 1.2.5 Test article illumination
    - 1.2.5.1 Ensuring full coverage of test area
    - 1.2.5.2 Multiple illumination sources (e.g., aligning multiple laser diodes)
    - 1.2.5.3 Parallax correction for camera-to-part distance
- 1.3 Test article imaging considerations
  - 1.3.1 Specular versus diffuse reflectors
  - 1.3.2 Overall reflectivity
  - 1.3.3 Transparent or translucent surface
  - 1.3.4 Orienting test article to minimize glare
  - 1.3.5 Possible surface preparations
- 1.4 Measurement standardization
  - 1.4.1 Mapping screen resolution to inspection area
    - 1.4.1.1 Defining pixels/unit distance
  - 1.4.2 Accounting for shear vector when making measurements
  - 1.4.3 Manual video scale and shear standardization
  - 1.4.4 Automatic video scale and shear standardization via spot projection
  - 1.4.5 Video scale and shear measurement limitations
- 1.5 Measuring indications
  - 1.5.1 Overall indication sizing (single lobe versus double lobe)
  - 1.5.2 Marking defect location on test articles
- 1.6 Image processor settings (defined within Level I technique description)
  - 1.6.1 Video signal optimization
  - 1.6.2 System specific settings
  - 1.6.3 Processing modes (review)
    - 1.6.3.1 Subtraction
    - 1.6.3.2 Phase stepping
      - 1.6.3.2.1 Wrapped versus unwrapped images
    - 1.6.3.3 Continuous phase stepping with variable reference
    - 1.6.3.4 Hybrid techniques (e.g., phase reversal, additive subtractive phase reversal, etc.)
  - 1.6.4 File saving
    - 1.6.4.1 File types
    - 1.6.4.2 Linking images to test data

2.0 Primary Stressing Methods (Introduction)
- 2.1 Mechanical loading
- 2.2 Thermal stressing
- 2.3 Vacuum (pressure reduction) stressing
  - 2.3.1 Chamber, whole body pressure reduction
  - 2.3.2 Vacuum hood or window, single-sided vacuum stressing
- 2.4 Pressurization stressing
- 2.5 Vibration stressing
  - 2.5.1 Acoustic versus mechanical
  - 2.5.2 Contact versus noncontact
  - 2.5.3 Frequency limitations acoustic versus mechanical
- 2.6 Combined stressing systems
  - 2.6.1 Sequential application
  - 2.6.2 Combined application

3.0 Test Standards
- 3.1 Importance of test standards
- 3.2 Representative/relevant standards
  - 3.2.1 Representative defects for test article
  - 3.2.2 Representative defects for chosen stressing method
- 3.3 Operational validation

4.0 Documentation
- 4.1 Introduction to documentation
- 4.2 Digital image files
  - 4.2.1 Generic image formats (JPEG, TIFF, BMP, AVI, MPEG)
  - 4.2.2 System specific file types
  - 4.2.3 Incorporation of images into reports
  - 4.2.4 File naming and storage (linking images to test part)
- 4.3 Report writing

Basic Applications Course

1.0 Basic Image Interpretation
- 1.1 Fringe generation (subtraction and phase stepped)
- 1.2 Basic quantitative interpretation of fringes
  - 1.2.1 Holography versus shearography (single-lobed versus double-lobed indications)
- 1.3 Wrapped versus unwrapped images
- 1.4 Common defect types and test results
  - 1.4.1 Disbonds
  - 1.4.2 Crushed core
  - 1.4.3 Impact damage in solid laminates
- 1.5 Depth, indication size, and stressing relationships
  - 1.5.1 Fringe growth versus stressing level
  - 1.5.2 Number of fringes versus depth for constant size defect and fixed stress
  - 1.5.3 Number of fringes versus defect size for constant depth and fixed stress
  - 1.5.4 Relative rate of growth for varying size defects a constant depth

2.0 Basic Stressing Methods
- 2.1 Mechanical loading
  - 2.1.1 Point loading
  - 2.1.2 Load cells or fatigue fixture
  - 2.1.3 Single-sided vacuum stressing
  - 2.1.4 Applications
- 2.2 Interpretation of example results
- 2.3 Thermal stressing
  - 2.3.1 Basic premise
    - 2.3.1.1 Thermal expansion effects
    - 2.3.1.2 Effects of internal structures or anomalies
    - 2.3.1.3 Simple disbond or delamination example
  - 2.3.2 Thermal stressing sources
    - 2.3.2.1 Infrared flash lamps
    - 2.3.2.2 Conventional heat lamps
    - 2.3.2.3 Heat gun
    - 2.3.2.4 Cold air generators
  - 2.3.3 Uniform application of heat
    - 2.3.3.1 Thermal gradient versus equilibrium temperature
  - 2.3.4 Basic thermal stressing techniques
    - 2.3.4.1 Refresh before heat
    - 2.3.4.2 Refresh after heat
  - 2.3.5 Time versus temperature effects
  - 2.3.6 Time versus depth effects
- 2.4 Applications
- 2.5 Interpretation of example results

3.0 Vacuum (Pressure Reduction) Stressing
- 3.1 Types of stressing
  - 3.1.1 Vacuum chamber (whole-body pressure reduction)
    - 3.1.1.1 Basic premise
    - 3.1.1.2 Basic requirements for effective stressing
    - 3.1.1.3 Potential problems with laminates
    - 3.1.1.4 Reference image capture at reduced pressure versus ambient pressure
  - 3.1.2 Portable single-sided vacuum stressing (vacuum hood/vacuum windows)
    - 3.1.2.1 Combined single-sided pressure reduction with mechanical stress
- 3.2 Applications
- 3.3 Interpretation of example results

4.0 Pressurization Stress
- 4.1 Types of stressing
  - 4.1.1 Increase of decrease in internal pressure of test article
- 4.2 Biaxial strain implications
  - 4.2.1 Directional sensitivity of shearography (hoop versus axial stress)
- 4.3 Applications
- 4.4 Interpretation of example results

5.0 Vibration Excitation: Acoustic and Mechanical
- 5.1 Basic premise
- 5.2 Physical contact requirements
  - 5.2.1 Acoustic (noncontact)
  - 5.2.2 Mechanical (contact required)
- 5.3 Frequency range limitation
- 5.4 Types of excitation
  - 5.4.1 White noise
  - 5.4.2 Single frequency
  - 5.4.3 Frequency sweep
- 5.5 Image-processing modes
  - 5.5.1 Phase reversal
- 5.6 General relationships
  - 5.6.1 Frequency response versus material stiffness
  - 5.6.2 Frequency response versus depth
  - 5.6.3 Frequency response versus defect size
- 5.7 Amplitude
- 5.8 Safety when using high-intensity sound
- 5.9 Applications
- 5.10 Interpretation of example results

6.0 Complex Structures
- 6.1 Types of constructions
- 6.2 Interpretation of example results

Holography/Shearography Level II Topical Outline

Intermediate Physics Course

1.0 Introduction to Holography and Shearography
- 1.1 Basic premise – looking for subsurface defects and anomalies by observing changes in the surface of a test article while it is acted on by an applied stress
- 1.2 Stress/strain relationship of a specific material or structure (Young's modulus)
- 1.3 Difference between holography and shearography systems

2.0 Physics of Light
- 2.1 Basic wave theory
- 2.2 White light versus monochromatic light
- 2.3 Coherence and interference
- 2.4 Interferometry
  - 2.4.1 Michelson interferometer example
  - 2.4.2 Wavelength of light used as a measuring stick

3.0 Lasers
- 3.1 Introduction to lasers
- 3.2 Properties of laser light
  - 3.2.1 Practical sources of monochromatic-coherent light
    - 3.2.1.1 Coherence length
  - 3.2.2 High-power densities
  - 3.2.3 Polarized outputs
- 3.3 Interference and the formation of laser speckle
- 3.4 Laser speckle and interferometry
- 3.5 Speckle interferometry
- 3.6 Commonly used lasers based on medium (basic properties)
  - 3.6.1 Solid-state lasers
  - 3.6.2 Gas lasers
  - 3.6.3 Ion lasers
  - 3.6.4 Diode lasers
  - 3.6.5 DPSS
- 3.7 Logistics and the choice of lasers
- 3.8 Use of multiple laser sources
  - 3.8.1 Multiple beams from single lasers (beam splitting)
  - 3.8.2 Independent laser sources (e.g., laser diodes)
  - 3.8.3 Coherence and phase limitations
- 3.9 Use of fiber-optic delivery systems

4.0 Laser Safety
- 4.1 Introduction
- 4.2 Potential dangers
  - 4.2.1 Eye exposure
  - 4.2.2 Skin exposure
  - 4.2.3 Potential ignitions source
  - 4.2.4 Hazardous material exposure from laser mediums
  - 4.2.5 Hazardous byproducts (e.g., ozone production)
- 4.3 Levels of laser classification 1–4 based on increasing level of potential danger
- 4.4 Laser classification based on:
  - 4.4.1 Pulsed or continuous wave emission
  - 4.4.2 Wavelength
  - 4.4.3 Power
  - 4.4.4 Expanded or collimated beam
    - 4.4.4.1 Power density reduction via inverse square law
  - 4.4.5 Extended source or point source
  - 4.4.6 Expanding beams – inverse square law
- 4.5 Dangers of intra-beam (collimated) viewing
- 4.6 Dangers of concentrated (magnified) viewing
- 4.7 Laser system classification
  - 4.7.1 Potential exposure during normal operation and maintenance versus service
  - 4.7.2 Safety during service operations
  - 4.7.3 Measurement locations
  - 4.7.4 Aversion response time (blink response)
  - 4.7.5 Enclosures and interlocks
- 4.8 Rules for the safe use of lasers
- 4.9 Keeping laser systems safe
- 4.10 Safety requirements for the laboratory
- 4.11 Safety requirements for production
- 4.12 Safety requirements for the workshop or field
- 4.13 Laser safety officers
- 4.14 Additional laser safety references

5.0 Holography and Speckle Interferometry (Historical Development)

- 5.1 Holographic recording process (wavefront recording and reconstruction)
  - 5.1.1 Stability requirements
- 5.2 Holography
- 5.3 Practical limitations of digital wavefront recording
- 5.4 Speckle photography and speckle correlation
- 5.5 Speckle interferometry
  - 5.5.1 Reference beam makes speckle image phase sensitive
- 5.6 Digital holography – basic image formation (correlation/subtraction)
  - 5.6.1 Stability requirements

6.0 Digital Holography and Shearography Optics

- 6.1 Holography camera optics
  - 6.1.1 Independent reference beam
    - 6.1.1.1 Beam ratio
  - 6.1.2 Sensitive to absolute displacement changes
  - 6.1.3 Single-lobed indications for a simple displacement
- 6.2 Shearography camera optics
  - 6.2.1 Laterally sheared object and reference beams (shearing interferometer)
  - 6.2.2 Sensitive to relative displacements between sheared points on test article
  - 6.2.3 Double-lobed indications showing first derivative of the displacement
    - 6.2.3.1 Strain gradient (lines of isostrain)
- 6.3 Shear vector
  - 6.3.1 Relationship between shear distance and camera shear angle
  - 6.3.2 Shear distance and system sensitivity
  - 6.3.3 Common shearing convention and its importance
  - 6.3.4 Nominal shear magnitudes
  - 6.3.5 Shear orientation and direction of maximum sensitivity
  - 6.3.6 Minimizing effect of part or camera motion
  - 6.3.7 Effect of shear vector orientation on indication appearance
- 6.4 Types of shearography cameras
  - 6.4.1 Fixed shear
  - 6.4.2 Adjustable shear
  - 6.4.3 Phase stepped
- 6.5 Phase stepping
  - 6.5.1 Correlation shearography review (subtraction)
  - 6.5.2 Phase stepping defined
  - 6.5.3 Image capture
  - 6.5.4 Phase map creation
  - 6.5.5 Phase map advantages
  - 6.5.6 Unwrapped phase maps
  - 6.5.7 Unwrapped phase map advantages

7.0 Physics of Materials

- 7.1 Stress strain and strain relationship (Young's modulus)
- 7.2 Deformation versus strain
- 7.3 Flat plate deformation equation
- 7.4 Thermal expansion
- 7.5 Vacuum stressing loads
- 7.6 Pressurization loads (biaxial strain/axial and hoop)
- 7.7 Vibrations stressing and resonance

Intermediate Operating Course

1.0 Holography and Shearography Systems

- 1.1 Automated inspection systems
- 1.2 Tripod-based systems
- 1.3 On vehicle inspections

2.0 Sources of Noise and Solutions

- 2.1 Stability
- 2.2 Vibration
- 2.3 Thermal currents
- 2.4 Air currents

3.0 Fixturing for Test Parts and Camera Systems

- 3.1 Simple forms
  - 3.1.1 Three-point mounting
  - 3.1.2 Use of preloads
- 3.2 Automated system requirements

4.0 Speckle Interferometry Camera

- 4.1 Field of view
- 4.2 Resolution versus field of view
- 4.3 Focus and iris settings
- 4.4 Sensitivity versus angles
- 4.5 In-plane and out-of-plane considerations
- 4.6 Effects of shear orientation
- 4.7 Specular reflections

5.0 Speckle Interferometry – Image Processor

- 5.1 Advanced processor adjustment
- 5.2 Advanced postprocessing techniques
- 5.3 Interface options
- 5.4 Documentation options

6.0 Stressing Systems, Setup, and Operation

- 6.1 Thermal stressing systems
- 6.2 Vacuum inspection systems
- 6.3 Pressurization systems
- 6.4 Vibration excitation

7.0 Method Development

- 7.1 Test standards
- 7.2 Effective research for optimum defect detection
- 7.3 Representative defect sample/confidence pieces
- 7.4 Method format/procedure/technique writing

8.0 Documentation
- 8.1 Digital image file management
- 8.2 Reporting
- 8.3 Archiving data

Intermediate Applications Course

1.0 Materials and Applications
- 1.1 Laminates
- 1.2 Honeycombs
- 1.3 Foam core materials
- 1.4 Advanced materials
- 1.5 Pressure vessel, piping, and tubing
- 1.6 Plasma spray and ceramics
- 1.7 Bonded metal

2.0 Fringe Interpretation
- 2.1 Quantitative fringe measurement
- 2.2 Defect measurement and characterization
- 2.3 Strain measurement

3.0 Mechanical Loading
- 3.1 Review of mechanical loading methods
- 3.2 Applications
  - 3.2.1 Cracks
  - 3.2.2 Material weaknesses
  - 3.2.3 Detection of ply wrinkling
  - 3.2.4 Interpretation of results
  - 3.2.5 Strain gradient versus loading

4.0 Thermal stressing
- 4.1 Review of thermal stressing methods
  - 4.1.1 Time versus temperature thermal gradient
  - 4.1.2 Time versus depth
  - 4.1.3 Multiple image analysis
- 4.2 Applications
  - 4.2.1 Delaminations
  - 4.2.2 Impact damage
  - 4.2.3 Composite repair evaluation
  - 4.2.4 Foreign material
- 4.3 Interpretation of results – phase change

5.0 Vacuum Stressing
- 5.1 Review of stressing methods
  - 5.1.1 The general-purpose method
  - 5.1.2 Depth versus fringes
- 5.2 Applications
  - 5.2.1 Near-side disbonds
  - 5.2.2 Far-side disbonds
  - 5.2.3 Composite repair evaluation
  - 5.2.4 Delaminations
- 5.3 Interpretations of results
  - 5.3.1 Effects of windows

6.0 Pressurization Stressing
- 6.1 Review of pressure stressing methods
- 6.2 Applications
  - 6.2.1 Piping and tubing
  - 6.2.2 Pressure vessels
  - 6.2.3 Aircraft fuselage
- 6.3 Interpretation of results

7.0 Vibration Excitation, Mechanical
- 7.1 Review of mechanical vibration excitation methods
  - 7.1.1 Frequency versus material stiffness
  - 7.1.2 Frequency versus material thickness/depth
  - 7.1.3 Frequency versus defect size
  - 7.1.4 Amplitude
  - 7.1.5 Sweep rate
  - 7.1.6 White noise
- 7.2 Applications
- 7.3 Interpretation of results

8.0 Vibration Excitation, Acoustic
- 8.1 Review of acoustic vibration excitation methods
  - 8.1.1 Frequency versus material stiffness
  - 8.1.2 Frequency versus material thickness/depth
  - 8.1.3 Frequency versus defect size
  - 8.1.4 Amplitude
  - 8.1.5 Sweep rate
  - 8.1.6 White noise
- 8.2 Applications
- 8.3 Interpretation of results
- 8.4 Safe use of acoustic exciters

9.0 Other Stressing Methods
- 9.1 Review of stress relaxation methods
- 9.2 Applications
- 9.3 Interpretations

Holography/Shearography Level III Topical Outline

1.0 Principles/Theory
- 1.1 Light
  - 1.1.1 Light and basic wave theory
  - 1.1.2 Coherence and inference
  - 1.1.3 Electronic speckle
  - 1.1.4 Speckle interferometry
- 1.2 Lasers
  - 1.2.1 Introduction to lasers
  - 1.2.2 Properties of laser light
    - 1.2.2.1 Practical sources of monochromatic-coherent light
      - 1.2.2.1.1 Coherence length
    - 1.2.2.2 High-power densities
    - 1.2.2.3 Polarized outputs
  - 1.2.3 Interference and the formation of laser speckle
  - 1.2.4 Laser speckle and interferometry
  - 1.2.5 Speckle interferometry

- 1.2.6 Commonly used lasers based on medium (basic properties)
  - 1.2.6.1 Solid-state lasers
  - 1.2.6.2 Gas lasers
  - 1.2.6.3 Ion lasers
  - 1.2.6.4 Diode lasers
  - 1.2.6.5 DPSS
- 1.2.7 Logistics and the choice of lasers
- 1.2.8 Use of multiple laser sources
  - 1.2.8.1 Multiple beams from single lasers (beam splitting)
  - 1.2.8.2 Independent laser sources (e.g., laser diodes)
  - 1.2.8.3 Coherence and phase limitations
- 1.2.9 Use of fiber-optic delivery systems

1.3 Optics
- 1.3.1 Holographic optics
- 1.3.2 Shearography optics
- 1.3.3 Polarization
- 1.3.4 Filters

1.4 Material properties
- 1.4.1 Stress/strain Young's modulus of elasticity
- 1.4.2 Plate deformation equation
- 1.4.3 Deformation versus strain
- 1.4.4 Specular versus diffuse reflections
- 1.4.5 Transparent coating and translucent materials
- 1.4.6 Thermal expansion of materials
- 1.4.7 Vacuum stress and out-of-plane strain
- 1.4.8 Pressurization, axial, hoop, and out-of-plane strain
- 1.4.9 Vibration and resonance
- 1.4.10 Stress relaxation

2.0 Equipment/Materials

2.1 Holographic systems
- 2.1.1 Fundamentals of holography
- 2.1.2 Holographic instruments
- 2.1.3 Interpreting results

2.2 Types of shearography systems
- 2.2.1 Image subtraction correlation systems
- 2.2.2 Phase-stepped systems
- 2.2.3 Phase-stepped systems with variable reference
- 2.2.4 Hybrid systems (e.g., phase reversal, additive subtractive phase modulation, etc.)
- 2.2.5 Production system examples
- 2.2.6 Automated system examples
- 2.2.7 Field inspection system examples

2.3 Basic setup
- 2.3.1 Illumination
  - 2.3.1.1 Illumination and field of view
  - 2.3.1.2 Laser speckle considerations
  - 2.3.1.3 In-plane and out-of-plane
- 2.3.2 Fixturing
  - 2.3.2.1 Simple forms
  - 2.3.2.2 Automated system requirements
- 2.3.3 Sensitivity shear vector (in portable systems as system rotational relationships)
  - 2.3.3.1 Sensitivity relating to shear magnitude
  - 2.3.3.2 Sensitivity relating to shear direction
  - 2.3.3.3 Sensitivity relation to in-plane/out-of-plane strain
- 2.3.4 Sensitivity shear vector (in manual systems and automated systems)
  - 2.3.4.1 Sensitivity relating to shear magnitude
  - 2.3.4.2 Sensitivity relating to shear direction
  - 2.3.4.3 Sensitivity relation to in-plane/out-of-plane strain
- 2.3.5 Sources of noise and solutions
  - 2.3.5.1 Stability
  - 2.3.5.2 Vibration
  - 2.3.5.3 Thermal gradients
  - 2.3.5.4 Air currents
  - 2.3.5.5 Shear vector settings

2.4 Instrument settings
- 2.4.1 Focus
- 2.4.2 Iris settings
- 2.4.3 Shear optics and vector
- 2.4.4 Sensitivity versus shear vector
- 2.4.5 Effects of shear orientation
- 2.4.6 Aligning illumination and camera axis
- 2.4.7 Field of view (FOV)
- 2.4.8 Resolution versus FOV
- 2.4.9 Phase stepping
  - 2.4.9.1 Single reference systems
  - 2.4.9.2 Multiple reference systems
- 2.4.10 Reflections

2.5 Measurement standardization
- 2.5.1 Mapping screen resolution to inspection area
  - 2.5.1.1 Defining pixels/unit distance
- 2.5.2 Accounting for shear vector when making measurements
- 2.5.3 Manual video scale and shear standardization
- 2.5.4 Automatic video scale and shear standardization via spot projection
- 2.5.5 Video scale and shear measurement limitations

2.6 Phase-step standardization

2.7 Advanced settings

2.8 Image processing
- 2.8.1 Image formation techniques
- 2.8.2 Advanced postprocessing
- 2.8.3 Effects of processing

3.0 Stressing Methods and Example Applications

3.1 Mechanical loading
- 3.1.1 Cracks
- 3.1.2 Revealing substructure
- 3.1.3 Material weakness
- 3.1.4 Ply wrinkling detection

- 3.2 Thermal stressing
  - 3.2.1 Thermal stressing methods
  - 3.2.2 Types of thermal energy application
    - 3.2.2.1 Time versus temperature
    - 3.2.2.2 Time versus depth
  - 3.2.3 Substructure
  - 3.2.4 Impact damage
  - 3.2.5 Composite repair evaluation
  - 3.2.6 Foreign material detection
  - 3.2.7 Near-side defects
  - 3.2.8 Far-side defects
- 3.3 Vacuum stressing
  - 3.3.1 Ambient pressure reduction (chamber)
  - 3.3.2 Contact vacuum stressing (vacuum hood/ window)
  - 3.3.3 Depth versus fringe density
- 3.4 Pressurization stressing
  - 3.4.1 Types of stressing
  - 3.4.2 Applications
    - 3.4.2.1 Piping
    - 3.4.2.2 Composite pressure vessels
    - 3.4.2.3 Biaxial strain implications
    - 3.4.2.4 Fuselage (barrel) inspections
    - 3.4.2.5 Control surfaces
- 3.5 Vibration stressing
  - 3.5.1 Mechanical vibration (MECAD)
    - 3.5.1.1 Applications
    - 3.5.1.2 Interpretation of results
  - 3.5.2 Acoustic vibrations (ACAD)
    - 3.5.2.1 Applications
    - 3.5.2.2 Interpretations
  - 3.5.3 Common excitation considerations
    - 3.5.3.1 Frequency versus material stiffness
    - 3.5.3.2 Frequency versus material thickness/ depth
    - 3.5.3.3 Frequency versus defect size
    - 3.5.3.4 Amplitude
    - 3.5.3.5 Sweep rate
    - 3.5.3.6 White noise
    - 3.5.3.7 Phase reversal
- 3.6 Other stressing methods
  - 3.6.1 Stress relaxation
  - 3.6.2 Tensile stressing (induction heat stress)

4.0 Interpretation/Evaluation

- 4.1 Fringe interpretation
- 4.2 Double-lobed fringe set
- 4.3 Shear orientation effects
- 4.4 Strain concentrations
- 4.5 Defect sizing
- 4.6 Strain measurement
- 4.7 In-plane/out-of-plane considerations

5.0 Procedures

- 5.1 Documentation
  - 5.1.1 Digital file management
  - 5.1.2 Reporting
  - 5.1.3 Postprocessing
  - 5.1.4 Data extraction
  - 5.1.5 Archiving data
  - 5.1.6 Developing procedures
- 5.2 Guideline standards
- 5.3 Standardized procedures
- 5.4 User-developed standards
- 5.5 Interpretation of codes, standards, and procedures
- 5.6 Responsibilities of a Level III
- 5.7 Responsibilities as an external Level III
- 5.8 Training and examining Level I and Level II NDT personnel

6.0 Laser Safety

- 6.1 Introduction
- 6.2 Potential dangers
  - 6.2.1 Eye exposure
  - 6.2.2 Skin exposure
  - 6.2.3 Potential ignitions source
  - 6.2.4 Hazardous material exposure from laser mediums
  - 6.2.5 Hazardous byproducts (e.g., ozone production)
- 6.3 Levels of laser classification 1–4 based on increasing level of potential danger
- 6.4 Laser classification based on:
  - 6.4.1 Pulsed or continuous wave emission
  - 6.4.2 Wavelength
  - 6.4.3 Power
  - 6.4.4 Expanded or collimated beam
    - 6.4.4.1 Power density reduction via inverse square law
  - 6.4.5 Extended source or point source
  - 6.4.6 Expanding beams – inverse square law
- 6.5 Dangers of intra-beam (collimated) viewing
- 6.6 Dangers of concentrated (magnified) viewing
- 6.7 Laser system classification
  - 6.7.1 Potential exposure during normal operation and maintenance versus service
  - 6.7.2 Safety during service operations
  - 6.7.3 Measurement locations
  - 6.7.4 Aversion response time (blink response)
  - 6.7.5 Enclosures and interlocks
- 6.8 Rules for the safe use of lasers
- 6.9 Keeping laser systems safe
- 6.10 Safety requirements for the laboratory
- 6.11 Safety requirements for production
- 6.12 Safety requirements for the workshop or field
- 6.13 Laser safety officers
- 6.14 Additional laser safety references

- **7.0 Applications**
  - 7.1 Holography applications
    - 7.1.1 Felt metal
    - 7.1.2 Plasma bonding
    - 7.1.3 Other inspections
  - 7.2 Shearography applications
    - 7.2.1 Aerospace applications
      - 7.2.1.1 Composite materials
      - 7.2.1.2 Laminates
      - 7.2.1.3 Sandwich constructions
        - 7.2.1.3.1 Honeycomb
        - 7.2.1.3.2 Foam cores
        - 7.2.1.3.3 Other core material
      - 7.2.1.4 Advanced materials
      - 7.2.1.5 Bonded metal
    - 7.2.2 Marine applications
      - 7.2.2.1 Solid laminates
      - 7.2.2.2 Sandwich structures
      - 7.2.2.3 Ply dropoff
      - 7.2.2.4 Bulkhead detection
      - 7.2.2.5 Repair evaluation
    - 7.2.3 Tire inspection
      - 7.2.3.1 Equipment
      - 7.2.3.2 Fault detection
    - 7.2.4 Encapsulated microcircuit leak detection
      - 7.2.4.1 Leak rate
      - 7.2.4.2 Volume
    - 7.2.5 Civil engineering applications
      - 7.2.5.1 Composite reinforcement
      - 7.2.5.2 Insulation panel inspection
    - 7.2.6 Complex structures
      - 7.2.6.1 Types of construction
      - 7.2.6.2 Interpretation of results

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LT
LEAK TESTING TOPICAL OUTLINES

Leak Testing Level I Topical Outline

Fundamentals in Leak Testing Course

1.0 Introduction
- 1.1 History of leak testing (LT)
- 1.2 Reasons for LT
  - 1.2.1 Material loss prevention
  - 1.2.2 Contamination
  - 1.2.3 Component/system reliability
  - 1.2.4 Pressure-differential maintenance
  - 1.2.5 Personnel/public safety
- 1.3 Functions of LT
  - 1.3.1 Categories
  - 1.3.2 Applications
- 1.4 Training and certification

2.0 LT Fundamentals
- 2.1 Terminology
  - 2.1.1 Leakage terms
  - 2.1.2 Leakage tightness
  - 2.1.3 Quantitative/semi-quantitative
  - 2.1.4 Sensitivity/calibration and standardization terms
- 2.2 LT units
  - 2.2.1 Mathematics in LT
  - 2.2.2 Exponential notation
  - 2.2.3 Basic and fundamental units
  - 2.2.4 The International System of Units (SI)
- 2.3 Physical units in LT
  - 2.3.1 Volume and pressure
  - 2.3.2 Time and temperature
  - 2.3.3 Absolute values
  - 2.3.4 Standard or atmospheric conditions
  - 2.3.5 Leakage measurement
- 2.4 LT standards
  - 2.4.1 Capillary or permeation, reservoir (capsule) and nonreservoir (open)
  - 2.4.2 National Institute of Standards and Technology (NIST) standards
  - 2.4.3 System calibration versus instrument standardization
  - 2.4.4 Inaccuracy of calibration
  - 2.4.5 Test procedure sensitivity (TPS), and its calculation
  - 2.4.6 Effective instrument sensitivity, and its calculation
  - 2.4.7 Test quality factor (TQF) test reliability – correlation of system calibration factors
- 2.5 Flow characteristics
  - 2.5.1 Gas flow
  - 2.5.2 Liquid flow
  - 2.5.3 Correlation of leakage rates
  - 2.5.4 Anomalous leaks
  - 2.5.5 Leak clogging
- 2.6 Vacuum fundamentals
  - 2.6.1 Introduction to vacuum
    - 2.6.1.1 Terminology
    - 2.6.1.2 Principles
    - 2.6.1.3 Units of pressure
  - 2.6.2 Characteristics of gases
    - 2.6.2.1 Kinetic theory
    - 2.6.2.2 Mean free path
  - 2.6.3 Gas laws
  - 2.6.4 Quantity, throughput, conductance calculations, and conductance of gas
    - 2.6.4.1 Quantity
      - 2.6.4.1.1 Comparison with an electric circuit
      - 2.6.4.1.2 Comparison with water flow
    - 2.6.4.2 Conductance analogy with electrical resistance
      - 2.6.4.2.1 Resistance connected in series
      - 2.6.4.2.2 Resistance connected in parallel
- 2.7 The four gas loads (types of leaks)
  - 2.7.1 "Real" or mechanical leak
  - 2.7.2 Permeation leak
  - 2.7.3 Virtual leak
  - 2.7.4 Outgassing/off-gassing
- 2.8 Vacuum system operation
  - 2.8.1 Effects of evacuating a vessel
  - 2.8.2 Pump-down time
- 2.9 Vacuum system characteristics
  - 2.9.1 General
    - 2.9.1.1 Operating limits
    - 2.9.1.2 Rate of pressure rise – measurement
  - 2.9.2 Vacuum pumps
    - 2.9.2.1 Pumping speed calculations
    - 2.9.2.2 Mechanical pumps (positive displacement)

2.9.2.2.1 Oil-sealed rotary pumps
- 2.9.2.2.1.1 Construction
- 2.9.2.2.1.2 Operation
- 2.9.2.2.1.3 Pump fluids
- 2.9.2.2.1.4 Difficulties with rotary pumps
- 2.9.2.2.1.5 Care of rotary pumps
- 2.9.2.2.1.6 Scroll pumps
- 2.9.2.2.1.7 Diaphragm pumps
- 2.9.2.2.1.8 Screw pumps
- 2.9.2.2.1.9 Rotary piston pumps
- 2.9.2.2.1.10 Root pumps and hook and claw pumps

2.9.2.3 Vapor (diffusion) pumps
- 2.9.2.3.1 Construction
- 2.9.2.3.2 Operation
- 2.9.2.3.3 Pump fluids
- 2.9.2.3.4 Difficulties with diffusion pumps
- 2.9.2.3.5 Diagnosis of diffusion pump trouble

2.9.2.4 Sublimation pumps (getter pumps)
2.9.2.5 Ion pumps
2.9.2.6 Turbomolecular pumps
2.9.2.7 Absorption pumps
2.9.2.8 Cryopumps

Safety in Leak Testing Course

Note: It is recommended that the trainee, as well as all other LT personnel, receive instruction in this course prior to performing work in LT.

1.0 Safety Considerations
- 1.1 Personnel and the public
- 1.2 Product serviceability
- 1.3 Test validity
- 1.4 Safe work practices

2.0 Safety Precautions
- 2.1 Explosive/implosive hazards
- 2.2 Flammability, ignitibility, combustibility hazards
- 2.3 Toxicity and asphyxiation hazards
- 2.4 Cleaning and electrical hazards

3.0 Pressure Precautions
- 3.1 Pressure test versus proof test
- 3.2 Preliminary LT
- 3.3 Pressurization check
- 3.4 Design limitations
- 3.5 Equipment and setup

4.0 Safety Devices
- 4.1 Pressure control valves and regulators
- 4.2 Pressure relief valves and vents
- 4.3 Flow rate of regulator and relief valves

5.0 Hazardous and Tracer-Gas Safety
- 5.1 Combustible gas detection and safety
- 5.2 Toxic gas detection and safety
- 5.3 Oxygen-deficiency detectors
- 5.4 Radioisotope detection

6.0 Types of Monitoring Equipment
- 6.1 Area monitors
- 6.2 Personnel monitors
- 6.3 Leak-locating devices

7.0 Safety Regulations
- 7.1 State and federal regulations
- 7.2 Safety codes/standards
- 7.3 Hazardous gas standards
- 7.4 Nuclear Regulatory Commission (NRC) radiation requirements

8.0 Vacuum Pump Safety Considerations
- 8.1 Positive displacement vacuum pumps
- 8.2 Capped exhaust hazard
- 8.3 Captured contaminants exposure during servicing
- 8.4 Contaminated oil (including radiological mixed waste)
- 8.5 Hazardous exhaust gases and oxygen depletion from inert gas exhaust
- 8.6 Unintentional pressurization due to misphasing of a three-phase pump
- 8.7 Turbomolecular pump safety
  - 8.7.1 Rotor failure and anchorage for inertia on failure
  - 8.7.2 Hazardous materials considerations for servicing the pump
- 8.8 Cryogenic pump safety
  - 8.8.1 Combustible vapors during warming
  - 8.8.2 Hazardous vapors during warming

Leak Testing Methods Course

1.0 The following LT methods may be incorporated as applicable.
- 1.1 Each of these methods can be further divided into major techniques as shown in the following examples:
  - 1.1.1 Bubble leak testing
    - 1.1.1.1 Immersion
    - 1.1.1.2 Solution film testing
  - 1.1.2 Ultrasonic testing
    - 1.1.2.1 Sonic/mechanical flow
    - 1.1.2.2 Sound generator
    - 1.1.2.3 Acoustic imagers
  - 1.1.3 Voltage discharge testing
    - 1.1.3.1 Voltage spark
    - 1.1.3.2 Color change

1.1.4 Pressure testing
1.1.4.1 Hydrostatic
1.1.4.2 Pneumatic
1.1.5 Ionization
1.1.5.1 Photoionization
1.1.5.2 Flame ionization
1.1.6 Non-mass spectrometer detector probe
1.1.6.1 Thermal conductivity
1.1.6.2 Solid state
1.1.6.3 Laser-based
1.1.7 Radiation absorption
1.1.7.1 Infrared
1.1.7.2 Ultraviolet
1.1.7.3 Laser
1.1.8 Chemical-based
1.1.8.1 Chemical penetrants
1.1.8.2 Chemical gas tracer (colorimetric)
1.1.9 Halogen detector
1.1.9.1 Halide torch
1.1.9.2 Electron capture
1.1.9.3 Halogen diode leak testing
1.1.9.4 Laser-based
1.1.9.5 Solid state
1.1.10 Pressure change leak testing
1.1.10.1 Absolute
1.1.10.2 Reference
1.1.10.3 Pressure rise
1.1.10.4 Flow measurement
1.1.10.5 Pressure decay
1.1.10.6 Volumetric
1.1.10.7 Pressure rise-calibrated leak technique
1.1.10.8 Pressure decay-calibrated leak technique
1.1.11 Mass spectrometer leak testing (MSLT) and detector probe "sniffer" leak testing
1.1.11.1 Magnetic sector helium or argon mass spectrometer leak detector
1.1.11.2 Residual gas analyzer
1.1.11.3 Helium detector probe leak detector
1.1.12 Radioisotope
1.1.13 Thermographic LT
1.1.14 Hydrogen LT

2.0 LT Method Course Outline

2.1 The following may be applied to any of the listed methods:
2.2 Terminology
2.3 Basic techniques and/or units
2.3.1 Leak location – measurement/monitoring
2.3.2 Visual and other sensing devices
2.3.3 Various techniques
2.4 The four gas loads (types of leaks)
2.4.1 "Real" or mechanical leak
2.4.2 Permeation leak
2.4.3 Virtual leak
2.4.4 Outgassing/off-gassing
2.5 Testing materials and equipment
2.5.1 Materials, gases/fluids used
2.5.2 Control devices and operation
2.5.3 Instrument/gauges used
2.5.4 Range and standardization of instrument/gauges
2.6 Testing principles and practices
2.6.1 Pressure/vacuum and control used
2.6.2 Principles of techniques used
2.6.3 Effects of temperature and other atmospheric conditions
2.6.4 Standardization for testing
2.6.5 Probing/scanning or measurement/monitoring
2.6.6 Leak interpretation evaluation
2.6.7 Calculation of the TQF– test reliablity
2.6.8 Calculation of TPS
2.6.9 Calculation of instrument sensitivity
2.6.10 Principles of test-system standardization/calibration, including the five major benefits of full system standardization
2.6.10.1 Objective leakage rate/signal amplitude standardization
2.6.10.2 Objective determination of test dwell timing or maximum travel speed from direct observation in each test cycle
2.6.10.3 Objective evidence of system function in each test cycle immediately bracketing the leak detection–leakage measurement function
2.6.10.4 Evidence in each test cycle that the system senses to the extremity of the test object
2.6.10.5 Objective determination of the TQF – test reliability, in each test cycle
2.7 Acceptance and rejection criteria
2.8 Safety concerns
2.9 Advantages and limitations
2.10 Codes/standards

Leak Testing Level II Topical Outline

Principles of Leak Testing Course

1.0 Introduction

1.1 LT fundamentals
1.1.1 Reasons for LT
1.1.1.1 Law of conservation of matter – leakage significance
1.1.2 Functions of LT
1.1.3 Terminology
1.1.4 LT units
1.1.5 Leak conductance
1.2 LT standards
1.2.1 Leak standards
1.2.2 NIST traceability and calibration
1.2.3 Instrument standardization versus system calibration

1.2.4 System calibration techniques
1.2.5 Inaccuracy of calibration
1.2.6 TPS and its calculation
1.2.7 Effective instrument sensitivity and its calculation
1.2.8 TQF, test reliability – correlation of system calibration factors
1.2.9 Tracer-gas leak rate/air-equivalent leak rate

1.3 LT safety
1.3.1 Safety considerations
1.3.2 Safety precautions
1.3.3 Pressure precautions
1.3.4 Tracer-gas safety and monitoring
1.3.5 Safety devices
1.3.6 Cleaning and electrical hazards
1.3.7 Safe work practices
1.3.8 Safety regulations

1.4 LT procedure
1.4.1 Basic categories and techniques
1.4.2 Leak location versus leakage measurement
1.4.3 Pressurization or evacuation
1.4.4 Sealed units with or without tracer gas
1.4.5 Units inaccessible from one or both sides
1.4.6 System at, above, or below atmospheric pressure

1.5 LT specifications
1.5.1 Design versus working conditions
1.5.2 Pressure and temperature control
1.5.3 Types of LT methods
1.5.4 Sensitivity of LT methods
1.5.5 Test method and sensitivity needed
1.5.6 Preparation of an LT specification
1.5.7 Qualification of LT specification (qualification of a procedure)

1.6 Detector/instrument performance factors
1.6.1 Design and use
1.6.2 Accuracy and precision
1.6.3 Linearity (straight/logarithmic scale)
1.6.4 Calibration and standardization frequency
1.6.5 Response and recovery time

2.0 Physical Principles in LT

2.1 Physical quantities
2.1.1 Fundamental units
2.1.2 Volume and pressure
2.1.3 Time and temperature
2.1.4 Absolute values
2.1.5 Standard versus atmospheric conditions
2.1.6 Leakage rates

2.2 Structure of matter
2.2.1 Atomic theory
2.2.2 Ionization and ion pairs
2.2.3 States of matter
2.2.4 Molecular structure
2.2.5 Diatomic and monatomic molecules
2.2.6 Molecular weight

2.3 Gas principles and laws
2.3.1 Brownian movement
2.3.2 Mean free path
2.3.3 Pressure and temperature effects on gases
2.3.4 Pascal's law of pressure
2.3.5 Charles's, Gay-Lussac's, and Boyle's gas laws
2.3.6 Ideal gas law
2.3.7 Dalton's law of partial pressure
2.3.8 Vapor pressure and effects in vacuum

2.4 Gas properties
2.4.1 Kinetic theory of gases
2.4.2 Graham's law of diffusion
2.4.3 Stratification
2.4.4 Avogadro's principle
2.4.5 Gas law relationship
2.4.6 General ideal gas law
2.4.7 Gas mixture and concentration
2.4.8 Gas velocity, density, and viscosity

2.5 The four gas loads (types of leaks)
2.5.1 "Real" or mechanical leak
2.5.2 Permeation leak
2.5.3 Virtual leak
2.5.4 Outgassing/off-gassing

3.0 Principles of Gas Flow

3.1 Standard leaks
3.1.1 Capillary
3.1.2 Permeation
3.1.3 Capsule (reservoir)
3.1.4 Open

3.2 Modes of gas flow
3.2.1 Molecular and viscous
3.2.2 Transitional
3.2.3 Laminar, turbulent, sonic

3.3 Factors affecting gas flow

3.4 Geometry of leakage path
3.4.1 Mean free flow of fluid
3.4.2 Clogging and check valve effects
3.4.3 Irregular aperture size
3.4.4 Leak rate versus cross section of flow
3.4.5 Temperature and atmospheric conditions
3.4.6 Velocity gradient versus viscosity
3.4.7 Reynolds number versus Knudsen number

Pressure and Vacuum Technology Course

1.0 Pressure Technology

1.1 Properties of a fluid
1.1.1 What is a fluid?
1.1.2 Liquid versus gas
1.1.3 Compressibility
1.1.4 Partial and vapor pressure
1.1.5 Critical pressure and temperature
1.1.6 Viscosity of a liquid
1.1.7 Surface tension and capillarity of a liquid

1.2 Gas properties
  1.2.1 Review of gas properties
  1.2.2 What is a perfect/ideal gas?
  1.2.3 Pressure and temperature effects on gases
  1.2.4 Viscosity of a gas
  1.2.5 Gas flow modes
  1.2.6 Gas flow conductance
  1.2.7 Dynamic flow measurements
  1.2.8 Factors affecting gas flow
1.3 Pressurization
  1.3.1 Pressure measurements
  1.3.2 Types of pressure gauges
    1.3.2.1 Bourdon or diaphragm
    1.3.2.2 Manometers
  1.3.3 Pressure control and procedure
  1.3.4 Mixing of gases
  1.3.5 Tracer gases and concentration
  1.3.6 Pressure hold time
  1.3.7 Pressure versus sensitivity
  1.3.8 Gauge calibration
    1.3.8.1 Working range
    1.3.8.2 Frequency
    1.3.8.3 Master gauge versus dead-weight tester
1.4 LT background/noise variables
  1.4.1 Atmospheric changes
  1.4.2 Liquid/air temperature correction
  1.4.3 Vapor pressure (evaporation/condensation)
  1.4.4 Vapor/moisture pockets
  1.4.5 Geometry/volume changes
  1.4.6 Surface/internal vibration waves
1.5 Detector/instrument performance variables
  1.5.1 Instrument standardization variables
  1.5.2 Limits of accuracy
  1.5.3 Intrinsic and inherent safety performance
  1.5.4 Protection for electromagnetic interference, radio-frequency interference, shock, etc.
  1.5.5 Flooding, poisoning, contamination
1.6 Measurement and data documentation
  1.6.1 Experimental, simulation and/or preliminary testing
  1.6.2 Analysis of background/noise variables
  1.6.3 Analysis of leakage indications/signals
  1.6.4 Validation and error analysis
  1.6.5 Interpretation and evaluation of results
  1.6.6 Documentation of data and test results

2.0 Vacuum Technology
2.1 Nature of vacuum
  2.1.1 What is a vacuum?
  2.1.2 Vacuum terminology
  2.1.3 Degrees of vacuum
  2.1.4 Mean free path in a vacuum
  2.1.5 Gas flow in a vacuum
2.2 Vacuum measurement
  2.2.1 Pressure units in a vacuum
  2.2.2 Absolute versus gauge pressure
  2.2.3 Mechanical force technology gauges
    2.2.3.1 Bourdon or diaphragm
    2.2.3.2 Manometer (U-tube or McLeod)
    2.2.3.3 Capacitance manometer
    2.2.3.4 Piezo-resistive
  2.2.4 Thermal conductivity technology gauges
    2.2.4.1 Thermocouple
    2.2.4.2 Pirani and convection Pirani
    2.2.4.3 Ionization
  2.2.5 Ionization
    2.2.5.1 Cold cathode – Penning
    2.2.5.2 Hot cathode – Bayard-Alpert
  2.2.6 Gauge calibration – full range
2.3 Vacuum pumps
  2.3.1 Types of vacuum pumps
  2.3.2 Mechanical pumps
    2.3.2.1 Reciprocating versus rotary
    2.3.2.2 Roots, turbomolecular, drag and dry scroll pumps
  2.3.3 Nonmechanical pumps
    2.3.3.1 Fluid entrainment or diffusion
    2.3.3.2 Condensation or sorption
  2.3.4 Pump oils
  2.3.5 Pumping speed and pump-down time
  2.3.6 Vacuum pump safety
    2.3.6.1 Positive displacement vacuum pumps
      2.3.6.1.1 Capped exhaust hazard
      2.3.6.1.2 Captured contaminants exposure during servicing
      2.3.6.1.3 Contaminated oil (including radiological mixed waste)
      2.3.6.1.4 Hazardous exhaust gases and oxygen depletion from inert gas exhaust
      2.3.6.1.5 Unintentional pressurization due to misphasing of a three-phase pump
    2.3.6.2 Turbomolecular pump safety
      2.3.6.2.1 Rotor failure and anchorage for inertia on failure
      2.3.6.2.2 Hazardous materials considerations for servicing the pump
    2.3.6.3 Cryogenic pump safety
      2.3.6.3.1 Combustible vapors during warming
      2.3.6.3.2 Hazardous vapors during warming

- 2.4 Vacuum materials
  - 2.4.1 Outgassing – vapor pressure
  - 2.4.2 Elastomers, gaskets, O-rings
  - 2.4.3 Metals, metal alloys, and nonmetals
    - 2.4.3.1 Carbon steel versus stainless steel
    - 2.4.3.2 Aluminum, copper, nickel, and alloys
  - 2.4.4 Nonmetals
    - 2.4.4.1 Glass, ceramics
    - 2.4.4.2 Plastics, tygon, etc.
  - 2.4.5 Joint design
    - 2.4.5.1 Sealed joint
    - 2.4.5.2 Welded/brazed joint
    - 2.4.5.3 Mechanical joint
    - 2.4.5.4 Double seals
  - 2.4.6 Vacuum greases and sealing materials
  - 2.4.7 Tracer-gas permeation through materials
  - 2.4.8 Vacuum conditioning of vacuum systems
- 2.5 Design of a vacuum system
  - 2.5.1 Production of a vacuum
    - 2.5.1.1 Removal of gas molecules
    - 2.5.1.2 Gas quantity or throughput
    - 2.5.1.3 Conductance
  - 2.5.2 Stages of vacuum pumping
    - 2.5.2.1 Various vacuum pumps
    - 2.5.2.2 Various traps and baffles
    - 2.5.2.3 Pumping stages or sequences
  - 2.5.3 Vacuum valve location
    - 2.5.3.1 Vacuum valve design and seat leakage
    - 2.5.3.2 Isolation and protection
    - 2.5.3.3 Automatic versus manual
    - 2.5.3.4 Venting
- 2.6 Maintenance and cleanliness
  - 2.6.1 Maintenance of vacuum equipment
    - 2.6.1.1 Under constant vacuum
    - 2.6.1.2 Dry gas (nitrogen)
    - 2.6.1.3 Hazardous contaminated surfaces and fluids in vacuum equipment maintenance
  - 2.6.2 Routing oil changes
  - 2.6.3 System cleanliness
    - 2.6.3.1 Initial cleanliness
    - 2.6.3.2 Cleaning procedures and effects on leak location and measurement
    - 2.6.3.3 Continued cleanliness
    - 2.6.3.4 Polarity of solvents and the removal of sealants and vacuum lubricants
- 2.7 Analysis and documentation
  - 2.7.1 Analysis of outgassing and background contamination
  - 2.7.2 Instrument standardization/system calibration
  - 2.7.3 Calculation of the TQF
  - 2.7.4 Calculation of TPS
  - 2.7.5 Calculation of instrument sensitivity
  - 2.7.6 Principles of test-system standardization/calibration, including the five major benefits of full system standardization
  - 2.7.7 Analysis of leakage indications/signals
  - 2.7.8 Interpretation and evaluation
  - 2.7.9 Documentation of standardization and test results

3.0 Techniques/Standardization and Calibration

- 3.1 Bubble leak testing
- 3.2 Pressure change leak testing
- 3.3 Halogen detector leak testing
- 3.4 Mass spectrometer leak testing

Leak Testing Level III Topical Outline

Leak Test Selection Course

1.0 Choice of LT Procedure

- 1.1 Basic categories of LT
  - 1.1.1 Leak location
  - 1.1.2 Leakage measurement
  - 1.1.3 Leakage monitoring
- 1.2 Types of LT methods
  - 1.2.1 Specifications
  - 1.2.2 Sensitivity
- 1.3 Basic techniques
  - 1.3.1 Pressurization or evacuation
  - 1.3.2 Sealed unit with or without tracer gases
  - 1.3.3 Probing or visual leak location
  - 1.3.4 Tracer or detector probing
  - 1.3.5 Accumulation techniques

2.0 Principles/Theory

- 2.1 Physical principles in LT
  - 2.1.1 Physical quantities
    - 2.1.1.1 Fundamental units
    - 2.1.1.2 Volume and pressure
    - 2.1.1.3 Time and temperature
    - 2.1.1.4 Absolute values
    - 2.1.1.5 Standard versus atmospheric conditions
    - 2.1.1.6 Leakage rates
  - 2.1.2 Structure of matter
    - 2.1.2.1 Atomic theory
    - 2.1.2.2 Ionization and ion pairs
    - 2.1.2.3 States of matter
    - 2.1.2.4 Molecular structure
    - 2.1.2.5 Diatomic and monatomic molecules
    - 2.1.2.6 Molecular weight
  - 2.1.3 Gas principles and law
    - 2.1.3.1 Brownian movement
    - 2.1.3.2 Mean free path
    - 2.1.3.3 Pressure and temperature effects on gases
    - 2.1.3.4 Pascal's law of pressure
    - 2.1.3.5 Charles's and Boyle's laws
    - 2.1.3.6 Ideal gas law
    - 2.1.3.7 Dalton's law of partial pressure
    - 2.1.3.8 Vapor pressure and effects in vacuum

2.1.4 Gas properties
2.1.4.1 Kinetic theory of gases
2.1.4.2 Graham's law of diffusion
2.1.4.3 Stratification
2.1.4.4 Avogadro's principle
2.1.4.5 Gas law relationship
2.1.4.6 General ideal gas law
2.1.4.7 Gas mixture and concentration
2.1.4.8 Gas velocity, density, and viscosity
2.2 Principles of gas flow
2.2.1 Standard leaks
2.2.1.1 Capillary
2.2.1.2 Permeation
2.2.2 Modes of gas flow
2.2.2.1 Molecular and viscous
2.2.2.2 Transitional
2.2.2.3 Laminar, turbulent, sonic
2.2.3 Factors affecting gas flow
2.2.4 Geometry of leakage path
2.2.4.1 Mean free flow of fluid
2.2.4.2 Clogging and check valve effects
2.2.4.3 Irregular aperture size
2.2.4.4 Leak rate versus viscosity
2.2.4.5 Temperature and atmospheric conditions
2.2.4.6 Velocity gradient versus viscosity
2.2.4.7 Reynolds number versus Knudsen number
2.2.5 Principles of mass spectrometer and detector probe leak detection
2.2.5.1 Vacuum and pressure technology
2.2.5.2 Outgassing of materials versus pressure
2.2.5.3 Vacuum pumping technology
2.3 Proper selection of LT as method of choice
2.3.1 Differences between LT and other methods
2.3.2 Complementary roles of LT and other methods
2.3.3 Potential for conflicting results between methods
2.3.4 Factors that qualify/disqualify the use of LT

3.0 Equipment/Material
3.1 LT standards
3.1.1 Capillary or permeation
3.1.2 Capsule (reservoir) or open
3.1.3 NIST standards
3.1.4 System calibration versus instrument standardization
3.1.5 Inaccuracy of calibration
3.2 Detector/instrument performance factors
3.2.1 Design and use
3.2.2 Accuracy and precision
3.2.3 Linearity (straight/logarithmic scale)
3.2.4 Calibration and standardization frequency
3.2.5 Response and recovery time
3.3 Vacuum pumps
3.3.1 Mechanical pumps (positive displacement, rough vacuum)
3.3.1.1 Dry versus wet technologies
3.3.1.2 Oil-sealed rotary pumps
3.3.1.2.1 Construction
3.3.1.2.2 Operation
3.3.1.2.3 Pump fluid
3.3.1.2.4 Difficulties with rotary pumps
3.3.1.2.5 Care of rotary pumps
3.3.1.3 Scroll pumps
3.3.1.4 Diaphragm pumps
3.3.1.5 Screw pumps
3.3.1.6 Rotary piston pumps
3.3.1.7 Roots pumps and hook and claw pumps
3.3.2 High and ultrahigh vacuum pumps
3.3.2.1 Vapor (diffusion) pumps
3.3.2.1.1 Construction
3.3.2.1.2 Operation
3.3.2.1.3 Pump fluid
3.3.2.1.4 Diagnosis of diffusion pump troubles
3.3.2.2 Sublimation pumps (getter pumps)
3.3.2.3 Ion pumps
3.3.2.4 Turbomolecular pumps
3.3.2.5 Absorption pumps
3.3.2.6 Cryopumps
3.4 Bubble leak testing practices and techniques
3.4.1 Immersion bubble leak tests
3.4.1.1 Pressurized object
3.4.1.2 Vacuum over solution
3.4.1.3 Heated solution
3.4.1.4 Choice of solution
3.4.2 Solution film bubble leak tests
3.4.2.1 Vacuum box
3.4.2.1.1 Applications
3.4.2.1.2 Limitations
3.4.2.2 Pressurized object
3.4.2.2.1 Sensitive pressurized object
3.4.2.2.2 Pressurized object for gross leaks and vacuum systems
3.4.2.2.3 Reinforcing pad plate testing
3.4.3 Qualifying bubble leak testing specifications (procedures)
3.4.4 Solutions
3.4.5 Solution applicators
3.4.6 Vacuum box design
3.4.7 Achievable test sensitivities
3.4.8 Lighting and visual acuity
3.4.9 Safety considerations
3.4.9.1 Pressurization and pressure safety
3.4.9.2 Slips, trips, and falls (solution film)
3.4.9.3 Eye safety (air ejector vacuum boxes)
3.4.9.4 Situational awareness in vacuum box testing

3.5 Absolute pressure testing equipment
- 3.5.1 Pressure measuring instruments
- 3.5.2 Temperature measuring instruments
- 3.5.3 Dew point measuring instruments
- 3.5.4 Accuracy of equipment
- 3.5.5 Calibration of equipment
- 3.5.6 Reference panel instruments
- 3.5.7 Reference system installation and testing

3.6 Absolute pressure hold testing of containers
- 3.6.1 Equation for determining pressure change
- 3.6.2 Temperature measuring

3.7 Absolute pressure leakage rate testing of containers
- 3.7.1 Equation(s) for determining percent loss
- 3.7.2 Positioning of temperature and dew point sensors for mean sampling accuracy
- 3.7.3 Analysis of temperature and dew point data

3.8 Pressure decay – calibrated leak method

3.9 Pressure rise – calibrated leak method

3.10 Analysis of data for determination of accurate results
- 3.10.1 Pressure change measurement LT
- 3.10.2 TPS and its calculation
- 3.10.3 TQF, test reliability – correlation of system calibration factors
- 3.10.4 Test resolution – calculation
- 3.10.5 Test accuracy (bias) calculation

3.11 Flow measurement leakage rate measurement
- 3.11.1 Limitations
- 3.11.2 Mass extraction
- 3.11.3 Pressure maintenance

3.12 Halogen detector leak testing equipment
- 3.12.1 Leak detector control unit
- 3.12.2 Gun detectors
- 3.12.3 Standard leaks
- 3.12.4 Refrigerant tracer gases

3.13 Helium detector probe "sniffer" and mass spectrometer testing equipment
- 3.13.1 Mechanical vacuum pump systems
- 3.13.2 Cryogenic pumps
- 3.13.3 Diffusion pumps
- 3.13.4 Vacuum gauges
- 3.13.5 Vacuum hose
- 3.13.6 Vacuum valves
- 3.13.7 Standard leaks
- 3.13.8 Vacuum sealing compounds
- 3.13.9 Vacuum connectors
- 3.13.10 Detector "sniffer" probes
- 3.13.11 Tracer probes

4.0 Techniques/Standardization and Calibration

4.1 Bubble leak testing
- 4.1.1 Bubble leak testing practices and techniques
  - 4.1.1.1 Vacuum box testing
  - 4.1.1.2 Pipe, nozzle, and pad plate testing
  - 4.1.1.3 Vessel testing
  - 4.1.1.4 Weather effects and lighting

4.2 Pressure change leak testing
- 4.2.1 Absolute pressure leak test and reference system test
  - 4.2.1.1 Principles of absolute pressure testing
    - 4.2.1.1.1 General gas law equation
    - 4.2.1.1.2 Effects of temperature change
    - 4.2.1.1.3 Effects of water vapor pressure change
    - 4.2.1.1.4 Effects of barometric pressure change
  - 4.2.1.2 Terminology related to absolute pressure testing

4.3 Halogen detector leak testing
- 4.3.1 Principles of halogen detector leak testing
- 4.3.2 Terminology related to halogen detector leak testing
- 4.3.3 Calibration of detectors for testing
  - 4.3.3.1 Standard leak settings
  - 4.3.3.2 Halogen mixture percentages
  - 4.3.3.3 Detection sensitivity versus test sensitivity
- 4.3.4 Halogen detector probe "sniffer" testing techniques and practices
  - 4.3.4.1 Detector probe "sniffer" speed
  - 4.3.4.2 Halogen background
  - 4.3.4.3 Effects of heat on refrigerant R-12
- 4.3.5 Halogen leak detector operation and servicing
  - 4.3.5.1 Operation of the probe
  - 4.3.5.2 Replacing the sensing element
  - 4.3.5.3 Cleaning the sensing element

4.4 Mass spectrometer leak testing
- 4.4.1 Terminology related to mass spectrometer leak testing
- 4.4.2 Helium mass spectrometer
  - 4.4.2.1 Operation
  - 4.4.2.2 Standardization
  - 4.4.2.3 Maintenance
- 4.4.3 Helium detector probe "sniffer" and mass spectrometer pressure leak testing
  - 4.4.3.1 Detector probe "sniffer" techniques
  - 4.4.3.2 Mixture percentage
  - 4.4.3.3 Pressure-differential techniques
  - 4.4.3.4 Bagging/accumulation techniques
  - 4.4.3.5 Standardization of helium mass spectrometer for "sniffer" testing

4.4.4 Helium mass spectrometer vacuum testing by dynamic method
- 4.4.4.1 Tracer probing
- 4.4.4.2 Bagging or hooding
- 4.4.4.3 System calibration
- 4.4.4.4 Helium mixture
- 4.4.4.5 Calculation of leakage rate
- 4.4.4.6 TPS and its calculation
- 4.4.4.7 Effective instrument sensitivity and its calculation
- 4.4.4.8 TQF, test reliability – correlation of system calibration factors

4.4.5 Helium mass spectrometer vacuum testing by static method
- 4.4.5.1 Static equation
- 4.4.5.2 System standardization
- 4.4.5.3 Helium mixture
- 4.4.5.4 System pressure
- 4.4.5.5 Calculation of leakage rate
- 4.4.5.6 TPS and its calculation
- 4.4.5.7 Effective instrument sensitivity and its calculation
- 4.4.5.8 TQF, test reliability – correlation of system calibration factors

5.0 Interpretation/Evaluation

5.1 Basic techniques and/or units
- 5.1.1 Leak location – measurement/monitoring
- 5.1.2 Visual and other sensing devices
- 5.1.3 Various techniques

5.2 Test materials and equipment effects
- 5.2.1 Materials, gases/fluids used
- 5.2.2 Liquid flow
  - 5.2.2.1 Effects of viscosity and surface tension
  - 5.2.2.2 Correlating liquid to gas flow
- 5.2.3 Control devices and responses
- 5.2.4 Instrumentation/gauges used
- 5.2.5 Range and standardization

5.3 Effects of temperature and other atmospheric conditions

5.4 Standardization for testing

5.5 Probing/scanning or measurement/monitoring

5.6 Leak interpretation evaluation

5.7 Acceptance and rejection criteria

6.0 Procedures

6.1 LT procedures
- 6.1.1 Basic categories and techniques
- 6.1.2 Leak location versus leakage measurement
- 6.1.3 Pressurization or evacuation
- 6.1.4 Sealed units with or without tracer gas
- 6.1.5 Units accessible from one or both sides
- 6.1.6 Systems at, above, or below atmospheric pressure
- 6.1.7 Qualification of LT procedures
- 6.1.8 TPS and its calculation
- 6.1.9 Effective instrument sensitivity and its calculation
- 6.1.10 TQF, test reliability – correlation of system calibration factors

6.2 LT specifications
- 6.2.1 Design versus working conditions
- 6.2.2 Pressures and temperature control
- 6.2.3 Types of leak testing methods
- 6.2.4 Sensitivity of leak testing methods
- 6.2.5 Test method and sensitivity needed
- 6.2.6 TPS and its calculation
- 6.2.7 Effective instrument sensitivity and its calculation
- 6.2.8 TQF, test reliability – correlation of system calibration factors
- 6.2.9 Preparation of an LT specification

7.0 Safety and Health

7.1 Safety considerations
- 7.1.1 Personnel and the public
- 7.1.2 Product serviceability
- 7.1.3 Test validity
- 7.1.4 Safe work practices

7.2 Safety precautions
- 7.2.1 Explosive/implosive hazards
- 7.2.2 Flammability, ignitibility, combustibility hazards
- 7.2.3 Toxicity and asphyxiation hazards
- 7.2.4 Cleaning and electrical hazards

7.3 Pressure precautions
- 7.3.1 Pressure test versus proof test
- 7.3.2 Preliminary leak test
- 7.3.3 Pressurization check
- 7.3.4 Design limitations
- 7.3.5 Equipment and setup

7.4 Safety devices
- 7.4.1 Pressure control valves and regulators
- 7.4.2 Pressure relief valves and vents
- 7.4.3 Flow rate of regulator and relief valves

7.5 Hazardous and tracer gas safety
- 7.5.1 Combustible gas detection and safety
- 7.5.2 Toxic gas detection and safety
- 7.5.3 Oxygen-deficiency detectors
- 7.5.4 Radioisotope detection

7.6 Types of monitoring equipment
- 7.6.1 Area monitors
- 7.6.2 Personnel monitors
- 7.6.3 Leak-locating devices

7.7 Safety
- 7.7.1 State and federal regulations
- 7.7.2 Safety codes/standards
- 7.7.3 Hazardous gas standards
- 7.7.4 NRC radiation requirements

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PT
LIQUID PENETRANT TESTING TOPICAL OUTLINES

Liquid Penetrant Testing Level I Topical Outline

1.0 Introduction
- 1.1 Brief history of nondestructive testing and liquid penetrant testing (PT)
- 1.2 Purpose of PT
- 1.3 Basic principles of PT
- 1.4 Types of liquid penetrants commercially available
- 1.5 Method of personnel qualification

2.0 Liquid Penetrant Processing
- 2.1 Preparation of parts
- 2.2 Adequate lighting
- 2.3 Application of penetrant to parts
- 2.4 Removal of surface penetrant
- 2.5 Developer application and drying
- 2.6 Inspection and evaluation
- 2.7 Postcleaning

3.0 Various PT Methods
- 3.1 Current ASTM and ASME standard methods – ASTM E165, E1208, E1209, E1210, and E1417
- 3.2 Characteristics of each method
- 3.3 General applications of each method

4.0 PT Equipment
- 4.1 PT units
- 4.2 Lighting for PT equipment and light meters
- 4.3 Materials for PT
- 4.4 Precautions in PT

Liquid Penetrant Testing Level II Topical Outline

1.0 Review
- 1.1 Basic principles
- 1.2 Process of various methods
- 1.3 Equipment

2.0 Selection of the Appropriate Liquid Penetrant Testing Method
- 2.1 Advantages of various methods
- 2.2 Disadvantages of various methods

3.0 Inspection and Evaluation of Indications
- 3.1 General
  - 3.1.1 Discontinuities inherent in various materials
  - 3.1.2 Reason for indications
  - 3.1.3 Appearance of indications
  - 3.1.4 Time for indications to appear
  - 3.1.5 Persistence of indications
  - 3.1.6 Effects of temperature and lighting [white to ultraviolet (UV)]
  - 3.1.7 Effects of metal smearing operations (shot peening, machining, etc.)
  - 3.1.8 Preferred sequence for penetrant inspection
  - 3.1.9 Part preparation (precleaning, stripping, etc.)
- 3.2 Factors affecting indications
  - 3.2.1 Precleaning
  - 3.2.2 Penetrant used
  - 3.2.3 Prior processing
  - 3.2.4 Technique used
- 3.3 Indications from cracks
  - 3.3.1 Cracks occurring during solidification
  - 3.3.2 Cracks occurring during processing
  - 3.3.3 Cracks occurring during service
- 3.4 Indications from porosity
- 3.5 Indications from specific material forms
  - 3.5.1 Forgings
  - 3.5.2 Castings
  - 3.5.3 Plate
  - 3.5.4 Welds
  - 3.5.5 Extrusions
- 3.6 Evaluation of indications
  - 3.6.1 Human factors
  - 3.6.2 Continuity of inspection
  - 3.6.3 True indications
  - 3.6.4 False indications
  - 3.6.5 Relevant indications
  - 3.6.6 Nonrelevant indications
  - 3.6.7 Process control
    - 3.6.7.1 Controlling process variables
    - 3.6.7.2 Testing and maintenance materials

4.0 Inspection Procedures and Standards
- 4.1 Inspection procedures (minimum requirements)
- 4.2 Documentation of inspection/test
- 4.3 Standards/codes
  - 4.3.1 Applicable methods/processes
  - 4.3.2 Acceptance criteria

Liquid Penetrant Testing Level III Topical Outline

1.0 Principles/Theory
- 1.1 Principles of PT process
  - 1.1.1 Process variables
  - 1.1.2 Effects of test object factors on process
- 1.2 Theory
  - 1.2.1 Physics of how penetrants work
  - 1.2.2 Control and measurement of penetrant process variables
    - 1.2.2.1 Surface tension, viscosity, and capillary entrapment
    - 1.2.2.2 Measurement of penetrability, washability, and emulsification
    - 1.2.2.3 Contrast, brightness, and fluorescence
    - 1.2.2.4 Contamination of materials
    - 1.2.2.5 Proper selection of penetrant levels for different testing (sensitivity)
- 1.3 Proper selection of PT as method of choice
  - 1.3.1 Difference between PT and other methods
  - 1.3.2 Complementary roles of PT and other methods
  - 1.3.3 Potential for conflicting results between methods
  - 1.3.4 Factors that qualify/disqualify the use of PT
  - 1.3.5 Selection of PT technique
- 1.4 Liquid penetrant processing
  - 1.4.1 Preparation of parts
  - 1.4.2 Applications of penetrants and emulsifiers to parts
  - 1.4.3 Removal of surface penetrants
  - 1.4.4 Developer application and drying
  - 1.4.5 Evaluation
  - 1.4.6 Postcleaning
  - 1.4.7 Precautions

2.0 Equipment/Materials
- 2.1 Methods of measurement
- 2.2 Lighting for PT
  - 2.2.1 White light intensity
  - 2.2.2 Ultraviolet radiation intensity, warm-up time, etc.
  - 2.2.3 Physics and physiological differences
- 2.3 Materials for PT
  - 2.3.1 Solvent-removable
  - 2.3.2 Water-washable
  - 2.3.3 Postemulsifiable
    - 2.3.3.1 Water base (hydrophilic)
    - 2.3.3.2 Oil base (lipophilic)
  - 2.3.4 Dual sensitivity
- 2.4 Testing and maintenance of materials

3.0 Interpretation/Evaluation
- 3.1 General
  - 3.1.1 Human factors
  - 3.1.2 Continuity of inspection
  - 3.1.3 Appearance of penetrant indications
  - 3.1.4 Persistence of indications
- 3.2 Factors affecting indications
  - 3.2.1 Preferred sequence for penetrant inspection
  - 3.2.2 Part preparation (precleaning, stripping, etc.)
  - 3.2.3 Environment (lighting, temperature, etc.)
  - 3.2.4 Effect of metal smearing operations (shot peening, machining, etc.)
- 3.3 Indications from discontinuities
  - 3.3.1 Metallic materials
  - 3.3.2 Nonmetallic materials
- 3.4 Relevant and nonrelevant indications
  - 3.4.1 True indications
  - 3.4.2 False indications

4.0 Procedures
- 4.1 Foreword (scope, reference documents)
- 4.2 Personnel
- 4.3 Apparatus to be used, including settings
- 4.4 Product (description or drawing, including area of interest and purpose of the test)
- 4.5 Test conditions, including preparation for testing
- 4.6 Detailed instructions for application of the test
- 4.7 Recording and classifying the results of the test
- 4.8 Reporting the results

5.0 Safety and Health
- 5.1 Toxicity
- 5.2 Flammability
- 5.3 Precautions for ultraviolet radiation
- 5.4 Material safety data sheets (MSDS)

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MFL
MAGNETIC FLUX LEAKAGE TESTING TOPICAL OUTLINES

Magnetic Flux Leakage Testing Level I Topical Outline

1.0 Magnetic Flux Leakage Testing (MFL)
- 1.1 Brief history of MFL
- 1.2 Basic principles of MFL

2.0 Principles of Magnetic Fields
- 2.1 Magnetic fields characteristics
- 2.2 Flux line characteristics

3.0 Magnetism by Means of Electric Current
- 3.1 Field around a conductor
- 3.2 Right-hand rule
- 3.3 Field in ferromagnetic conductors

4.0 Indirect Magnetization
- 4.1 Circular fields
- 4.2 Longitudinal fields
- 4.3 Transverse fields

5.0 Magnetization Variables
- 5.1 Type of magnetizing current
- 5.2 Alloy magnetic properties
  - 5.2.1 Hysteresis curve
  - 5.2.2 Permeability
  - 5.2.3 Factors affecting permeability

6.0 Flux Leakage
- 6.1 Flux leakage theory
- 6.2 Normal component of flux leakage

7.0 Search Coils
- 7.1 Rate of change in the normal component of flux leakage
- 7.2 Faraday's law (rate of change versus induced voltage)
- 7.3 Factors that affect the voltage induced in a search coil

8.0 Hall Effect Search Units
- 8.1 Hall effect principles
- 8.2 Factors that affect the output voltage of Hall effect element

9.0 Signal Processing
- 9.1 Rectification
- 9.2 Filtering

10.0 Readout Mechanism
- 10.1 Displays
- 10.2 Strip-chart recorder
- 10.3 Computerized data acquisition

Magnetic Flux Leakage Testing Level II Topical Outline

Magnetic Flux Leakage Evaluation Course

1.0 Review of Magnetic Theory
- 1.1 Flux leakage theory
- 1.2 Types of flux leakage sensing probes

2.0 Factors that Affect Flux Leakage Fields
- 2.1 Degree of magnetization
- 2.2 Defect geometry
- 2.3 Defect location
- 2.4 Defect orientation
- 2.5 Distance between adjacent defects

3.0 Signal-to-Noise Ratio
- 3.1 Definition
- 3.2 Relationship to flux leakage testing
- 3.3 Methods of improving signal-to-noise ratio (SNR)

4.0 Selection of Method of Magnetization for Flux Leakage Testing
- 4.1 Magnetization characteristics for various magnetic materials
- 4.2 Magnetization by means of electric fields
  - 4.2.1 Circular field
  - 4.2.2 Longitudinal field
  - 4.2.3 Value of flux density
- 4.3 Magnetization by means of permanent magnets
  - 4.3.1 Permanent magnet relationship and theory
  - 4.3.2 Permanent magnet materials
- 4.4 Selection of proper magnetization method

5.0 Coupling
- 5.1 Liftoff in MFL

6.0 Signal Processing Considerations
- 6.1 Amplification
- 6.2 Filtering

7.0 Applications
- 7.1 General
  - 7.1.1 Flaw detection
  - 7.1.2 Sorting for properties related to permeability
  - 7.1.3 Measurement of magnetic-characteristic values
- 7.2 Specific
  - 7.2.1 Tank floor and side inspection
  - 7.2.2 Wire rope inspection
  - 7.2.3 Tube inspection
  - 7.2.4 "Intelligent" pigs
  - 7.2.5 Bar inspection

8.0 User Standards and Operating Procedures
- 8.1 Explanation of standards and specifications used in MFL
- 8.2 Explanation of operating procedures used in MFL

Magnetic Flux Leakage Testing Level III Topical Outline

1.0 Principles/Theory
- 1.1 Flux leakage theory
- 1.2 Förster and other theories
- 1.3 Finite element methods
- 1.4 DC flux leakage/AC flux leakage

2.0 Equipment/Materials
- 2.1 Detectors
  - 2.1.1 Advantages/limitations
- 2.2 Coils
  - 2.2.1 Advantages/limitations
- 2.3 Factors affecting choice of sensing elements
  - 2.3.1 Type of part to be inspected
  - 2.3.2 Type of discontinuity to be detected
  - 2.3.3 Speed of testing required
  - 2.3.4 Amount of testing required
  - 2.3.5 Probable location of discontinuity
  - 2.3.6 Applications other than discontinuity detection
- 2.4 Readout selection
  - 2.4.1 Oscilloscope and other monitor displays
  - 2.4.2 Alarm, lights, etc.
  - 2.4.3 Marking system
  - 2.4.4 Sorting gates and tables
  - 2.4.5 Cutoff saw or shears
  - 2.4.6 Automation and feedback
  - 2.4.7 Strip-chart recorder
  - 2.4.8 Computerized data acquisition
- 2.5 Instrument design considerations
  - 2.5.1 Amplification
  - 2.5.2 Filtering
  - 2.5.3 Sensor configuration

3.0 Techniques/Standardization
- 3.1 Consideration affecting choice of test
  - 3.1.1 SNR
  - 3.1.2 Response speed
  - 3.1.3 Skin effect
- 3.2 Coupling
  - 3.2.1 Fill factor
  - 3.2.2 Liftoff
- 3.3 Field strength
  - 3.3.1 Permeability changes
  - 3.3.2 Saturation
- 3.4 Comparison of techniques
- 3.5 Standardization
  - 3.5.1 Techniques
  - 3.5.2 Reference standards
- 3.6 Techniques – general
  - 3.6.1 Surface or subsurface flaw detection
  - 3.6.2 Noise suppression

4.0 Interpretation/Evaluation
- 4.1 Flaw detection
- 4.2 Process control
- 4.3 General interpretations
- 4.4 Defect characterization

5.0 Standards
- 5.1 ASME Section V Article 16, Magnetic Flux Leakage
- 5.2 API 653 Appendix G

6.0 Procedures

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MT
MAGNETIC PARTICLE TESTING TOPICAL OUTLINES

Magnetic Particle Testing Level I Topical Outline

1.0 Introduction
- 1.1 History of magnetic particle testing (MT)
- 1.2 Purpose of NDT
- 1.3 Overview of basic NDT methods
- 1.4 Training and certification process
- 1.5 MT process overview
- 1.6 Advantages and limitations
- 1.7 Terminology

2.0 Principles of Magnets and Magnetic Fields
- 2.1 Theory of magnetic fields
  - 2.1.1 Earth's magnetic field
  - 2.1.2 Electrical fields
  - 2.1.3 Magnetic fields around magnetized materials
- 2.2 Theory of magnetism
  - 2.2.1 Domain theory
  - 2.2.2 Materials
  - 2.2.3 Flux density
  - 2.2.4 Law of magnetism
  - 2.2.5 Properties of magnetic fields
  - 2.2.6 Flux leakage
  - 2.2.7 Hysteresis of magnetism
  - 2.2.8 Measurement of fields

3.0 Material Preparation and Discontinuities
- 3.1 Component preparation
- 3.2 Precleaning
  - 3.2.1 Solvent
  - 3.2.2 Chemical
  - 3.2.3 Mechanical
- 3.3 Coatings
- 3.4 Surface discontinuities
- 3.5 Subsurface discontinuities

4.0 Circular Magnetism
- 4.1 Field produce in a straight conductor
- 4.2 Field within materials
  - 4.2.1 Right-hand rule
  - 4.2.2 Fields in materials
  - 4.2.3 Indication geometry
- 4.3 Direct magnetic induction
- 4.4 Indirect magnetic induction
- 4.5 Strength of fields
- 4.6 Field detection
- 4.7 Advantages/disadvantages

5.0 Longitudinal Magnetism
- 5.1 Field produced by coil
- 5.2 Field within materials
  - 5.2.1 Magnetic field induction
  - 5.2.2 Fields in materials
  - 5.2.3 Indication geometry
- 5.3 Strength of fields
- 5.4 Field detection
- 5.5 Advantages/disadvantages

6.0 Sources of Magnetic Fields
- 6.1 Magnets
- 6.2 Types of magnetizing current
  - 6.2.1 Direct current
  - 6.2.2 Alternating current
  - 6.2.3 Advantages/disadvantages

7.0 Inspection Materials
- 7.1 Wet particles
- 7.2 Dry particles
- 7.3 Application of particles

8.0 Principles of Demagnetization
- 8.1 Residual magnetism
- 8.2 Reasons for demagnetization
- 8.3 Longitudinal and circular residual fields
- 8.4 Basic principles of demagnetization
- 8.5 Residual and coercive force
- 8.6 Methods of demagnetization
- 8.7 Measurement of results

9.0 Equipment
- 9.1 Equipment selection considerations
  - 9.1.1 Type of magnetizing current
  - 9.1.2 Location and nature of test
  - 9.1.3 Test materials
  - 9.1.4 Purpose of test
  - 9.1.5 Area inspected

9.2 Stationary equipment
9.3 Mobile equipment
9.4 Portable equipment
9.5 Multidirectional equipment
9.6 Automatic equipment
9.7 Demagnetization equipment
9.8 Light sources
 9.8.1 White light
 9.8.2 UV-A light
 9.8.3 Ambient light
 9.8.4 Measurement devices
9.9 Field measurement devices
 9.9.1 Field direction
 9.9.2 Field strength
9.10 Materials
 9.10.1 Visible particles
 9.10.2 Fluorescent particles
 9.10.3 Wet versus dry
 9.10.4 Bath properties and concentrations
 9.10.5 Portable materials

10.0 Selecting Proper Methods of Magnetization
10.1 Procedure and written instructions
10.2 Industry standards
 10.2.1 Welds
 10.2.2 Pipe
 10.2.3 Aerospace
 10.2.4 Other users

11.0 Product and flaw stages
11.1 Inherent stage
11.2 Primary processing stage
11.3 Secondary processing stage
11.4 In-service stage
11.5 Types of discontinuities detected by MT
 11.5.1 Inclusions
 11.5.2 Blowholes
 11.5.3 Porosity
 11.5.4 Flakes
 11.5.5 Cracks
 11.5.6 Pipes
 11.5.7 Laminations
 11.5.8 Laps
 11.5.9 Forging bursts
 11.5.10 Voids

12.0 Magnetic Particle Testing Indications and Interpretations
12.1 Human factors
12.2 View conditions
12.3 Continuity of inspection
12.4 Lighting
12.5 Component use and condition
12.6 Purpose of test
12.7 Interpretation of indication
 12.7.1 Relevant versus nonrelevant
 12.7.2 Surface versus subsurface indications
12.8 Evaluation
 12.8.1 Criteria
 12.8.2 Acceptance/rejection
12.9 Postcleaning and protection
12.10 Documentation of test

Magnetic Particle Testing Level II Topical Outline

1.0 Review
1.1 Basic principles
1.2 Basic magnetic particle process
1.3 Equipment
1.4 Terminology
1.5 Test factors
1.6 Safety
1.7 Training and certification processes

2.0 Principles
2.1 Theory
 2.1.1 Flux patterns
 2.1.2 Frequency and voltage factors
 2.1.3 Current calculations
 2.1.4 Surface flux strength
 2.1.5 Subsurface effects
2.2 Magnets and magnetism
 2.2.1 Distance factors versus strength of flux
 2.2.2 Internal and external flux patterns
 2.2.3 Phenomenon action at the discontinuity
 2.2.4 Heat effects on magnetism
 2.2.5 Material hardness versus magnetic retention
2.3 Flux fields
 2.3.1 Direct current
  2.3.1.1 Depth of penetration factors
  2.3.1.2 Source of current
 2.3.2 Direct pulsating current
  2.3.2.1 Similarity to direct current
  2.3.2.2 Advantages
  2.3.2.3 Typical fields
 2.3.3 Alternating current
  2.3.3.1 Cyclic effects
  2.3.3.2 Surface strength characteristics
  2.3.3.3 Safety precautions
  2.3.3.4 Voltage and current factors
  2.3.3.5 Source of current
2.4 Effects of discontinuities on materials
 2.4.1 Design factors
  2.4.1.1 Mechanical properties
  2.4.1.2 Part use
 2.4.2 Relationship to load-carrying ability
 2.4.3 Cyclic loading issues

- 2.5 Magnetization by means of electric current
  - 2.5.1 Circular techniques
    - 2.5.1.1 Current determinations
    - 2.5.1.2 Depth-factor considerations
    - 2.5.1.3 Precautions – safety and overheating
    - 2.5.1.4 Contact prods and plates
      - 2.5.1.4.1 Requirements for prods and plates
      - 2.5.1.4.2 Current-carrying capabilities
    - 2.5.1.5 Discontinuities commonly detected
  - 2.5.2 Longitudinal technique
    - 2.5.2.1 Principles of induced flux fields
    - 2.5.2.2 Geometry of part to be inspected
    - 2.5.2.3 Shapes and sizes of coils
    - 2.5.2.4 Use of coils and cables
      - 2.5.2.4.1 Strength of field
      - 2.5.2.4.2 Current directional flow versus flux field
      - 2.5.2.4.3 Shapes, sizes, and current capacities
    - 2.5.2.5 Determining proper current
      - 2.5.2.5.1 Types of current required
      - 2.5.2.5.2 Current evaluation
      - 2.5.2.5.3 Measuring current
    - 2.5.2.6 Yokes
    - 2.5.2.7 Discontinuities commonly detected
- 2.6 Selecting the proper method of magnetization
  - 2.6.1 Alloy, shape, and condition of part
  - 2.6.2 Type of magnetizing current
  - 2.6.3 Direction of magnetic field
  - 2.6.4 Sequence of operations
  - 2.6.5 Value of flux density

3.0 Principles of Demagnetization

- 3.1 Residual magnetism
- 3.2 Reasons for demagnetization
- 3.3 Longitudinal and circular residual fields
- 3.4 Basic principles of demagnetization
- 3.5 Residual and coercive force
- 3.6 Methods of demagnetization
- 3.7 Measurement of results

4.0 Equipment

- 4.1 Portable equipment
  - 4.1.1 Reason for portable equipment
  - 4.1.2 Capabilities of portable equipment
  - 4.1.3 Similarity to stationary equipment
- 4.2 Mobile equipment
  - 4.2.1 Reason for mobile equipment
  - 4.2.2 Capabilities of mobile equipment
  - 4.2.3 Similarity to stationary equipment
- 4.3 Stationary equipment
  - 4.3.1 Requirements for automation
  - 4.3.2 Sequential operations
  - 4.3.3 Control and operation factors
  - 4.3.4 Alarm and rejection mechanisms
- 4.4 Multidirectional equipment
  - 4.4.1 Capabilities
  - 4.4.2 Sequential operations
  - 4.4.3 Control and operation factors
  - 4.4.4 Applications
- 4.5 Liquids and powders
  - 4.5.1 Liquid requirements as a particle vehicle
  - 4.5.2 Safety precautions
  - 4.5.3 Temperature needs
  - 4.5.4 Powder and paste contents
  - 4.5.5 Mixing procedures
  - 4.5.6 Need for accurate proportions
  - 4.5.7 Bath-strength factors and strength quality
- 4.6 Light sources
  - 4.6.1 White light
  - 4.6.2 UV-A light
  - 4.6.3 Ambient light
  - 4.6.4 Measurement devices
  - 4.6.5 Safety

5.0 Types of Discontinuities

- 5.1 In castings
- 5.2 In ingots
- 5.3 In wrought sections and parts
- 5.4 In welds

6.0 Magnetic Particle Testing Indications and Interpretations

- 6.1 Human factors
- 6.2 Continuity of inspection
- 6.3 View conditions
- 6.4 Lighting
- 6.5 Component use and condition
- 6.6 Purpose of test
- 6.7 Interpretation of indication
  - 6.7.1 Relevant versus nonrelevant
  - 6.7.2 Surface versus subsurface indications
- 6.8 Evaluation
  - 6.8.1 Criteria
  - 6.8.2 Acceptance/rejection
- 6.9 Postcleaning and protection
- 6.10 Documentation of test

7.0 Procedure Development

- 7.1 Use of standards (e.g., ASTM, ASME)
- 7.2 Need for standards and references
- 7.3 Part considerations
  - 7.3.1 History of part
  - 7.3.2 Manufacturing process
  - 7.3.3 Possible causes of defect
  - 7.3.4 Usage of part
  - 7.3.5 Acceptance/rejection criteria
- 7.4 Use of tolerances
- 7.5 Validation processes

8.0 Quality Control of Equipment and Processes
- 8.1 Equipment preventive maintenance
- 8.2 Vehicle bath maintenance and strength checks
  - 8.2.1 Settling test process
  - 8.2.2 Other bath-strength tests
- 8.3 Lighting intensity checks
  - 8.3.1 UV light
  - 8.3.2 White light
- 8.4 Field strength validation devices
- 8.5 Residual field checking devices

Magnetic Particle Testing Level III Topical Outline

1.0 Principles/Theory
- 1.1 Principles of magnets and magnetic fields
  - 1.1.1 Theory of magnetic fields
  - 1.1.2 Theory of magnetism
  - 1.1.3 Terminology associated with MT
- 1.2 Characteristics of magnetic fields
  - 1.2.1 Bar magnet
  - 1.2.2 Ring magnet

2.0 Equipment/Materials
- 2.1 MT equipment
  - 2.1.1 Equipment selection considerations
  - 2.1.2 Manual inspection equipment
  - 2.1.3 Medium- and heavy-duty equipment
  - 2.1.4 Stationary equipment
  - 2.1.5 Mechanized inspection equipment
- 2.2 Inspection materials
  - 2.2.1 Wet particle technique
  - 2.2.2 Dry particle technique

3.0 Technique
- 3.1 Magnetization by means of electric current
  - 3.1.1 Circular field
    - 3.1.1.1 Field around a straight conductor
    - 3.1.1.2 Right-hand rule
    - 3.1.1.3 Field in parts through which current flows
    - 3.1.1.4 Methods of inducing current flow in parts
    - 3.1.1.5 Discontinuities commonly indicated by circular field
    - 3.1.1.6 Applications of circular magnetization
  - 3.1.2 Longitudinal field
    - 3.1.2.1 Field direction
    - 3.1.2.2 Discontinuities commonly indicated by longitudinal techniques
    - 3.1.2.3 Applications of longitudinal magnetization
- 3.2 Selecting the proper method of magnetization
  - 3.2.1 Alloy, shape, and condition of part
  - 3.2.2 Type of magnetizing field
  - 3.2.3 Direction of magnetic field
  - 3.2.4 Sequence of operation
  - 3.2.5 Value of flux density
- 3.3 Demagnetization
  - 3.3.1 Reasons for requiring demagnetization
  - 3.3.2 Methods of demagnetization

4.0 Interpretation/Evaluation
- 4.1 Continuity of inspection
- 4.2 MT indications and interpretations
- 4.3 Effects of discontinuities on materials and types of discontinuities indicated by MT
- 4.4 MT procedures, codes, standards, and specifications

5.0 Procedures
- 5.1 Foreword (scope, reference documents)
- 5.2 Personnel
- 5.3 Apparatus to be used, including settings
- 5.4 Product (description or drawing, including area of interest and purpose of the test)
- 5.5 Test conditions, including preparation for testing
- 5.6 Detailed instructions for application of the test
- 5.7 Recording and classifying the results of the test
- 5.8 Reporting the results

6.0 Safety and Health
- 6.1 Toxicity
- 6.2 Electrical
- 6.3 Flammability
- 6.4 Precautions for UV
- 6.5 Material safety data sheets (MSDS)
- 6.6 Precautions for UV

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MW
MICROWAVE TECHNOLOGY TESTING TOPICAL OUTLINES

Microwave Technology Testing Level I Topical Outline

Theory Course

1.0 Introduction to Microwave Theory
- 1.1 Microwave scanning
- 1.2 Reflection interference pattern
- 1.3 Differential amplitude measurement
- 1.4 Basic math review

2.0 Basic Electromagnetic Theory
- 2.1 Nature of electromagnetic waves
- 2.2 Wave generation
- 2.3 Propagation and modes

3.0 Material Properties
- 3.1 Conductors and dielectrics

4.0 Scanning Microwave Basics
- 4.1 Interferometric techniques
  - 4.1.1 Interference pattern
  - 4.1.2 Compare amplitude of reflected signal with standing wave
  - 4.1.3 Emitter and receiver in probe fixed physical relationship
  - 4.1.4 Fixed phase relationship
  - 4.1.5 Phase relation
  - 4.1.6 Two-channel receiver
  - 4.1.7 Standoff
- 4.2 Multifrequency techniques
  - 4.2.1 Phase and magnitude relationship
- 4.3 Frequency modulated continuous wave
  - 4.3.1 Time-based display
- 4.4 Antenna and sensor pattern
  - 4.4.1 Reflected signal pattern
- 4.5 Image creation
  - 4.5.1 Signal voltage – position
  - 4.5.2 Interference map – phase relationship
  - 4.5.3 Channel relationship
- 4.6 Signal differential and phase relationship
- 4.7 Interference pattern image
  - 4.7.1 Line and circle patterns
  - 4.7.2 Nodes and depth
- 4.8 Measurement
  - 4.8.1 Interference patterns
  - 4.8.2 Wavelength in material

Technique Course

1.0 Basic Scanning Microwave Instrument
- 1.1 System hardware

2.0 Instrument Electronics
- 2.1 Transmitter
- 2.2 Receiver
- 2.3 Antennas
- 2.4 Probe
- 2.5 Position interface
- 2.6 Sampling
- 2.7 Data format

3.0 Position Control
- 3.1 Positioning interface
- 3.2 Position feedback

4.0 Antenna Design
- 4.1 Phase relation
- 4.2 Two-channel receiver
- 4.3 Standoff

5.0 Display System
- 5.1 Hardware
- 5.2 Display software

6.0 Control Functions
- 6.1 Power
- 6.2 Signal penetration
- 6.3 Response amplification

7.0 Data Collection and Formatting

Microwave Technology Testing Level II Topical Outline

Theory Course

1.0 Measure Electromagnetic Wave Interaction with Part
- 1.1 Attenuation
- 1.2 Reflection

2.0 Maxwell's Equations

3.0 Microwave Propagation
- 3.1 Wave propagation
- 3.2 Wave interaction with matter

4.0 Microwave Energy and Excitation Energies

5.0 Microwave Energy Field – Charge Interaction
- 5.1 Dipole moment
- 5.2 Polarity
- 5.3 Interference
- 5.4 Anisotropic material properties

6.0 Material Properties
- 6.1 Conductors and dielectrics
- 6.2 Excitation energies

7.0 Wave Propagation
- 7.1 Attenuation and impedance
- 7.2 Reflection
- 7.3 Refraction
- 7.4 Beam spread

8.0 Microwave Antennas
- 8.1 Antenna pattern polarization
  - 8.1.1 Energy profile
- 8.2 Near field versus far field
- 8.3 Lenses, horns, and SAR

9.0 Phase Relationship

Technique Course

1.0 Data Format and Collection
- 1.1 Metadata (heading and reference data)
- 1.2 Scan pattern
- 1.3 Sample density
- 1.4 File management
- 1.5 Transportability

2.0 Instrument Characteristics
- 2.1 Interferometric techniques
  - 2.1.1 Interference pattern
  - 2.1.2 Compare amplitude of reflected signal with standing wave
  - 2.1.3 Emitter and receiver in probe fixed physical relationship
  - 2.1.4 Fixed phase relationship
- 2.2 Multifrequency techniques
  - 2.2.1 FFT/IFFT
  - 2.2.2 SAR
- 2.3 Frequency modulated continuous wave
  - 2.3.1 Time-based signal processing
- 2.4 Probe design
  - 2.4.1 Phase relation
  - 2.4.2 Multichannel receiver
  - 2.4.3 Standoff
- 2.5 Antenna pattern
  - 2.5.1 Optimizing beam geometry
- 2.5 Image creation
  - 2.5.1 Multiple channels
  - 2.5.2 Variable frequency
- 2.6 Signal differential and phase relationship
  - 2.6.1 Phase vector
  - 2.6.2 Unit circle
  - 2.6.3 Analysis

3.0 Standardization
- 3.1 Frequency selection
- 3.2 Standoff
- 3.3 Power
- 3.4 Gain
- 3.5 Sample rate
- 3.6 Scan separation

4.0 Reference Blocks
- 4.1 Nominal specimen
- 4.2 Fault examples
- 4.3 Measurement
  - 4.3.1 Dielectric constant
  - 4.3.2 Wavelength in material
  - 4.3.3 Loss tangent

5.0 Imaging Surfaces at a Distance

6.0 Thickness Measurement

7.0 Density Measurement

8.0 Evaluation
- 8.1 Codes and standards
- 8.2 Determination of relative condition
  - 8.2.1 Comparison of tested part to prototype
  - 8.2.2 Comparison to nominal
  - 8.2.3 Evaluation of artifacts
- 8.3 Validation of results

Microwave Technology Testing Level III Topical Outline

Theory Course

1.0 Microwave Signal Phase Relationship

2.0 Wave Behavior in Complex Structures

3.0 Lenses, Horns, and SAR

4.0 Microwave Interaction with Matter

Technique Course

5.0 Basic Materials, Fabrication, and Product Technology
- 5.1 Fundamentals of material technology
  - 5.1.1 Properties of materials
    - 5.1.1.1 Strength and elastic properties
    - 5.1.1.2 Physical properties
    - 5.1.1.3 Material properties testing
  - 5.1.2 Origin of discontinuities and failure modes
    - 5.1.2.1 Inherent discontinuities
    - 5.1.2.2 Process-induced discontinuities
    - 5.1.2.3 Service-induced discontinuities
  - 5.1.3 Statistical nature of detecting and characterizing discontinuities
- 5.2 Dielectric materials susceptible to microwave examination
  - 5.2.1 Plastics
  - 5.2.2 Rubber
  - 5.2.3 Resins
  - 5.2.4 Fiberglass reinforced plastic
  - 5.2.5 Ceramics
  - 5.2.6 Composites
  - 5.2.7 Advanced materials
  - 5.2.8 Glass
  - 5.2.9 Coatings
- 5.3 In-process materials
- 5.4 Structures of interest

6.0 Origin of Discontinuities and Failure Modes
- 6.1 Process-induced discontinuities
  - 6.1.1 Placement geometry
  - 6.1.2 Contaminants
  - 6.1.3 Bonds between layers
  - 6.1.4 Voids
  - 6.1.5 Porosity
  - 6.1.6 Incorrect chemistry
  - 6.1.7 Incorrect material phase/crystallization
- 6.2 Service-induced discontinuities
  - 6.2.1 Inherent discontinuities
  - 6.2.2 Mechanical disruption
  - 6.2.3 Thermal damage
    - 6.2.3.1 Material phase differences
    - 6.2.3.2 Crystal structure changes

7.0 Statistical Nature of Detecting and Characterizing Discontinuities

8.0 Thickness Measurements
- 8.1 Coatings
- 8.2 Parts

9.0 Evaluation
- 9.1 Codes and standards
- 9.2 Design and construction specifications
- 9.3 Development of reference standards
- 9.4 Determination of relative condition
  - 9.4.1 Comparison of tested part to prototype
  - 9.4.2 Comparison to nominal
- 9.5 Evaluation of artifacts
- 9.6 Validation

10.0 Responsibilities of Levels of Certification
- 10.1 Level III
- 10.2 Level II
- 10.3 Level I trainee

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* Available from The American Society for Nondestructive Testing Inc., Columbus, OH.

NR

NEUTRON RADIOGRAPHIC TESTING TOPICAL OUTLINES

Neutron Radiographic Testing Level I Topical Outline

Note: Independent of the training recommended for Level I and Level II certification, a trainee is required to receive radiation safety training as required by the regulatory jurisdiction. A Radiation Safety Topical Outline is available in Appendix A and can be used as guidance.

Basic Neutron Radiographic Physics Course

1.0 Introduction
- 1.1 History of industrial neutron radiographic testing (NR)
- 1.2 General principles of examination of materials by penetrating radiation
- 1.3 Relationship of penetrating neutron radiation, radiography, and radiometry
- 1.4 Comparison with other NDT methods, particularly with X-rays and gamma rays
- 1.5 General areas of application
  - 1.5.1 Imaging
  - 1.5.2 Metrology
  - 1.5.3 Product

2.0 Physical Principles
- 2.1 Sources for NR (general description)
  - 2.1.1 Isotopes
  - 2.1.2 Nuclear reactors
  - 2.1.3 Accelerators
- 2.2 Interaction between neutrons and matter
  - 2.2.1 Absorption
    - 2.2.1.1 Thermal neutrons
    - 2.2.1.2 Resonance neutrons
    - 2.2.1.3 Fast neutrons
  - 2.2.2 Scatter
    - 2.2.2.1 Elastic
    - 2.2.2.2 Inelastic
- 2.3 NR techniques
  - 2.3.1 Film imaging techniques
  - 2.3.2 Nonfilm imaging techniques
- 2.4 Glossary of terms and units of measure

3.0 Radiation Sources for Neutrons (Specific Description)
- 3.1 Reactors
  - 3.1.1 Principle of fission chain reactions
  - 3.1.2 Neutron thermalization (slowing down)
  - 3.1.3 Thermal-neutron flux
- 3.2 Accelerators
  - 3.2.1 Types of accelerators
  - 3.2.2 Neutron-producing reactions
- 3.3 Isotopic sources
  - 3.3.1 Radioisotope + Be
    - 3.3.1.1 α – Be
    - 3.3.1.2 ϒ – Be
  - 3.3.2 Radioisotope + D
    - 3.3.2.1 ϒ – D
  - 3.3.3 Spontaneous fission
    - 3.3.3.1 ^{252}Cf

4.0 Personnel Safety Radiation Protection Review
- 4.1 Hazards of excessive exposure
  - 4.1.1 General – beta, gamma radiation
  - 4.1.2 Specific neutron hazards
    - 4.1.2.1 Relative biological effectiveness
    - 4.1.2.2 Neutron activation
- 4.2 Methods of controlling radiation dose
  - 4.2.1 Time
  - 4.2.2 Distance
  - 4.2.3 Shielding
- 4.3 Specific equipment requirements
  - 4.3.1 Neutron monitoring dosimeters
  - 4.3.2 Gamma ray monitoring dosimeters
  - 4.3.3 Radiation survey equipment
    - 4.3.3.1 Beta/gamma
    - 4.3.3.2 Neutron
  - 4.3.4 Recording/recordkeeping
- 4.4 Radiation work procedures
- 4.5 Federal, state, and local regulations

Basic Neutron Radiographic Testing Technique Course

1.0 Radiation Detection Imaging
- 1.1 Converter screens
  - 1.1.1 Principles of operation
  - 1.1.2 Direct-imaging screens
  - 1.1.3 Transfer-imaging screens
- 1.2 Film – principles, properties, and uses with neutron converter screens
  - 1.2.1 Radiation response
  - 1.2.2 Vacuum/contact considerations
  - 1.2.3 Radiographic speed
  - 1.2.4 Radiographic contrast
- 1.3 Track-etch
  - 1.3.1 Radiation response
  - 1.3.2 Vacuum/contact considerations
  - 1.3.3 Radiographic speed
  - 1.3.4 Radiographic contrast

2.0 Neutron Radiographic Testing Process: Basic Imaging Considerations
- 2.1 Definition of sensitivity (including image quality indicators)
- 2.2 Contrast and definition
  - 2.2.1 Neutron energy and neutron screen relationship
  - 2.2.2 Effect of scattering in object
- 2.3 Geometric principles
- 2.4 Generation and control of scatter
- 2.5 Choice of neutron source
- 2.6 Choice of film
- 2.7 Use of exposure curves
- 2.8 Cause of correction of unsatisfactory radiographs
  - 2.8.1 High film density
  - 2.8.2 Low film density
  - 2.8.3 High contrast
  - 2.8.4 Low contrast
  - 2.8.5 Poor definition
  - 2.8.6 Excessive film fog
  - 2.8.7 Light leaks
  - 2.8.8 Artifacts
- 2.9 Arithmetic of exposure

3.0 Test Result Interpretation
- 3.1 Relationship between X-ray and N-ray
- 3.2 Effects on measurement and interpretation of test
- 3.3 Administrative control of test quality by interpreter
- 3.4 Familiarization with image

Neutron Radiographic Testing Level II Topical Outline

Neutron Radiographic Testing Physics Course

1.0 Introduction
- 1.1 General principles of examination of materials by penetrating radiation
- 1.2 Relationship of penetrating neutron radiation, radiography, and radiometry
- 1.3 Comparison with other methods, particularly with X-rays and gamma rays
- 1.4 Specific areas of application in industry

2.0 Review of Physical Principles
- 2.1 Nature of penetrating radiation (all types)
  - 2.1.1 Particles
  - 2.1.2 Wave properties
  - 2.1.3 Electromagnetic waves
  - 2.1.4 Fundamentals of radiation physics
  - 2.1.5 Sources of radiation
    - 2.1.5.1 Electronic sources
    - 2.1.5.2 Isotopic sources
    - 2.1.5.3 Nuclear reactors
    - 2.1.5.4 Accelerators
- 2.2 Interaction between penetrating radiation and matter (neutron and gamma ray)
  - 2.2.1 Absorption
  - 2.2.2 Scatter
  - 2.2.3 Other interactions
- 2.3 Glossary of terms and units of measure

3.0 Radiation Sources for Neutrons
- 3.1 Neutron sources – general
  - 3.1.1 Reactors
    - 3.1.1.1 Principle of fission chain reactions
    - 3.1.1.2 Fast-neutron flux – energy and spatial distribution
    - 3.1.1.3 Neutron thermalization
    - 3.1.1.4 Thermal-neutron flux – energy and spatial distribution
  - 3.1.2 Accelerators
    - 3.1.2.1 Types of accelerators
    - 3.1.2.2 Neutron-producing reactions
    - 3.1.2.3 Available yields and energy spectra
  - 3.1.3 Isotopic sources
    - 3.1.3.1 Radioisotope + Be
    - 3.1.3.2 Radioisotope + D
    - 3.1.3.3 Spontaneous fission ^{252}Cf
  - 3.1.4 Beam design
    - 3.1.4.1 Source placement
    - 3.1.4.2 Collimation
    - 3.1.4.3 Filtering
    - 3.1.4.4 Shielding

4.0 Radiation Detection
- 4.1 Imaging
  - 4.1.1 Converter screens
    - 4.1.1.1 Principles of operations
    - 4.1.1.2 Types of screens
      - 4.1.1.2.1 Direct exposure
      - 4.1.1.2.2 Transfer exposure
      - 4.1.1.2.3 Track-etch process
      - 4.1.1.2.4 Spectral sensitivity (each process)

- 4.1.2 Film – principles, properties, use with neutron converter screens
  - 4.1.2.1 Material examination
  - 4.1.2.2 Monitoring
- 4.1.3 Fluoroscopy
  - 4.1.3.1 Fluorescent screen
  - 4.1.3.2 Image amplification
  - 4.1.3.3 Cine techniques
- 4.1.4 Direct TV viewing
- 4.1.5 Special instrumentation associated with above techniques

4.2 Nonimaging devices
- 4.2.1 Solid-state
  - 4.2.1.1 Scintillometer
  - 4.2.1.2 Photo-resistive devices
  - 4.2.1.3 Other
- 4.2.2 Gaseous
  - 4.2.2.1 Proportional counters
  - 4.2.2.2 Geiger counters
  - 4.2.2.3 Ionization chambers
  - 4.2.2.4 Other
- 4.2.3 Neutron detectors
  - 4.2.3.1 Boron-based gas counters
  - 4.2.3.2 Fission counters
  - 4.2.3.3 Helium-3 detectors
  - 4.2.3.4 Lithium-based scintillator
  - 4.2.3.5 Instrumentation
    - 4.2.3.5.1 Rate meters
    - 4.2.3.5.2 Counters
    - 4.2.3.5.3 Amplifiers and preamplifiers
    - 4.2.3.5.4 Recording readouts
    - 4.2.3.5.5 Other

5.0 Radiological Safety Principles Review

5.1 Controlling personnel exposure

5.2 Time, distance, shielding concepts

5.3 As low as reasonably achievable (ALARA) concept

5.4 Radiation detection equipment

5.5 Exposure device operating characteristics

Neutron Radiographic Testing Technique Course

1.0 Neutron Radiographic Testing Process

1.1 Basic neutron imaging considerations
- 1.1.1 Definition of sensitivity (including image quality indicators)
- 1.1.2 Contrast and definition
  - 1.1.2.1 Neutron energy and neutron screen relationship
  - 1.1.2.2 Effect of scattering in object
  - 1.1.2.3 Exposure versus foil thickness
- 1.1.3 Geometric principles
- 1.1.4 Intensifying screens
  - 1.1.4.1 Fluorescent (neutron-sensitive)
  - 1.1.4.2 Metallic (neutron-sensitive)
- 1.1.5 Generation and control of scatter
- 1.1.6 Choice of source
- 1.1.7 Choice of film/detector
- 1.1.8 Use of exposure curves and process by which they are generated
- 1.1.9 Fluoroscopic inspection
  - 1.1.9.1 Theory of operation
  - 1.1.9.2 Applications
  - 1.1.9.3 Limitations
- 1.1.10 Film processing
  - 1.1.10.1 Darkroom procedures
  - 1.1.10.2 Darkroom equipment and chemicals
  - 1.1.10.3 Film processing do's and don'ts
- 1.1.11 Viewing of radiographs
  - 1.1.11.1 Illuminator requirements (intensity)
  - 1.1.11.2 Background lighting
  - 1.1.11.3 Judging quality of neutron radiographs
- 1.1.12 Causes and correction of unsatisfactory radiographs
  - 1.1.12.1 High film density
  - 1.1.12.2 Insufficient film density
  - 1.1.12.3 High contrast
  - 1.1.12.4 Low contrast
  - 1.1.12.5 Poor definition
  - 1.1.12.6 Excessive neutron scatter
  - 1.1.12.7 Fog
  - 1.1.12.8 Light leaks
  - 1.1.12.9 Artifacts
- 1.1.13 Arithmetic of exposure and of other factors affecting neutron radiographs

1.2 Miscellaneous applications
- 1.2.1 Blocking and filtering
- 1.2.2 Multifilm techniques
- 1.2.3 Enlargement and projection
- 1.2.4 Stereoradiography
- 1.2.5 Triangulation methods
- 1.2.6 Autoradiography
- 1.2.7 Flash neutron radiography
- 1.2.8 "In-motion" radiography and fluoroscopy
- 1.2.9 Backscatter NR
- 1.2.10 Neutron tomography
- 1.2.11 Micro-neutron radiography
- 1.2.12 Causes of "diffraction" effects and minimization of interference with test
- 1.2.13 Determination of focal spot size
- 1.2.14 Panoramic techniques
- 1.2.15 Altering film contrast and density
- 1.2.16 Gauging and control processes

2.0 Test Result Interpretation

2.1 Basic factors
- 2.1.1 General aspects (relationship between X-ray and neutron radiographs)
- 2.1.2 Effects on measurement and interpretation of test
- 2.1.3 Administrative control of test quality by interpreter
- 2.1.4 Familiarization with image

- 2.2 Material considerations
  - 2.2.1 Metallurgy or other material consideration as it affects use of item and test results
  - 2.2.2 Materials-processing effects on use of item and test results
  - 2.2.3 Discontinuities – their causes and effects
  - 2.2.4 Radiographic appearance of discontinuities
- 2.3 Codes, standards, specifications, and procedures
  - 2.3.1 Thermal NR
  - 2.3.2 Resonance neutron radiography
  - 2.3.3 Other applicable codes, etc.

Neutron Radiographic Testing Level III Topical Outline

1.0 Principles/Theory

- 1.1 Nature of penetrating radiation
- 1.2 Interaction between penetrating radiation and matter
- 1.3 NR
  - 1.3.1 Imaging by film
  - 1.3.2 Imaging by fluorescent materials
  - 1.3.3 Imaging by electronic devices
- 1.4 Radiometry

2.0 Equipment/Materials

- 2.1 Sources of neutrons
  - 2.1.1 Reactors
  - 2.1.2 Accelerators
  - 2.1.3 Isotopic sources
  - 2.1.4 Beam control factors
- 2.2 Radiation detectors
  - 2.2.1 Imaging
    - 2.2.1.1 Converter screens
    - 2.2.1.2 Film principles, properties, use with neutron converter screens
    - 2.2.1.3 Fluoroscopy
    - 2.2.1.4 TV and optical systems
- 2.3 Nonimaging devices
  - 2.3.1 Solid-state detectors
  - 2.3.2 Gaseous ionization detectors
  - 2.3.3 Neutron detectors
  - 2.3.4 Instrumentation
  - 2.3.5 Gauging and control processes

3.0 Techniques/Standardization

- 3.1 Blocking and filtering
- 3.2 Multifilm technique
- 3.3 Enlargement and projection
- 3.4 Stereoradiography
- 3.5 Triangulation methods
- 3.6 Autoradiography
- 3.7 Flash radiography
- 3.8 In-motion radiography
- 3.9 Fluoroscopy
- 3.10 Electron emission radiography
- 3.11 Microradiography
- 3.12 Laminography (tomography)
- 3.13 Control of diffraction effects
- 3.14 Panoramic exposures
- 3.15 Gauging
- 3.16 Real-time imaging
- 3.17 Image analysis techniques

4.0 Interpretation/Evaluation

- 4.1 Radiographic interpretation
  - 4.1.1 Image-object relationships
  - 4.1.2 Material considerations
    - 4.1.2.1 Material processing as it affects use of item and test results
    - 4.1.2.2 Discontinuities – their cause and effects
    - 4.1.2.3 Radiographic appearance of discontinuities
  - 4.1.3 Codes, standards, and specifications

5.0 Procedures

- 5.1 The radiographic process
  - 5.1.1 Imaging considerations
    - 5.1.1.1 Sensitivity
    - 5.1.1.2 Contrast and definition
    - 5.1.1.3 Geometric factors
    - 5.1.1.4 Intensifying screens
    - 5.1.1.5 Scattered radiation
    - 5.1.1.6 Source factors
    - 5.1.1.7 Detection media
    - 5.1.1.8 Exposure curves
- 5.2 Film processing
  - 5.2.1 Darkroom procedures
  - 5.2.2 Darkroom equipment and chemicals
  - 5.2.3 Film processing
- 5.3 Viewing of radiographs
  - 5.3.1 Illuminator requirements
  - 5.3.2 Background lighting
  - 5.3.3 Optical aids
- 5.4 Judging radiographic quality
  - 5.4.1 Density
  - 5.4.2 Contrast
  - 5.4.3 Definition
  - 5.4.4 Artifacts
  - 5.4.5 Image quality indicators (IQIs)
  - 5.4.6 Causes and corrections of unsatisfactory radiographs

6.0 Safety and Health

- 6.1 Personnel safety and radiation hazards
  - 6.1.1 Exposure hazards
    - 6.1.1.1 General – beta, gamma
    - 6.1.1.2 Specific neutron hazards
  - 6.1.2 Methods of controlling radiation exposure
  - 6.1.3 Operation and emergency procedures

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RT

RADIOGRAPHIC TESTING TOPICAL OUTLINES

Radiography Level I Topical Outline

Note: Independent of the training recommended for Level I and Level II certification, a trainee is required to receive radiation safety training as required by the regulatory jurisdiction. A Radiation Safety Topical Outline is available in Appendix A and can be used as guidance.

Basic Radiographic Testing Physics Course

1.0 Introduction
- 1.1 History and discovery of radioactive materials
- 1.2 Definition of industrial radiographic testing (RT)
- 1.3 Radiation protection – why?
- 1.4 Basic math review – exponents, square root, etc.

2.0 Fundamental Properties of Matter
- 2.1 Elements and atoms
- 2.2 Molecules and compounds
- 2.3 Atomic particles – properties of protons, electrons, and neutrons
- 2.4 Atomic structure
- 2.5 Atomic number and weight
- 2.6 Isotope versus radioisotope

3.0 Radioactive Materials
- 3.1 Production
  - 3.1.1 Neutron activation
  - 3.1.2 Nuclear fission
- 3.2 Stable versus unstable (radioactive) atoms
- 3.3 Becquerel – the unit of activity
- 3.4 Half-life of radioactive materials
- 3.5 Plotting of radioactive decay
- 3.6 Specific activity – becquerels/gram

4.0 Types of Radiation
- 4.1 Particulate radiation – properties: alpha, beta, neutron
- 4.2 Electromagnetic radiation – X-ray, gamma ray
- 4.3 X-ray production
- 4.4 Gamma ray production
- 4.5 Gamma ray energy
- 4.6 Energy characteristics of common radioisotope sources
- 4.7 Energy characteristics of X-ray machines

5.0 Interaction of Radiation with Matter
- 5.1 Ionization
- 5.2 Radiation interaction with matter
  - 5.2.1 Photoelectric effect
  - 5.2.2 Compton scattering
  - 5.2.3 Pair production
- 5.3 Unit of radiation exposure – coulomb per kilogram (C/kg)
- 5.4 Emissivity of commonly used radiographic sources
- 5.5 Emissivity of X-ray exposure devices
- 5.6 Attenuation of electromagnetic radiation – shielding
- 5.7 Half-value layers (HVL), tenth-value layers (TVL)
- 5.8 Inverse square law

6.0 Exposure Devices and Radiation Sources
- 6.1 Radioisotope sources
  - 6.1.1 Sealed-source design and fabrication
  - 6.1.2 Gamma ray sources
  - 6.1.3 Beta and bremsstrahlung sources
  - 6.1.4 Neutron sources
- 6.2 Radioisotope exposure device characteristics
- 6.3 Electronic radiation sources – 500 keV and less, low energy
  - 6.3.1 Generator – high-voltage rectifiers
  - 6.3.2 X-ray tube design and fabrication
  - 6.3.3 X-ray control circuits
  - 6.3.4 Accelerating potential
  - 6.3.5 Target material and configuration
  - 6.3.6 Heat dissipation
  - 6.3.7 Duty cycle
  - 6.3.8 Beam filtration
- 6.4* Electronic radiation sources – medium- and high-energy
  - 6.4.1* Resonance transformer
  - 6.4.2* Van de Graaff accelerator
  - 6.4.3* Linear accelerator
  - 6.4.4* Betatron
  - 6.4.5* Coulomb per kilogram (C/kg) output
  - 6.4.6* Equipment design and fabrication
  - 6.4.7* Beam filtration
- 6.5* Fluoroscopic radiation sources
  - 6.5.1* Fluoroscopic equipment design
  - 6.5.2* Direct-viewing screens
  - 6.5.3* Image amplification
  - 6.5.4* Special X-ray tube considerations and duty cycle
  - 6.5.5* Screen unsharpness
  - 6.5.6* Screen conversion efficiency

7.0 Radiographic Safety Principles Review
- 7.1 Controlling personnel exposure
- 7.2 Time, distance, shielding concepts
- 7.3 ALARA concept
- 7.4 Radiation detection equipment
- 7.5 Exposure device operating characteristics

* Topics may be deleted if the employer does not use these methods and techniques.

Radiography Technique Course

1.0 Introduction
- 1.1 Process of radiography
- 1.2 Types of electromagnetic radiation sources
- 1.3 Electromagnetic spectrum
- 1.4 Penetrating ability or "quality" of X-rays and gamma rays
- 1.5 Spectrum of X-ray tube source
- 1.6 Spectrum of gamma radioisotope source
- 1.7 X-ray tube – change of mA or kVp effect on "quality" and intensity

2.0 Basic Principles of Radiography
- 2.1 Geometric exposure principles
  - 2.1.1 "Shadow" formation and distortion
  - 2.1.2 Shadow enlargement calculation
  - 2.1.3 Shadow sharpness
  - 2.1.4 Geometric unsharpness
  - 2.1.5 Finding discontinuity depth
- 2.2 Radiography screens
  - 2.2.1 Lead intensifying screens
  - 2.2.2 Fluorescent intensifying screens
  - 2.2.3 Intensifying factors
  - 2.2.4 Importance of screen-to-film contact
  - 2.2.5 Importance of screen cleanliness and care
  - 2.2.6 Techniques for cleaning screens
- 2.3 Radiography cassettes
- 2.4 Composition of industrial radiography film
- 2.5 The "heel effect" with X-ray tubes

3.0 Radiographs
- 3.1 Formation of the latent image on film
- 3.2 Inherent unsharpness
- 3.3 Arithmetic of radiography exposure
  - 3.3.1 Milliamperage – distance-time relationship
  - 3.3.2 Reciprocity law
  - 3.3.3 Photographic density
  - 3.3.4 X-ray exposure charts – material thickness, kV, and exposure
  - 3.3.5 Gamma ray exposure chart
  - 3.3.6 Inverse square law considerations
  - 3.3.7 Calculation of exposure time for gamma and X-ray sources
- 3.4 Characteristic (Hurter and Driffield) curve
- 3.5 Film speed and class descriptions
- 3.6 Selection of film for particular purpose

4.0 Radiographic Image Quality
- 4.1 Radiographic sensitivity
- 4.2 Radiographic contrast
- 4.3 Film contrast
- 4.4 Subject contrast
- 4.5 Definition
- 4.6 Film graininess and screen mottle effects
- 4.7 Image quality indicators (IQIs)

5.0 Film Handling, Loading, and Processing
- 5.1 Safelight and darkroom practices
- 5.2 Loading bench and cleanliness
- 5.3 Opening of film boxes and packets
- 5.4 Loading of film and sealing cassettes
- 5.5 Handling techniques for "green film"
- 5.6 Elements of manual film processing

6.0 Exposure Techniques – Radiography
- 6.1 Single-wall radiography
- 6.2 Double-wall radiography
  - 6.2.1 Viewing two walls simultaneously
  - 6.2.2 Offset double-wall exposure single-wall viewing
  - 6.2.3 Elliptical techniques
- 6.3 Panoramic radiography
- 6.4 Use of multiple-film loading
- 6.5 Specimen configuration

7.0 Fluoroscopic Techniques
- 7.1 Dark adaptation and eye sensitivity
- 7.2 Special scattered radiation techniques
- 7.3 Personnel protection
- 7.4 Sensitivity
- 7.5 Limitations
- 7.6 Direct-screen viewing
- 7.7 Indirect- and remote-screen viewing

Radiography Level II Topical Outline

Film Quality and Manufacturing Processes Course

1.0 Review of Basic Radiographic Principles
- 1.1 Interaction of radiation with matter
- 1.2 Math review
- 1.3 Exposure calculations
- 1.4 Geometric exposure principles
- 1.5 Radiographic image quality parameters

2.0 Darkroom Facilities, Techniques, and Processing
- 2.1 Facilities and equipment
  - 2.1.1 Automatic film processor versus manual processing
  - 2.1.2 Safelights
  - 2.1.3 Viewer lights
  - 2.1.4 Loading bench
  - 2.1.5 Miscellaneous equipment

2.2 Film loading
  2.2.1 General rules for handling unprocessed film
  2.2.2 Types of film packaging
  2.2.3 Cassette loading techniques for sheet and roll
2.3 Protection of radiography film in storage
2.4 Processing of film – manual
  2.4.1 Developer and replenishment
  2.4.2 Stop bath
  2.4.3 Fixer and replenishment
  2.4.4 Washing
  2.4.5 Prevention of water spots
  2.4.6 Drying
2.5 Automatic film processing
2.6 Film filing and storage
  2.6.1 Retention-life measurements
  2.6.2 Long-term storage
  2.6.3 Filing and separation techniques
2.7 Unsatisfactory radiographs – causes and cures
  2.7.1 High film density
  2.7.2 Insufficient film density
  2.7.3 High contrast
  2.7.4 Low contrast
  2.7.5 Poor definition
  2.7.6 Fog
  2.7.7 Light leaks
  2.7.8 Artifacts
2.8 Film density
  2.8.1 Step-wedge comparison film
  2.8.2 Densitometers

3.0 Indications, Discontinuities, and Defects
3.1 Indications
3.2 Discontinuities
  3.2.1 Inherent
  3.2.2 Processing
  3.2.3 Service
3.3 Defects

4.0 Manufacturing Processes and Associated Discontinuities
4.1 Casting processes and associated discontinuities
  4.1.1 Ingots, blooms, and billets
  4.1.2 Sand casting
  4.1.3 Centrifugal casting
  4.1.4 Investment casting
4.2 Wrought processes and associated discontinuities
  4.2.1 Forgings
  4.2.2 Rolled products
  4.2.3 Extruded products
4.3 Welding processes and associated discontinuities
  4.3.1 Submerged arc welding (SAW)
  4.3.2 Shielded metal arc welding (SMAW)
  4.3.3 Gas metal arc welding (GMAW)
  4.3.4 Flux cored arc welding (FCAW)
  4.3.5 Gas tungsten arc welding (GTAW)
  4.3.6 Resistance welding
  4.3.7 Special welding processes – electron-beam, electroslag, electrogas, etc.

5.0 Radiographic Safety Principles Review
5.1 Controlling personnel exposure
5.2 Time, distance, shielding concepts
5.3 ALARA concept
5.4 Radiation detection equipment
5.5 Exposure device operating characteristics

Radiographic Interpretation and Evaluation Course

1.0 Radiograph Viewing
1.1 Film-illuminator requirements
1.2 Background lighting
1.3 Multiple-composite viewing
1.4 IQI placement
1.5 Personnel dark adaptation and visual acuity
1.6 Film identification
1.7 Location markers
1.8 Film density measurement
1.9 Film artifacts

2.0 Application Techniques
2.1 Multiple-film techniques
  2.1.1 Thickness variation parameters
  2.1.2 Film speed
  2.1.3 Film latitude
2.2 Enlargement and projection
2.3 Geometrical relationships
  2.3.1 Geometrical unsharpness
  2.3.2 IQI sensitivity
  2.3.3 Source-to-film distance
  2.3.4 Focal spot size
2.4 Triangulation methods for discontinuity location
2.5 Localized magnification
2.6 Film handling techniques

3.0 Evaluation of Castings
3.1 Casting method review
3.2 Casting discontinuities
3.3 Origin and typical orientation of discontinuities
3.4 Radiographic appearance
3.5 Casting codes/standards – applicable acceptance criteria
3.6 Reference radiographs

4.0 Evaluation of Weldments
4.1 Welding method review
4.2 Welding discontinuities
4.3 Origin and typical orientation of discontinuities
4.4 Radiographic appearance
4.5 Welding codes/standards – applicable acceptance criteria
4.6 Reference radiographs or pictograms

5.0 Standards, Codes, and Procedures for Radiography
5.1 ASTM standards
5.2 Acceptable radiography techniques and setups
5.3 Applicable employer procedures
5.4 Procedure for radiograph parameter verification
5.5 Radiography reports

Computed Radiography Level I Topical Outline

Note: Independent of the training recommended for Level I and Level II certification, a trainee is required to receive radiation safety training as required by the regulatory jurisdiction. A Radiation Safety Topical Outline is available in Appendix A and can be used as guidance.

Basic Radiographic Testing Physics Course

1.0 Introduction
- 1.1 History and discovery of radioactive materials
- 1.2 Definition of industrial radiography
- 1.3 Radiation protection – why?
- 1.4 Basic math review – exponents, square root, etc.

2.0 Fundamental Properties of Matter
- 2.1 Elements and atoms
- 2.2 Molecules and compounds
- 2.3 Atomic particles – properties of protons, electrons, and neutrons
- 2.4 Atomic structure
- 2.5 Atomic number and weight
- 2.6 Isotope versus radioisotope

3.0 Radioactive Materials
- 3.1 Production
  - 3.1.1 Neutron activation
  - 3.1.2 Nuclear fission
- 3.2 Stable versus unstable (radioactive) atoms
- 3.3 Becquerel – the unit of activity
- 3.4 Half-life of radioactive materials
- 3.5 Plotting of radioactive decay
- 3.6 Specific activity – becquerels/gram

4.0 Types of Radiation
- 4.1 Particulate radiation – properties: alpha, beta, neutron
- 4.2 Electromagnetic radiation – X-ray, gamma ray
- 4.3 X-ray production
- 4.4 Gamma ray production
- 4.5 Gamma ray energy
- 4.6 Energy characteristics of common radioisotope sources
- 4.7 Energy characteristics of X-ray machines

5.0 Interaction of Radiation with Matter
- 5.1 Ionization
- 5.2 Radiation interaction with matter
  - 5.2.1 Photoelectric effect
  - 5.2.2 Compton scattering
  - 5.2.3 Pair production
- 5.3 Unit of radiation exposure – coulomb per kilogram (C/kg)
- 5.4 Emissivity of commonly used radiographic sources
- 5.5 Emissivity of X-ray exposure devices
- 5.6 Attenuation of electromagnetic radiation – shielding
- 5.7 HVL, TVL
- 5.8 Inverse square law

6.0 Exposure Devices and Radiation Sources
- 6.1 Radioisotope sources
  - 6.1.1 Sealed-source design and fabrication
  - 6.1.2 Gamma ray sources
  - 6.1.3 Beta and bremsstrahlung sources
  - 6.1.4 Neutron sources
- 6.2 Radioisotope exposure device characteristics
- 6.3 Electronic radiation sources – 500 keV and less; low energy
  - 6.3.1 Generator – high-voltage rectifiers
  - 6.3.2 X-ray tube design and fabrication
  - 6.3.3 X-ray control circuits
  - 6.3.4 Accelerating potential
  - 6.3.5 Target material and configuration
  - 6.3.6 Heat dissipation
  - 6.3.7 Duty cycle
  - 6.3.8 Beam filtration
- 6.4* Electronic radiation sources – medium- and high-energy
  - 6.4.1* Resonance transformer
  - 6.4.2* Van de Graaff accelerator
  - 6.4.3* Linear accelerator
  - 6.4.4* Betatron
  - 6.4.5* Coulomb per kilogram (C/kg) output
  - 6.4.6* Equipment design and fabrication
  - 6.4.7* Beam filtration

7.0 Radiographic Safety Principles Review
- 7.1 Controlling personnel exposure
- 7.2 Time, distance, shielding concepts
- 7.3 ALARA concept
- 7.4 Radiation detection equipment
- 7.5 Exposure device operating characteristics

* Topics may be deleted if the employer does not use these methods and techniques.

Computed Radiography Technique Course

1.0 Computed Radiography (CR) Overview
- 1.1 Photostimulable luminescence (PSL)
- 1.2 Comparison of radiography and CR
- 1.3 Digital images
  - 1.3.1 Bits
  - 1.3.2 Bytes
  - 1.3.3 Pixels/voxels
  - 1.3.4 Image file formats and compression
- 1.4 Advantages
- 1.5 Disadvantages
- 1.6 Examples

2.0 System Components
- 2.1 Imaging plates (IP)
- 2.2 IP readout devices
- 2.3 Monitors
- 2.4 Computers

3.0 Basic CR Techniques
- 3.1 Image acquisition
- 3.2 IQIs
- 3.3 Display of acquired images
- 3.4 Optimization of displayed image
- 3.5 Storage of acquired and optimized image

4.0 Digital Image Processing
- 4.1 Enhanced images
- 4.2 Signal-to-noise ratio (SNR)
- 4.3 Artifacts and anomalies

Computed Radiography Level II Topical Outline

Advanced Computed Radiography Course

1.0 CR Overview
- 1.1 Photostimulable luminescence (PSL)
- 1.2 Image acquisition
- 1.3 Image presentation
- 1.4 Artifacts

2.0 Image Display Characteristics
- 2.1 Image definition
- 2.2 Filtering techniques
- 2.3 SNR
- 2.4 Modulation transfer function (MTF)
- 2.5 Grayscale adjustments
- 2.6 IQIs

3.0 Image Viewing
- 3.1 Image monitor requirements
- 3.2 Background lighting
- 3.3 IQI placement
- 3.4 Personnel dark adaptation and visual acuity
- 3.5 Image identification
- 3.6 Location markers

4.0 Evaluation of CR Images
- 4.1 Pixel value
- 4.2 IQI
- 4.3 Artifact mitigation
- 4.4 System performance
- 4.5 Conformance to specifications
- 4.6 Image storage and transmission

5.0 Application Techniques
- 5.1 Multiple-view techniques
  - 5.1.1 Thickness variation parameters
- 5.2 Enlargement and projection
- 5.3 Geometric relationships
  - 5.3.1 Geometric unsharpness
  - 5.3.2 IQI sensitivity
  - 5.3.3 Source-to-image plate distance
  - 5.3.4 Focal spot size
- 5.4 Localized magnification
- 5.5 Plate handling techniques

6.0 Evaluation of Castings
- 6.1 Casting method review
- 6.2 Casting discontinuities
- 6.3 Origin and typical orientation of discontinuities
- 6.4 Radiographic appearance
- 6.5 Casting codes/standards – applicable acceptance criteria
- 6.6 Reference radiographs or images

7.0 Evaluation of Weldments
- 7.1 Welding method review
- 7.2 Welding discontinuities
- 7.3 Origin and typical orientation of discontinuities
- 7.5 Welding codes/standards – applicable acceptance criteria
- 7.6 Reference radiographs or images

8.0 Standards, Codes, and Procedures for Computed Radiography
- 8.1 ASTM/ASME standards
- 8.2 Acceptable computed radiography techniques and setups
- 8.3 Applicable employer procedures

9.0 Radiographic Safety Principles Review
- 9.1 Controlling personnel exposure
- 9.2 Time, distance, shielding concepts
- 9.3 ALARA concept
- 9.4 Radiation detection equipment
- 9.5 Exposure device operating characteristics

Computed Tomography Level I Topical Outline

Note: Independent of the training recommended for Level I and Level II certification, a trainee is required to receive radiation safety training as required by the regulatory jurisdiction. A Radiation Safety Topical Outline is available in Appendix A and can be used as guidance.

Basic Radiographic Physics Course

1.0 Introduction
- 1.1 History and discovery of radioactive materials
- 1.2 Definition of industrial radiography
- 1.3 Radiation protection – why?
- 1.4 Basic math review – exponents, square root, etc.

2.0 Fundamental Properties of Matter
- 2.1 Elements and atoms
- 2.2 Molecules and compounds
- 2.3 Atomic particles – properties of protons, electrons, and neutrons
- 2.4 Atomic structure
- 2.5 Atomic number and weight
- 2.6 Isotope versus radioisotope

3.0 Radioactive Materials
- 3.1 Production
  - 3.1.1 Neutron activation
  - 3.1.2 Nuclear fission
- 3.2 Stable versus unstable (radioactive) atoms
- 3.3 Becquerel – the unit of activity
- 3.4 Half-life of radioactive materials
- 3.5 Plotting of radioactive decay
- 3.6 Specific activity – becquerels/gram

4.0 Types of Radiation
- 4.1 Particulate radiation – properties: alpha, beta, neutron
- 4.2 Electromagnetic radiation – X-ray, gamma ray
- 4.3 X-ray production
- 4.4 Gamma ray production
- 4.5 Gamma ray energy
- 4.6 Energy characteristics of common radioisotope sources
- 4.7 Energy characteristics of X-ray machines

5.0 Interaction of Radiation with Matter
- 5.1 Ionization
- 5.2 Radiation interaction with matter
  - 5.2.1 Photoelectric effect
  - 5.2.2 Compton scattering
  - 5.2.3 Pair production
- 5.3 Unit of radiation exposure – coulomb per kilogram (C/kg)
- 5.4 Emissivity of commonly used radiographic sources
- 5.5 Emissivity of X-ray exposure devices
- 5.6 Attenuation of electromagnetic radiation – shielding
- 5.7 HVL, TVL
- 5.8 Inverse square law

6.0 Exposure Devices and Radiation Sources
- 6.1 Radioisotope sources
  - 6.1.1 Sealed-source design and fabrication
  - 6.1.2 Gamma ray sources
  - 6.1.3 Beta and bremsstrahlung sources
  - 6.1.4 Neutron sources
- 6.2 Radioisotope exposure device characteristics
- 6.3 Electronic radiation sources – 500 keV and less, low energy
  - 6.3.1 Generator – high-voltage rectifiers
  - 6.3.2 X-ray tube design and fabrication
  - 6.3.3 X-ray control circuits
  - 6.3.4 Accelerating potential
  - 6.3.5 Target material and configuration
  - 6.3.6 Heat dissipation
  - 6.3.7 Duty cycle
  - 6.3.8 Beam filtration
- 6.4* Electronic radiation sources – medium- and high-energy
  - 6.4.1* Resonance transformer
  - 6.4.2* Van de Graaff accelerator
  - 6.4.3* Linear accelerator
  - 6.4.4* Betatron
  - 6.4.5* Coulomb per kilogram (C/kg) output
  - 6.4.6* Equipment design and fabrication
  - 6.4.7* Beam filtration

7.0 Radiographic Safety Principles Review
- 7.1 Controlling personnel exposure
- 7.2 Time, distance, shielding concepts
- 7.3 ALARA concept
- 7.4 Radiation detection equipment
- 7.5 Exposure device operating characteristics

* Topics may be deleted if the employer does not use these methods and techniques.

Basic Computed Tomography Technique Course

1.0 Computed Tomography (CT) Overview
- 1.1 Difference between CT and conventional radiography
- 1.2 Benefits and advantages
- 1.3 Limitations
- 1.4 Industrial imaging examples

2.0 Basic Hardware Configuration
- 2.1 Scan geometries – general configurations by generation
- 2.2 Radiation sources
- 2.3 Detection systems
- 2.4 Manipulation/mechanical system
- 2.5 Computer system
- 2.6 Image reconstruction
- 2.7 Image display
- 2.8 Data storage
- 2.9 Operator interface

3.0 Fundamental CT Performance Parameters
- 3.1 Fundamental scan plan parameters
- 3.2 Basic system tradeoffs for spatial resolution/noise/slice thickness

4.0 Basic Image Interpretation and Processing
- 4.1 Artifacts – definitions, detection, and basic causes
- 4.2 CT density measurements

Computed Tomography Level II Topical Outline

Computed Tomography Technique Course

1.0 General Principles of CT and Terminology
- 1.1 CT technical background
- 1.2 Physical basis – X-ray interactions with material properties
- 1.3 Mathematical basis – line integrals
- 1.4 Data sampling principles
- 1.5 Physical limitations of the sampling process
- 1.6 Reconstruction algorithms
  - 1.6.1 Convolution/backprojections
  - 1.6.2 Fourier reconstructions
  - 1.6.3 Fan/cone beam

2.0 CT System Performance – Characterizing System Performance
- 2.1 CT system performance parameters overview
- 2.2 Spatial resolution
- 2.3 Contrast sensitivity
- 2.4 Artifacts
  - 2.4.1 Beam hardening, streak, under-sampling, etc.
- 2.5 Noise
- 2.6 Effective X-ray energy
- 2.7 System performance measurement techniques
- 2.8 Spatial resolution
- 2.9 Contrast sensitivity
  - 2.9.1 Standardizing CT density
  - 2.9.2 Measuring CT density
  - 2.9.3 Performance measurement intervals

3.0 Image Interpretation and Processing
- 3.1 Use of phantoms to monitor CT system performance
- 3.2 Evaluation of CT system performance parameters
- 3.3 Determination of artifacts
- 3.4 Artifact mitigation techniques

4.0 Advanced Image-Processing Algorithms
- 4.1 Modulation transfer function calculation
- 4.2 Effective energy calculation
- 4.3 Application of image-processing algorithms
- 4.4 Artifact mitigation techniques application

5.0 Radiographic Safety Principles Review
- 5.1 Controlling personnel exposure
- 5.2 Time, distance, shielding concepts
- 5.3 ALARA concept
- 5.4 Radiation detection equipment
- 5.5 Exposure device operating characteristics

Radiographic Interpretation and Evaluation Course

1.0 Evaluation of Castings
- 1.1 Casting method review
- 1.2 Casting discontinuities
- 1.3 Origin and typical orientation of discontinuities
- 1.4 Radiographic appearance
- 1.5 Casting codes/standards – applicable acceptance criteria

2.0 Evaluation of Weldments
- 2.1 Welding method review
- 2.2 Welding discontinuities
- 2.3 Origin and typical orientation of discontinuities
- 2.4 Welding codes/standards – applicable acceptance criteria

3.0 Standards, Codes, and Procedures for Radiography
- 3.1 ASTM standards
- 3.2 Acceptable computed tomography techniques and setups
- 3.3 Applicable employer procedures

Digital Radiography Level I Topical Outline

Note: Independent of the training recommended for Level I and Level II certification, a trainee is required to receive radiation safety training as required by the regulatory jurisdiction. A Radiation Safety Topical Outline is available in Appendix A and can be used as guidance.

Basic Radiographic Testing Physics Course

1.0 Introduction
- 1.1 History and discovery of radioactive materials
- 1.2 Definition of industrial radiography
- 1.3 Radiation protection – why?
- 1.4 Basic math review – exponents, square root, etc.

2.0 Fundamental Properties of Matter
- 2.1 Elements and atoms
- 2.2 Molecules and compounds
- 2.3 Atomic particles – properties of protons, electrons, and neutrons
- 2.4 Atomic structure
- 2.5 Atomic number and weight
- 2.6 Isotope versus radioisotope

3.0 Radioactive Materials
- 3.1 Production
  - 3.1.1 Neutron activation
  - 3.1.2 Nuclear fission
- 3.2 Stable versus unstable (radioactive) atoms
- 3.3 Becquerel – the unit of activity
- 3.4 Half-life of radioactive materials
- 3.5 Plotting of radioactive decay
- 3.6 Specific activity – becquerels/gram

4.0 Types of Radiation
- 4.1 Particulate radiation – properties: alpha, beta, neutron
- 4.2 Electromagnetic radiation – X-ray, gamma ray
- 4.3 X-ray production
- 4.4 Gamma ray production
- 4.5 Gamma ray energy
- 4.6 Energy characteristics of common radioisotope sources
- 4.7 Energy characteristics of X-ray machines

5.0 Interaction of Radiation with Matter
- 5.1 Ionization
- 5.2 Radiation interaction with matter
  - 5.2.1 Photoelectric effect
  - 5.2.2 Compton scattering
  - 5.2.3 Pair production
- 5.3 Unit of radiation exposure – coulomb per kilogram (C/kg)
- 5.4 Emissivity of commonly used radiographic sources
- 5.5 Emissivity of X-ray exposure devices
- 5.6 Attenuation of electromagnetic radiation – shielding
- 5.7 HVL, TVL
- 5.8 Inverse square law

6.0 Exposure Devices and Radiation Sources

- 6.1 Radioisotope sources
  - 6.1.1 Sealed-source design and fabrication
  - 6.1.2 Gamma ray sources
  - 6.1.3 Beta and bremsstrahlung sources
  - 6.1.4 Neutron sources
- 6.2 Radioisotope exposure device characteristics
- 6.3 Electronic radiation sources – 500 keV and less; low-energy
  - 6.3.1 Generator – high-voltage rectifiers
  - 6.3.2 X-ray tube design and fabrication
  - 6.3.3 X-ray control circuits
  - 6.3.4 Accelerating potential
  - 6.3.5 Target material and configuration
  - 6.3.6 Heat dissipation
  - 6.3.7 Duty cycle
  - 6.3.8 Beam filtration
- 6.4* Electronic radiation sources – medium- and high-energy
  - 6.4.1* Resonance transformer
  - 6.4.2* Van de Graaff accelerator
  - 6.4.3* Linear accelerator
  - 6.4.4* Betatron
  - 6.4.5* Coulomb per kilogram (C/kg) output
  - 6.4.6* Equipment design and fabrication
  - 6.4.7* Beam filtration

7.0 Radiographic Safety Principles Review

- 7.1 Controlling personnel exposure
- 7.2 Time, distance, shielding concepts
- 7.3 ALARA concept
- 7.4 Radiation detection equipment
- 7.5 Exposure device operating characteristics

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* Topics may be deleted if the employer does not use these methods and techniques.

Basic Digital Radiography (DR) Technique Course

1.0 DR Overview

- 1.1 DR
- 1.2 Digital images
  - 1.2.1 Bits/bytes
  - 1.2.2 Pixels/voxels
- 1.3 Image file formats and compression
- 1.4 DR system overview
- 1.5 DR system capabilities
  - 1.5.1 DR versus film procedural steps
  - 1.5.2 Cost and environmental issues

2.0 DR System Components

- 2.1 Detector(s) used in the radiography shop
  - 2.1.1 Operating procedures to use the equipment

3.0 Image Fidelity Indicators (System Characterization)

- 3.1 IQIs: hole and wire types
- 3.2 Line pair gauges
- 3.3 Phantoms
- 3.4 Reference quality indicators (RQIs)
- 3.5 TV test patterns

4.0 Detector Issues

- 4.1 Scatter sensitivity
- 4.2 Radiation exposure tolerance
- 4.3 Portability
- 4.4 Detector handling

5.0 Technique Sheets

Digital Radiography Level II Topical Outline

Digital Radiography Technique Course

1.0 Basic DR versus Film Principles

- 1.1 Film versus DR images
  - 1.1.1 Linearity and latitude
  - 1.1.2 Contrast and resolution

2.0 DR System Components

- 2.1 X-ray and gamma ray sources
  - 2.1.1 Energy, mA, focal spot
  - 2.1.2 Stability
  - 2.1.3 Open and closed X-ray tubes
  - 2.1.4 Filtration
- 2.2 Computer
  - 2.2.1 Operator interface
  - 2.2.2 System controller
  - 2.2.3 Image processor
- 2.3 Monitors
  - 2.3.1 CRT
  - 2.3.2 LCD
- 2.4 Data archive
  - 2.4.1 Removable media (CD, DVD, tape)
  - 2.4.2 Redundant array of inexpensive disks (RAID)
  - 2.4.3 Central archive

3.0 Image Fidelity

- 3.1 Measuring image fidelity
  - 3.1.1 Contrast and resolution
  - 3.1.2 SNR
- 3.2 Image fidelity indicators (system characterization)

4.0 Image Processing (Postprocessing)

- 4.1 Grayscale adjustments
  - 4.1.1 Windowing and leveling
  - 4.1.2 Lookup tables (LUTs)
  - 4.1.3 Thresholding
  - 4.1.4 Histogram equalization
  - 4.1.5 Pseudo color

4.2 Arithmetic
    4.2.1 Addition (integration)
    4.2.2 Subtraction
    4.2.3 Division
    4.2.4 Multiplication
    4.2.5 Averaging
4.3 Filtering (kernels)
    4.3.1 Convolution
    4.3.2 Low pass (smoothing)
    4.3.3 High pass (edge enhancement)
    4.3.4 Median
    4.3.5 Unsharp mask
4.4 Region of interest (ROI)

5.0 Detector Issues for the Detector(s) Used
5.1 Frame rate
5.2 Resolution (pixel pitch, pixel size, etc.)
5.3 Blooming, bleed over
5.4 Ghosting/latent image/lag
5.5 Scatter sensitivity
5.6 Bit depth
5.7 Dynamic range and SNR
5.8 Fabrication anomalies (bad pixels, chip grades, etc.)
5.9 Radiation exposure tolerance
5.10 Portability
5.11 Detector handling

6.0 Detector Calibrations for the Detector(s) Used
6.1 Gain and offset
6.2 Detector-specific calibration

7.0 Monitor and Viewing Environment
7.1 Limited bit-depth display
7.2 Monitor resolution
7.3 Monitor brightness and contrast
7.4 Monitor testing
    7.4.1 Test patterns
    7.4.2 Luminance – cd/m^2
    7.4.3 Contrast – min:max, digital driving level (DDL)
7.5 Monitor calibration
7.6 Viewing area

8.0 Technique Development Considerations
8.1 Image unsharpness and geometric magnification
    8.1.1 Determining required geometric magnification
    8.1.2 Geometry and geometric unsharpness
    8.1.3 Focal spot size measurement method
    8.1.4 Total image unsharpness
8.2 SNR compensation for spatial resolution
    8.2.1 Frame averaging
    8.2.2 Binning
    8.2.3 X-ray spectrum optimization
        8.2.3.1 Filtering
        8.2.3.2 Beam collimation
        8.2.3.3 Beam energy
8.3 Image processing
    8.3.1 Understanding of cost and benefits of common image-processing techniques – windowing, filtering, subtraction, etc.

9.0 Detector Monitoring
10.0 Detector Maintenance
11.0 Use of Digital Reference Images
11.1 ASTM standards review
11.2 Use of reference images and contrast normalization

12.0 Radiographic Safety Principles Review
12.1 Controlling personnel exposure
12.2 Time, distance, shielding concepts
12.3 ALARA concept
12.4 Radiation detection equipment
12.5 Exposure device operating characteristics

Interpretation and Evaluation Course

1.0 Image Viewing
1.1 Image display requirements
1.2 Background lighting
1.3 Multiple-composite viewing
1.4 IQI placement
1.5 Personnel dark adaptation and visual acuity
1.6 Image identification
1.7 Location markers

2.0 Application Techniques
2.1 Multiple-view techniques
    2.1.1 Thickness variation parameters
2.2 Enlargement and projection
2.3 Geometric relationships
    2.3.1 Geometric unsharpness
    2.3.2 IQI sensitivity
    2.3.3 Source-to-detector distance
    2.3.4 Focal spot size
2.4 Triangulation methods for discontinuity location
2.5 Localized magnification

3.0 Evaluation of Castings
3.1 Casting method review
3.2 Casting discontinuities
3.3 Origin and typical orientation of discontinuities
3.4 Casting codes/standards – applicable acceptance criteria
3.5 Reference radiographs or images

4.0 Evaluation of Weldments
4.1 Welding method review
4.2 Welding discontinuities
4.3 Origin and typical orientation of discontinuities
4.4 Welding codes/standards – applicable acceptance criteria
4.5 Reference radiographs or images

5.0 Standards, Codes, and Procedures for Radiography
5.1 ASTM standards
5.2 Acceptable techniques and setups
5.3 Applicable employer procedures

Radiographic Testing Level III Topical Outline

Basic Radiographic Topics

1.0 Principles/Theory
- 1.1 Nature of penetrating radiation
- 1.2 Interaction between penetrating radiation and matter
- 1.3 Radiology overview
  - 1.3.1 Film radiography
  - 1.3.2 CR
  - 1.3.3 CT
  - 1.3.4 DR
    - 1.3.4.1 Radioscopy

2.0 Equipment/Materials
- 2.1 Electrically generated sources
  - 2.1.1 X-ray sources
    - 2.1.1.1 Generators and tubes as an integrated system
    - 2.1.1.2 Sources of electrons
    - 2.1.1.3 Electron accelerating methods
    - 2.1.1.4 Target materials and characteristics
    - 2.1.1.5 Equipment design considerations
    - 2.1.1.6 Microfocus sources
- 2.2 Isotope sources
  - 2.2.1 Exposure devices
  - 2.2.2 Source changers
  - 2.2.3 Remote handling equipment
  - 2.2.4 Collimators
  - 2.2.5 Specific characteristics
    - 2.2.5.1 Half-lives
    - 2.2.5.2 Energy levels
    - 2.2.5.3 HVL
    - 2.2.5.4 TVL
- 2.3 Radiation detection overview
  - 2.3.1 Direct imaging
    - 2.3.1.1 Film overview
    - 2.3.1.2 Radioscopy overview
    - 2.3.1.3 X-ray image intensifier system
  - 2.3.2 Digital data acquisition/detectors
    - 2.3.2.1 Film digitizers
    - 2.3.2.2 CR
    - 2.3.2.3 CT
    - 2.3.2.4 DR
- 2.4 Manipulators
  - 2.4.1 Manual versus automated
  - 2.4.2 Multiple axis
  - 2.4.3 Weight capacity
  - 2.4.4 Precision
- 2.5 Visual perception
  - 2.5.1 Spatial frequency
  - 2.5.2 Contrast
  - 2.5.3 Displayed brightness
  - 2.5.4 SNR
  - 2.5.5 Probability of detection (single versus multiple locations, scanning)
  - 2.5.6 Receiver operator characteristic curves

3.0 Safety and Health
- 3.1 Exposure hazards
  - 3.1.1 Occupational dose limits
- 3.2 Methods of controlling radiation exposure
  - 3.2.1 Time
  - 3.2.2 Distance
    - 3.2.2.1 Inverse square law
  - 3.2.3 Shielding
    - 3.2.3.1 HVL
    - 3.2.3.2 TVL
- 3.3 Operational and emergency procedures
- 3.4 Dosimetry and film badges
- 3.5 Gamma leak testing
- 3.6 Transportation regulations

Radiographic Testing

1.0 Techniques/Standardization
- 1.1 Imaging considerations
  - 1.1.1 Sensitivity
  - 1.1.2 Contrast and definition
  - 1.1.3 Geometric factors
  - 1.1.4 Intensifying screens
  - 1.1.5 Scattered radiation
  - 1.1.6 Source factors
  - 1.1.7 Detection media
  - 1.1.8 Exposure curves
- 1.2 Film processing
  - 1.2.1 Darkroom procedures
  - 1.2.2 Darkroom equipment and chemicals
  - 1.2.3 Film processing
- 1.3 Viewing of radiographs
  - 1.3.1 Illuminator requirements
  - 1.3.2 Background lighting
  - 1.3.3 Optical aids
- 1.4 Judging radiographic quality
  - 1.4.1 Density
  - 1.4.2 Contrast
  - 1.4.3 Definition
  - 1.4.4 Artifacts
  - 1.4.5 IQIs
  - 1.4.6 Causes and correction of unsatisfactory radiographs
- 1.5 Exposure calculations
- 1.6 Radiographic techniques
  - 1.6.1 Blocking and filtering
  - 1.6.2 Multifilm techniques
  - 1.6.3 Enlargement and projection
  - 1.6.4 Stereoradiography
  - 1.6.5 Triangulation methods
  - 1.6.6 Autoradiography
  - 1.6.7 Flash radiography
  - 1.6.8 In-motion radiography
  - 1.6.9 Control of diffraction effects

1.6.10 Pipe welding exposures
- 1.6.10.1 Contact
- 1.6.10.2 Elliptical
- 1.6.10.3 Panoramic

1.6.11 Gauging
1.6.12 Real-time imaging
1.6.13 Image analysis techniques
1.6.14 Image-object relationship

2.0 Interpretation/Evaluation

2.1 Material considerations
- 2.1.1 Materials processing as it affects use of item and test results
- 2.1.2 Discontinuities, their causes and effects
- 2.1.3 Radiographic appearance of discontinuities
- 2.1.4 Nonrelevent indications
- 2.1.5 Film artifacts
- 2.1.6 Code considerations

3.0 Procedures

Common Digital System Elements and Digital Image Properties

1.0 Digital Image Properties

1.1 Bits/bytes
1.2 Pixels/voxels
1.3 Image file formats and compression (JPEG, TIFF, DICONDE)
- 1.3.1 Advantages/disadvantages
- 1.3.2 Lossy versus lossless

1.4 Sampling theory (digitizing)
- 1.4.1 Pixel size (aperture)
- 1.4.2 Pixel pitch
- 1.4.3 Bit depth
- 1.4.4 Nyquist theory

2.0 Digital System Specific: Components

2.1 Computer
- 2.1.1 Operator interface
- 2.1.2 System controller
- 2.1.3 Image processor

2.2 Monitor and viewing environment
- 2.2.1 Type of monitors/displays
- 2.2.2 Limited bit-depth display
- 2.2.3 Monitor resolution
- 2.2.4 Monitor brightness and contrast
- 2.2.5 Monitor testing
- 2.2.6 Monitor calibration
- 2.2.7 Viewing area and ergonomics

2.3 Data archive
- 2.3.1 Removable media – single media (CD, DVD, tape)
- 2.3.2 Redundant array of inexpensive disks (RAID)
- 2.3.3 Central archive
- 2.3.4 Image retrieval

3.0 Digital System Specific: Image Processing Topics

3.1 ROI and measurements
- 3.1.1 Line profiles
- 3.1.2 Histograms (mean/standard deviations)
- 3.1.3 Discontinuity sizing
  - 3.1.3.1 Length
  - 3.1.3.2 Area
  - 3.1.3.3 Wall thickness
  - 3.1.3.4 Blob/cluster analysis

3.2 Grayscale display adjustments
- 3.2.1 Window width and level
- 3.2.2 LUTs
- 3.2.3 Thresholding
- 3.2.4 Histogram equalization
- 3.2.5 Pseudo color

3.3 Filtering (kernels)
- 3.3.1 Convolution
- 3.3.2 Low pass
- 3.3.3 High pass
- 3.3.4 Median
- 3.3.5 Unsharp mask

4.0 Acquisition System Considerations

4.1 Portability
4.2 Access requirements for detectors
4.3 High-energy applications

Computed Radiography (CR)

1.0 CR System Capabilities

1.1 CR system overview
1.2 CR versus film procedural steps
1.3 Cost and environmental issues
1.4 Film versus CR images
1.5 Linearity and latitude
1.6 Contrast and resolution

2.0 Measuring Image Fidelity

2.1 Contrast and resolution
2.2 MTF
2.3 SNR

3.0 Image Fidelity Indicators (System Characterization)

3.1 IQIs: hole and wire types
3.2 Line pair gauges
3.3 Phantoms
3.4 RQIs
3.5 TV test patterns

4.0 CR Technical Requirements

4.1 Qualification of CR systems
4.2 Classification of CR systems
4.3 Maintenance of CR systems
4.4 Technical requirements for inspection

5.0 CR Technical Development
- 5.1 Hardware development
  - 5.1.1 Hard/soft cassette usage
  - 5.1.2 Image plate wear and damage
  - 5.1.3 Image plate artifacts
- 5.2 Software development
- 5.3 CR image optimization
  - 5.3.1 Laser spot size optimization
  - 5.3.2 Use of lead screens

6.0 Use of Digital Reference Images
- 6.1 ASTM standards review
- 6.2 Digital reference images installation
  - 6.2.1 Include reference image resolutions/pixel size
- 6.3 Use of reference images and contrast normalization

7.0 Review of DR Industry Standards (e.g., ASTM)

Computed Tomography (CT)

1.0 Practice of CT
- 1.1 Capabilities of CT
- 1.2 Cost-effective application areas
- 1.3 Digital laminographic and fan beam CT methods

2.0 Principals of CT
- 2.1 Historical background
- 2.2 Principals of CT operation
- 2.3 Tomographic reconstruction/filtered backprojection
- 2.3 Hounsfield number
- 2.4 Resolution and contrast in CT

3.0 CT Systems
- 3.1 CT system configurations
  - 3.1.1 First generation
  - 3.1.2 Second generation
  - 3.1.3 Third generation
  - 3.1.4 Fourth generation
  - 3.1.5 Cone beam CT
- 3.2 CT system elements
- 3.3 CT system attributes and ramifications

4.0 Image Quality Measurement and System Characterization
- 4.1 Resolution and MTF
  - 4.1.1 Line pair gauges
- 4.2 Contrast sensitivity and contrast discrimination curves
- 4.3 Material density phantoms
- 4.4 Geometrical evaluation phantoms
- 4.5 Artifacts

5.0 CT Visualization, Advanced CT Analysis and Tools
- 5.1 Single slice rendering
- 5.2 3D/volume rendering
- 5.3 Rolled view
- 5.4 Porosity inclusion analysis
- 5.5 Coordinate measurement
- 5.6 Nominal actual comparison
- 5.7 Fiber composite material analysis
- 5.8 Foam analysis

6.0 Qualification of CT Procedures
- 6.1 Qualification plan
- 6.2 System performance characterization
  - 6.2.1 Process controls
- 6.3 Technique documentation
- 6.4 Technique validation

Digital Radiography (DR)

1.0 DR System Capabilities
- 1.1 DR system overview
- 1.2 DR versus film procedural steps
- 1.3 Cost and environmental issues
- 1.4 Film versus DR images
- 1.5 Linearity and latitude
- 1.6 Contrast and resolution

2.0 Measuring Image Fidelity
- 2.1 Contrast and resolution
- 2.2 MTF
- 2.3 SNR

3.0 Image Fidelity Indicators (System Characterization)
- 3.1 IQIs: hole and wire types
- 3.2 Line pair gauges
- 3.3 Phantoms
- 3.4 RQIs
- 3.5 TV test patterns

4.0 Detector Selection
- 4.1 ASTM E 2597 data interpretation
  - 4.1.1 Frame rate, resolution, ghosting/lag, bit depth
  - 4.1.2 Basic spatial resolution
  - 4.1.3 Bad pixel characterization
  - 4.1.4 Contrast sensitivity
  - 4.1.5 Efficiency
  - 4.1.6 Specific material thickness
  - 4.1.7 MTF
  - 4.1.8 SNR
- 4.2 Additional detector selection criteria/parameters
  - 4.2.1 Frame rate
  - 4.2.2 Blooming
  - 4.2.3 Ghosting/latent image/lag
  - 4.2.4 Scatter sensitivity
  - 4.2.5 Bit depth
  - 4.2.6 Fabrication anomalies (e.g., bad pixels, chip grades, etc.)
  - 4.2.7 Radiation exposure tolerance

5.0 DR Image Quality Topics
- 5.1 Standardization optimization
- 5.2 Setting bad pixel limits versus application
- 5.3 Image unsharpness and geometric magnification
  - 5.3.1 Determining required geometric magnification
  - 5.3.2 Geometry and geometric unsharpness
  - 5.3.3 Focal spot size measurement method
  - 5.3.4 Total image unsharpness
- 5.4 SNR compensation for spatial resolution
  - 5.4.1 Frame averaging
  - 5.4.2 Binning
  - 5.4.3 X-ray spectrum optimization
    - 5.4.3.1 Filtering
    - 5.4.3.2 Beam collimation
    - 5.4.3.3 Beam energy
- 5.5 Radiation damage management

6.0 Qualification of DR Procedures
- 6.1 Qualification plan
- 6.2 System performance characterization
  - 6.2.1 Process controls
- 6.3 Technique documentation
- 6.4 Technique validation

7.0 Use of Digital Reference Images
- 7.1 ASTM standards review
- 7.2 Digital reference images installation
  - 7.2.1 Include reference image resolutions/pixel size
- 7.3 Use of reference images and contrast normalization

Limited Certification for Radiographic Testing Interpretation Topical Outlines

Note: Independent of the training recommended for Level I and Level II certification, a trainee is required to receive radiation safety training as required by the regulatory jurisdiction. A Radiation Safety Topical Outline is available in Appendix A and can be used as guidance.

Film Interpretation Technique Course

1.0 Introduction
- 1.1 Process of radiography
- 1.2 Types of electromagnetic radiation sources
- 1.3 Electromagnetic spectrum
- 1.4 Penetrating ability or "quality" of X-rays and gamma rays
- 1.5 X-ray tube – change of mA or kVp effect on "quality" and intensity

2.0 Basic Principles of Radiography
- 2.1 Geometric exposure principles
  - 2.1.1 "Shadow" formation and distortion
  - 2.1.2 Shadow enlargement calculation
  - 2.1.3 Shadow sharpness
  - 2.1.4 Geometric unsharpness
  - 2.1.5 Finding discontinuity depth
- 2.2 Radiography screens
  - 2.2.1 Lead intensifying screens
  - 2.2.2 Fluorescent intensifying screens
  - 2.2.3 Intensifying factors
  - 2.2.4 Importance of screen-to-film contact
  - 2.2.5 Importance of screen cleanliness and care
- 2.3 Radiography cassettes
- 2.4 Composition of industrial radiography film

3.0 Radiographs
- 3.1 Formation of the latent image on film
- 3.2 Inherent unsharpness
- 3.3 Arithmetic of radiography exposure
  - 3.3.1 Milliamperage – distance-time relationship
  - 3.3.2 Reciprocity law
  - 3.3.3 Photographic density
  - 3.3.4 Inverse square law considerations
- 3.4 Characteristic (Hurter and Driffield) curve
- 3.5 Film speed and class descriptions
- 3.6 Selection of film for particular purpose

4.0 Radiographic Image Quality
- 4.1 Radiographic sensitivity
- 4.2 Radiographic contrast
- 4.3 Film contrast
- 4.4 Subject contrast
- 4.5 Definition
- 4.6 Film graininess and screen mottle effects
- 4.7 IQIs

5.0 Exposure Techniques – Radiography
- 5.1 Single-wall radiography
- 5.2 Double-wall radiography
  - 5.2.1 Viewing two walls simultaneously
  - 5.2.2 Offset double-wall exposure single-wall viewing
  - 5.2.3 Elliptical techniques
- 5.3 Panoramic radiography
- 5.4 Use of multiple-film loading
- 5.5 Specimen configuration

Film Quality and Manufacturing Processes Course

1.0 Darkroom Facilities, Techniques, and Processing
- 1.1 Facilities and equipment
  - 1.1.1 Automatic film processor versus manual processing
- 1.2 Protection of radiography film in storage
- 1.3 Processing of film – manual
  - 1.3.1 Developer and replenishment
  - 1.3.2 Stop bath
  - 1.3.3 Fixer and replenishment
  - 1.3.4 Washing
  - 1.3.5 Prevention of water spots
  - 1.3.6 Drying

- 1.4 Automatic film processing
- 1.5 Film filing and storage
  - 1.5.1 Retention-life measurements
  - 1.5.2 Long-term storage
  - 1.5.3 Filing and separation techniques
- 1.6 Unsatisfactory radiographs – causes and cures
  - 1.6.1 High film density
  - 1.6.2 Insufficient film density
  - 1.6.3 High contrast
  - 1.6.4 Low contrast
  - 1.6.5 Poor definition
  - 1.6.6 Fog
  - 1.6.7 Light leaks
  - 1.6.8 Artifacts
- 1.7 Film density
  - 1.7.1 Step-wedge comparison film
  - 1.7.2 Densitometers

2.0 Indications, Discontinuities, and Defects
- 2.1 Indications
- 2.2 Discontinuities
  - 2.2.1 Inherent
  - 2.2.2 Processing
  - 2.2.3 Service
- 2.3 Defects

3.0 Manufacturing Processes and Associated Discontinuities
- 3.1 Casting processes and associated discontinuities
  - 3.1.1 Ingots, blooms, and billets
  - 3.1.2 Sand casting
  - 3.1.3 Centrifugal casting
  - 3.1.4 Investment casing
- 3.2 Wrought processes and associated discontinuities
  - 3.2.1 Forgings
  - 3.2.2 Rolled products
  - 3.2.3 Extruded products
- 3.3 Welding processes and associated discontinuities
  - 3.3.1 Submerged arc welding (SAW)
  - 3.3.2 Shielded metal arc welding (SMAW)
  - 3.3.3 Gas metal arc welding (GMAW)
  - 3.3.4 Flux cored arc welding (FCAW)
  - 3.3.5 Gas tungsten arc welding (GTAW)

Radiography Interpretation and Evaluation Course

1.0 Radiograph Viewing
- 1.1 Film-illuminator requirements
- 1.2 Background lighting
- 1.3 Multiple-composite viewing
- 1.4 IQI placement
- 1.5 Personnel dark adaptation and visual acuity
- 1.6 Film identification
- 1.7 Location markers
- 1.8 Film density measurement
- 1.9 Film artifacts

2.0 Application Techniques
- 2.1 Multiple-film techniques
  - 2.1.1 Thickness variation parameters
  - 2.1.2 Film speed
  - 2.1.3 Film latitude
- 2.2 Enlargement and projection
- 2.3 Geometric relationships
  - 2.3.1 Geometric unsharpness
  - 2.3.2 IQI sensitivity
  - 2.3.3 Source-to-film distance
  - 2.3.4 Focal spot size
- 2.4 Triangulation methods for discontinuity location
- 2.5 Localized magnification
- 2.6 Film handling techniques

3.0 Evaluation of Castings
- 3.1 Casting method review
- 3.2 Casting discontinuities
- 3.3 Origin and typical orientation of discontinuities
- 3.4 Radiographic appearance
- 3.5 Casting codes/standards – applicable acceptance criteria
- 3.6 Reference radiographs

4.0 Evaluation of Weldments
- 4.1 Welding method review
- 4.2 Welding discontinuities
- 4.3 Origin and typical orientation of discontinuities
- 4.4 Radiographic appearance
- 4.5 Welding codes/standards – applicable acceptance criteria
- 4.6 Reference radiographs or pictograms

5.0 Standards, Codes, and Procedures for Radiography
- 5.1 Acceptable radiography techniques and setups
- 5.2 Applicable employer procedures
- 5.3 Procedure for radiograph parameter verification
- 5.4 Radiography reports

Limited Certification for Digital Radiography and Computed Radiography Interpretation

Digital Radiography and Computed Radiography Technique Course

1.0 Introduction
- 1.1 Process of radiography testing
- 1.2 Types of electromagnetic radiation sources
- 1.3 Electromagnetic spectrum
- 1.4 Penetrating ability or "quality" of X-rays and gamma rays
- 1.5 X-ray tube – change of mA or kVp effect on "quality" and intensity

2.0 Basic Principles of Digital Radiography (DR)
- 2.1 Geometric exposure principles
  - 2.1.1 "Shadow" formation and distortion
  - 2.1.2 Shadow enlargement calculation
  - 2.1.3 Shadow sharpness
  - 2.1.4 Geometric unsharpness
  - 2.1.5 Finding discontinuity depth
- 2.2 Digital detector array types and operating principles
- 2.3 Computed radiography (CR) systems and image plate types

3.0 Digital Radiographs
- 3.1 Inherent unsharpness
- 3.2 Arithmetic of radiography exposure
  - 3.2.1 Milliamperage – distance-time relationship
  - 3.2.2 Reciprocity law
  - 3.2.3 Detector gray value
  - 3.2.4 Inverse square law considerations

4.0 Digital Radiography Image Quality and Indicators
- 4.1 Contrast and resolution
- 4.2 MTF
- 4.3 SNR
- 4.4 IQIs – hole and wire types
- 4.5 Line pair gauges
- 4.6 Phantoms
- 4.7 RQIs
- 4.8 Monitor test patterns

5.0 Exposure Techniques – Digital Radiography
- 5.1 Single-wall technique
- 5.2 Double-wall technique
  - 5.2.1 Viewing two walls simultaneously
  - 5.2.2 Offset double-wall exposure single-wall viewing
  - 5.2.3 Elliptical techniques
- 5.3 Panoramic technique
- 5.4 Specimen configuration

Digital Radiography and Computed Radiography Interpretation and Evaluation Course

1.0 Use of Digital Reference Images
- 1.1 ASTM standards review
- 1.2 Digital reference images installation
  - 1.2.1 Include reference image resolutions/pixel size
- 1.3 Use of reference images and contrast normalization
- 1.4 Background lighting
- 1.5 IQI placement
- 1.6 Personnel dark adaptation and visual acuity
- 1.7 Image identification
- 1.8 Location markers

2.0 Application Techniques
- 2.1 Enlargement and projection
- 2.2 Geometric relationships
  - 2.2.1 Geometric unsharpness
  - 2.2.2 IQI sensitivity
  - 2.2.3 Source-to-detector/image plate distance
  - 2.2.4 Focal spot size
- 2.3 Triangulation methods for discontinuity location
- 2.4 Localized magnification
- 2.5 Detector and image plate handling techniques

3.0 Evaluation of Castings
- 3.1 Casting method review
- 3.2 Casting discontinuities
- 3.3 Origin and typical orientation of discontinuities
- 3.4 DR appearance
- 3.5 Casting codes/standards – applicable acceptance criteria
- 3.6 Digital reference radiographs

4.0 Evaluation of Weldments
- 4.1 Welding method review
- 4.2 Welding discontinuities
- 4.3 Origin and typical orientation of discontinuities
- 4.4 DR appearance
- 4.5 Welding codes/standards – applicable acceptance criteria
- 4.6 Digital reference radiographs or pictograms

5.0 Standards, Codes, and Procedures for Radiography
- 5.1 Acceptable DR techniques and setups
- 5.2 Applicable employer procedures
- 5.3 Procedure for radiograph parameter verification
- 5.4 DR reports

Limited Certification for Computed Tomography Interpretation

Computed Tomography Technique Course

1.0 Introduction
- 1.1 Types of electromagnetic radiation sources
- 1.3 Electromagnetic spectrum
- 1.4 Penetrating ability or "quality" of X-rays and gamma rays
- 1.5 X-ray tube – change of mA or kVp effect on "quality" and intensity

2.0 Basic Principles of Computed Tomography
- 2.1 Principals of computed tomography (CT) operation
- 2.2 Tomographic reconstruction/filtered backprojection
- 2.3 Hounsfield number
- 2.4 Resolution and contrast in CT

3.0 CT Systems
- 3.1 CT system configurations
  - 3.1.1 Cone beam CT
- 3.2 CT system elements
- 3.3 CT system attributes and ramifications

4.0 Image Quality Measurement and System Characterization
- 4.1 Resolution and MTF
  - 4.1.1 Line pair gauges
- 4.2 Contrast sensitivity and contrast discrimination curves
- 4.3 Material density phantoms
- 4.4 Geometrical evaluation phantoms
- 4.5 Artifacts

Computed Tomography Interpretation and Evaluation Course

1.0 Image Interpretation and Processing
- 1.1 Use of phantoms to monitor CT system performance
- 1.2 Evaluation of CT system performance parameters
- 1.3 Determination of artifacts
- 1.4 Artifact mitigation techniques

2.0 CT Visualization, Advanced CT Analysis and Tools
- 2.1 Single slice (2D) rendering
- 2.2 3D/volume rendering
- 2.3 Rolled view
- 2.4 Porosity inclusion analysis
- 2.5 Coordinate measurement
- 2.6 Nominal actual comparison
- 2.7 Fiber composite material analysis

3.0 Evaluation of Castings
- 3.1 Casting method review
- 3.2 Casting discontinuities
- 3.3 Origin and typical orientation of discontinuities
- 3.4 CT image appearance and artifacts
- 3.5 Casting codes/standards – applicable acceptance criteria

4.0 Evaluation of Weldments
- 4.1 Welding method review
- 4.2 Welding discontinuities
- 4.3 Origin and typical orientation of discontinuities
- 4.4 CT image appearance and artifacts
- 4.5 Welding codes/standards – applicable acceptance criteria
- 4.6 Digital reference radiographs or pictograms

RADIOGRAPHIC TESTING LEVEL I, II, AND III TRAINING REFERENCES

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Note: Technical papers on much of the subject material can be found in the journal of ASNT, *Materials Evaluation*. For specific topics, search the archive of *Materials Evaluation*, on the ASNT website (source.asnt.org).

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* Available from The American Society for Nondestructive Testing Inc., Columbus, OH.

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THERMAL/INFRARED TESTING TOPICAL OUTLINES

Thermal/Infrared Testing Level I Topical Outline

Basic Thermal/Infrared Physics Course

1.0 The Nature of Heat – What Is It and How Is It Measured/Expressed?
- 1.1 Instrumentation
- 1.2 Scales and conversions

2.0 Temperature – What Is It and How Is It Measured/Expressed?
- 2.1 Instrumentation
- 2.2 Scales and conversions

3.0 Heat Transfer Modes Familiarization
- 3.1 Heat conduction fundamentals
  - 3.1.1 Fourier's law of heat conduction (concept)
  - 3.1.2 Conductivity/resistance basics
- 3.2 Heat convection fundamentals
  - 3.2.1 Newton's law of cooling (concept)
  - 3.2.2 Film coefficient/film resistance basics
- 3.3 Heat radiation fundamentals
  - 3.3.1 Stefan-Boltzmann law (concept)
  - 3.3.2 Emissivity/absorptivity/reflectivity/transmissivity basics (Kirchhoff's law)

4.0 Radiosity Concepts Familiarization
- 4.1 Reflectivity
- 4.2 Transmissivity
- 4.3 Absorptivity
- 4.4 Emissivity
- 4.5 Infrared radiometry and imaging
- 4.6 Spatial resolution concepts
  - 4.6.1 Field of view (FOV)
  - 4.6.2 Instantaneous field of view (IFOV)
  - 4.6.3 Spatial resolution for temperature measurement – the split response function (SRF)
  - 4.6.4 Measurement instantaneous field of view (MIFOV)
- 4.7 Error potential in radiant measurements (an overview)

Basic Thermal/Infrared Operating Course

1.0 Introduction
- 1.1 Thermography defined
- 1.2 How infrared imagers work
- 1.3 Differences among imagers and alternative equipment
- 1.4 Operation of infrared thermal imager
  - 1.4.1 Selecting the best perspective
  - 1.4.2 Image area and lens selection for required details
  - 1.4.3 Optimizing the image
  - 1.4.4 Basic temperature measurement
  - 1.4.5 Basic emissivity measurement
- 1.5 Operation of support equipment for infrared surveys

2.0 Checking Equipment Calibration with Blackbody References

3.0 Infrared Image and Documentation Quality
- 3.1 Elements of a good infrared image
  - 3.1.1 Clarity (focus)
  - 3.1.2 Dynamic range of the image
  - 3.1.3 Recognizing and dealing with reflections
  - 3.1.4 Recognizing and dealing with spurious convection
- 3.2 Recording thermographic and corresponding visual data

4.0 Support Data Collection
- 4.1 Environmental data
- 4.2 Emissivity
  - 4.2.1 Measurement
  - 4.2.2 Estimation
  - 4.2.3 Surface modification
- 4.3 Surface reference temperatures
- 4.4 Identification and other

Basic Thermal/Infrared Applications Course

1.0 Detecting Thermal Anomalies Resulting from Differences in Thermal Resistance (Quasi-Steady-state Heat Flow)
- 1.1 Large surface-to-ambient temperature difference
- 1.2 Small surface-to-ambient temperature difference

2.0 Detecting Thermal Anomalies Resulting from Differences in Thermal Capacitance, Using System or Environmental Heat Cycles

3.0 Detecting Thermal Anomalies Resulting from Differences in Physical State

4.0 Detecting Thermal Anomalies Resulting from Fluid Flow Problems

5.0 Detecting Thermal Anomalies Resulting from Friction

6.0 Detecting Thermal Anomalies Resulting from Nonhomogeneous Exothermic or Endothermic Conditions

7.0 Field Quantification of Point Temperatures
- 7.1 Simple techniques for emissivity
- 7.2 Typical (high-emissivity) applications
- 7.3 Special problem of low-emissivity applications

Thermal/Infrared Testing Level II Topical Outline

Intermediate Thermal/Infrared Physics Course

1.0 Basic Calculations in the Three Modes of Heat Transfer
- 1.1 Conduction – principles and elementary calculation
  - 1.1.1 Thermal resistance – principles and elementary calculations
  - 1.1.2 Heat capacitance – principles and elementary calculations
- 1.2 Convection – principles and elementary calculations
- 1.3 Radiation – principles and elementary calculations

2.0 The Infrared Spectrum
- 2.1 Planck's law/curves
  - 2.1.1 Typical detected bands
  - 2.1.2 Spectral emissivities of real surfaces
  - 2.1.3 Effects due to semitransparent windows and/or gases
  - 2.1.4 Filters

3.0 Radiosity Problems
- 3.1 Blackbodies – theory and concepts
- 3.2 Emissivity problems
  - 3.2.1 Blackbody emissivity
  - 3.2.2 The graybody and the non-graybody
  - 3.2.3 Broadband and narrow-band emitter targets
  - 3.2.4 Specular and diffuse emitters
  - 3.2.5 Lambertian and non-Lambertian emitters (the angular sensitivity of emissivity)
  - 3.2.6 Effects of emissivity errors
- 3.3 Calculation of emissivity, reflectivity, and transmissivity (practical use of Kirchoff's law)
- 3.4 Reflectivity problem
  - 3.4.1 Quantifying effects of unavoidable reflections
  - 3.4.2 Theoretical corrections
- 3.5 Transmissivity problem
  - 3.5.1 Quantified effects of partial transmittance
  - 3.5.2 Theoretical corrections

4.0 Resolution Tests and Calculations
- 4.1 IFOV, FOV, and MIFOV measurements and calculations
- 4.2 MRTD measurements and calculations
- 4.3 Slit response function – measurement, calculations, interpretations, and comparisons
- 4.4 Resolution versus lens and distance
- 4.5 Dynamic range
- 4.6 Data acquisition rate/data density
- 4.7 Frame rate and field rate
- 4.8 Image data density
  - 4.8.1 Lines of resolution
  - 4.8.2 IFOVs/line
  - 4.8.3 Computer pixels/line

Intermediate Thermal/Infrared Operating Course

1.0 Operating for Infrared Measurements (Quantification)
- 1.1 Simple infrared energy measurement
- 1.2 Quantifying the emissivity of the target surface
- 1.3 Quantifying temperature profiles
  - 1.3.1 Use of blackbody temperature references in the image
  - 1.3.2 Use of temperature measurement devices for reference surface temperatures
  - 1.3.3 Common sources of temperature measurement errors
- 1.4 Computer processing to enhance imager data

2.0 Operating for High-Speed Data Collection
- 2.1 Producing accurate images of transient processes
- 2.2 Recording accurate images of transient processes
- 2.3 Equipment selection and operation for imaging from moving vehicles

3.0 Operating Special Equipment for "Active" Techniques
- 3.1 Hot or cold fluid energy sources
- 3.2 Heat lamp energy sources
- 3.3 Flash lamp energy sources
- 3.4 Electromagnetic induction
- 3.5 Laser energy sources

4.0 Reports and Documentation
- 4.1 Standardization requirements and records
- 4.2 Report data requirements
- 4.3 Preparing reports

Intermediate Thermal/Infrared Applications Course

1.0 Temperature Measurement Applications
- 1.1 Isotherms/alarm levels – personnel safety audits, etc.
- 1.2 Profiles

2.0 Energy Loss Analysis Applications

2.1 Conduction losses through envelopes

2.1.1 Basic envelope heat-flow quantification

2.1.2 Recognizing and dealing with wind effects

2.2 Mass-transfer heat exchange (air or other flows into or out of the system)

2.2.1 Location

2.2.2 Quantification

3.0 "Active" Applications

3.1 Insulation flaws

3.2 Delamination of composites

3.3 Bond quality of coatings

3.4 Location of high heat-capacity components

4.0 Filtered Applications

4.1 Sunlight

4.2 Furnace interiors

4.3 Semitransparent targets

5.0 Transient Applications

5.1 Imaging a rapidly moving process

5.2 Imaging from a vehicle

Thermal/Infrared Level II Building Diagnostics Physics Course

1.0 Basic Calculations in the Three Modes of Heat Transfer

1.1 Conduction – principles and elementary calculation

1.1.1 Thermal resistance – principles and elementary calculations

1.1.2 Heat capacitance – principles and elementary calculations

1.2 Convection – principles and elementary calculations

1.3 Radiation – principles and elementary calculations

1.4 Building envelope pressurization/depressurization effects on measurement of heat transfer

2.0 The Infrared Spectrum

2.1 Planck's law/curves

2.1.1 Typical detection bands

2.1.2 Spectral emissivities of real surfaces

2.1.3 Effects due to semitransparent windows and/or materials

3.0 Radiosity – Basic Theory and Building Applications

3.1 Blackbodies – theory and concepts

3.2 Emissivity problems

3.2.1 Blackbody, graybody, and non-graybody

3.2.2 Specular and diffuse emitters in building materials

3.2.3 Lambertian and non-Lambertian emitters (the angular sensitivity of emissivity)

3.2.4 Effects of emissivity errors

3.2.5 Calculation of emissivity, reflectivity, and transmissivity (practical use of Kirchoff's law)

3.2.6 Quantifying effects of unavoidable reflections in buildings

4.0 Understanding Infrared Camera Specifications for Buildings

4.1 Resolution tests and calculations

4.2 IFOV and FOV measurements and calculations

4.3 Resolution versus lens and distance

4.4 Dynamic range

4.5 Data acquisition rate/data density

4.6 Frame rate and field rate

Thermal/Infrared Level II Building Diagnostics Operating Course

1.0 Operating for Qualitative Analysis

1.1 Simple infrared energy measurement

1.2 Quantifying the relative emissivity variations of the target surface

1.3 Quantifying thermal profiles

1.4 Producing accurate images of qualitative thermal scenes

1.5 Qualitative and quantitative analysis during pressurization/depressurization testing of building envelopes

2.0 Operating for Quantitative Measurements and Analysis

2.1 Infrared energy measurement

2.2 Quantifying the emissivity of the target surface

2.3 Quantifying temperature profiles

2.3.1 Use of blackbody temperature references in the image

2.3.2 Use of temperature measurement devices for reference surface temperatures

2.3.3 Common sources of temperature measurement errors

2.4 Computer processing to enhance imager data

2.5 Producing accurate images of quantitative thermal scenes

3.0 Reports and Documentation

3.1 Standardization requirements and records

3.2 Report data requirements

3.3 Preparing reports

Thermal/Infrared Level II Building Diagnostics Applications Course

1.0 Qualitative versus Quantitative Measurement Applications

1.1 Isotherms/alarm levels – personnel safety audits, etc.

2.0 Energy Loss Analysis Applications

2.1 Conduction losses through envelopes

2.1.1 Basic envelope heat-flow quantification

2.1.2 Recognizing and dealing with wind effects

2.2 Mass-transfer heat exchange (air or other flows into or out of the system)

2.2.1 Location

2.2.2 Quantification

3.0 Building Applications
- 3.1 Thermal bridges and insulation flaws
- 3.2 Air infiltration in building envelopes
- 3.3 Envelope pressurization/depressurization (blower door) testing
- 3.4 Moisture in building materials
- 3.5 Sunlight
- 3.6 Semitransparent targets
- 3.7 Transient applications

Electrical and Mechanical Equipment Intermediate Thermal/Infrared Physics Course

1.0 Basic Calculations in the Three Modes of Heat Transfer
- 1.1 Conduction – principles and elementary calculation
  - 1.1.1 Thermal resistance – principles and elementary calculations
  - 1.1.2 Heat capacitance – principles and elementary calculations
- 1.2 Convection – principles and elementary calculations
- 1.3 Radiation – principles and elementary calculations

2.0 The Infrared Spectrum
- 2.1 Planck's law/curves
  - 2.1.1 Typical detected bands
  - 2.1.2 Spectral emissivities of real surfaces
  - 2.1.3 Effects due to semitransparent windows and/or gases
  - 2.1.4 Filters

3.0 Radiosity Problems
- 3.1 Blackbodies – theory and concepts
- 3.2 Emissivity problems
  - 3.2.1 Blackbody emissivity
  - 3.2.2 The graybody and the non-graybody
  - 3.2.3 Broadband and narrow-band emitter targets
  - 3.2.4 Specular and diffuse emitters
  - 3.2.5 Lambertian and non-Lambertian emitters (the angular sensitivity of emissivity)
  - 3.2.6 Effects of emissivity errors
- 3.3 Calculation of emissivity, reflectivity, and transmissivity (practical use of Kirchoff's law)
- 3.4 Reflectivity problem
  - 3.4.1 Quantifying effects of unavoidable reflections
  - 3.4.2 Theoretical corrections
- 3.5 Transmissivity problem
  - 3.5.1 Quantified effects of partial transmittance
  - 3.5.2 Theoretical corrections

4.0 Resolution Tests and Calculations
- 4.1 IFOV, FOV, and MIFOV measurements and calculations
- 4.2 MRTD measurements and calculations
- 4.3 Slit response function – measurement, calculations, interpretations, and comparisons
- 4.4 Resolution versus lens and distance
- 4.5 Dynamic range
- 4.6 Data acquisition rate/data density
- 4.7 Frame rate and field rate
- 4.8 Image data density
  - 4.8.1 Lines of resolution
  - 4.8.2 IFOVs/line
  - 4.8.3 Computer pixels/line

Intermediate Thermal/Infrared Operating Course

1.0 Operating for Infrared Measurements (Quantification)
- 1.1 Simple infrared energy measurement
- 1.2 Quantifying the emissivity of the target surface
- 1.3 Quantifying temperature profiles
  - 1.3.1 Use of blackbody temperature references in the image
  - 1.3.2 Use of temperature measurement devices for reference surface temperatures
  - 1.3.3 Common sources of temperature measurement errors
- 1.4 Computer processing to enhance imager data

2.0 Operating for High-Speed Data Collection
- 2.1 Producing accurate images of transient processes
- 2.2 Recording accurate images of transient processes
- 2.3 Equipment selection and operation for imaging from moving vehicles

3.0 Operating Special Equipment for "Active" Techniques
- 3.1 Hot or cold fluid energy sources
- 3.2 Heat lamp energy sources
- 3.3 Flash lamp energy sources
- 3.4 Electromagnetic induction
- 3.5 Laser energy sources

4.0 Reports and Documentation
- 4.1 Standardization requirements and records
- 4.2 Report data requirements
- 4.3 Preparing reports

Intermediate Thermal/Infrared Applications Course

1.0 Temperature Measurement Applications
- 1.1 Isotherms/alarm levels – personnel safety audits, etc.
- 1.2 Profiles

2.0 Energy Loss Analysis Applications
- 2.1 Conduction losses through enclosures and housings
  - 2.1.1 Basic envelope heat-flow quantification
  - 2.1.2 Recognizing and dealing with wind effects
- 2.2 Mass-transfer heat exchange (air or other flows into or out of the system)
  - 2.2.1 Location
  - 2.2.2 Quantification

3.0 "Active" Applications
- 3.1 Insulation flaws in electrical and mechanical equipment
- 3.2 Location of high heat-capacity components

4.0 Filtered Applications
- 4.1 Sunlight
- 4.2 Furnace interiors
- 4.3 Semitransparent targets

5.0 Transient Applications
- 5.1 Imaging a rapidly moving process
- 5.2 Imaging from a vehicle

Thermal/Infrared Level II NDT of Materials Physics Course

1.0 Basic Calculations in the Three Modes of Heat Transfer
- 1.1 Conduction – principles and elementary calculation
  - 1.1.1 Thermal resistance – principles and elementary calculations
  - 1.1.2 Heat capacitance – principles and elementary calculations
- 1.2 Convection – principles and elementary calculations
- 1.3 Radiation – principles and elementary calculations

2.0 The Infrared Spectrum
- 2.1 Planck's law/curves
  - 2.1.1 Typical detection bands
  - 2.1.2 Spectral emissivities of real surfaces
  - 2.1.3 Effects due to semitransparent windows and materials

3.0 Radiosity – Basic Theory and Application to Characteristics of Materials
- 3.1 Blackbodies – theory and concepts
- 3.2 Emissivity problems
  - 3.2.1 Blackbody, graybody, and non-graybody
  - 3.2.2 Specular and diffuse emitters in materials
  - 3.2.3 Lambertian and non-Lambertian emitters (the angular sensitivity of emissivity)
  - 3.2.4 Effects of emissivity errors
  - 3.2.5 Calculation of emissivity, reflectivity, and transmissivity (practical use of Kirchoff's law)
  - 3.2.6 Quantifying effects of unavoidable reflections

4.0 Understanding Infrared Camera Specifications Required for Determining Size, Depth, and Type of Flaws in Materials
- 4.1 Resolution tests and calculations
- 4.2 IFOV and FOV measurements and calculations
- 4.3 Resolution versus lens and distance
- 4.4 Dynamic range
- 4.5 Data acquisition rate/data density
- 4.6 Frame rate and field rate

Thermal/Infrared Level II NDT of Materials Operating Course

1.0 Operating for Qualitative Analysis
- 1.1 Simple infrared energy measurement – active versus passive
- 1.2 Quantifying the relative emissivity variations of the target surface
- 1.3 Quantifying thermal profiles
- 1.4 Producing accurate images of qualitative thermal scenes
- 1.5 Qualitative and quantitative analysis
- 1.6 Understanding thermal differences from the rate of change of thermal differences

2.0 Operating for Quantitative Measurements and Analysis
- 2.1 Infrared energy measurement
- 2.2 Quantifying the emissivity of the target surface
- 2.3 Quantifying temperature profiles
  - 2.3.1 Use of blackbody temperature references in the image
  - 2.3.2 Use of temperature measurement devices for reference surface temperatures
  - 2.3.3 Common sources of temperature measurement errors
- 2.4 Computer processing to enhance imager data
- 2.5 Producing accurate images of quantitative thermal scenes

3.0 Reports and Documentation
- 3.1 Standardization requirements and records
- 3.2 Report data requirements
- 3.3 Preparing reports

Thermal/ Infrared Level II NDT of Materials Applications Course

1.0 Qualitative versus Quantitative Measurement Applications
- 1.1 Isotherms/alarm levels – personnel safety audits, etc.

2.0 Energy Loss Analysis Applications
- 2.1 Conduction losses through envelopes
  - 2.1.1 Basic envelope heat-flow quantification
  - 2.1.2 Recognizing and dealing with wind effects
- 2.2 Mass-transfer heat exchange (air or other flows into or out of the system)
  - 2.2.1 Location
  - 2.2.2 Quantification

3.0 Composite Material Applications
- 3.1 Thermal bridges and insulation flaws
- 3.2 Behavior of water, moisture, and foreign object debris (FOD) in test materials
- 3.3 Differentiating materials and flaws/contaminants based on differences in the rate of change of surface temperature
- 3.4 Effects of sunlight and background heat sources
- 3.5 Semitransparent targets
- 3.6 Transient applications

Thermal/Infrared Testing Level III Topical Outline

1.0 Principles/Theory
- 1.1 Conduction
- 1.2 Convection
- 1.3 Radiation
- 1.4 The nature of heat and flow
  - 1.4.1 Exothermic or endothermic conditions
  - 1.4.2 Friction
  - 1.4.3 Variations in fluid flow
  - 1.4.4 Variations in thermal resistance
  - 1.4.5 Thermal capacitance
- 1.5 Temperature measurement principles
- 1.6 Proper selection of thermal/infrared testing (IR) as technique of choice
  - 1.6.1 Differences between IR and other techniques
  - 1.6.2 Complementary roles of IR and other methods
  - 1.6.3 Potential for conflicting results between methods
  - 1.6.4 Factors that qualify/disqualify the use of IR

2.0 Equipment/Materials
- 2.1 Temperature measurement equipment
  - 2.1.1 Liquid – in glass thermometers
  - 2.1.2 Vapor – pressure thermometers
  - 2.1.3 Bourdon tube thermometers
  - 2.1.4 Bi-metallic thermometers
  - 2.1.5 Melting-point indicators
  - 2.1.6 Thermochromic G crystal materials
  - 2.1.7 (Irreversible) thermochromic change materials
  - 2.1.8 Thermocouples
  - 2.1.9 Resistance thermometers
    - 2.1.9.1 RTDs
    - 2.1.9.2 Thermistors
  - 2.1.10 Optical pyrometers
  - 2.1.11 Infrared pyrometers
  - 2.1.12 Two-color infrared pyrometers
  - 2.1.13 Laser/infrared pyrometers
  - 2.1.14 Integrating hemisphere radiation pyrometers
  - 2.1.15 Fiber-optic thermometers
  - 2.1.16 Infrared photographic films and cameras
  - 2.1.17 Infrared line scanners
  - 2.1.18 Thermal/infrared imagers
- 2.2 Heat flux indicators
- 2.3 Performance parameters of noncontact devices
  - 2.3.1 Absolute precision and accuracy
  - 2.3.2 Repeatability
  - 2.3.3 Sensitivity
  - 2.3.4 Spectral response limits
  - 2.3.5 Response time
  - 2.3.6 Drift
  - 2.3.7 Spot size ratio
  - 2.3.8 IFOV
  - 2.3.9 Minimum resolvable temperature difference
  - 2.3.10 Slit response function

3.0 Techniques
- 3.1 Contact temperature indicators
  - 3.1.1 Standardization
- 3.2 Noncontact pyrometers
  - 3.2.1 Standardization of equipment
  - 3.2.2 Quantifying emissivity
  - 3.2.3 Evaluating background radiation
  - 3.2.4 Measuring (or mapping) radiant energy
  - 3.2.5 Measuring (or mapping) surface temperatures
  - 3.2.6 Measuring (or mapping) surface heat flows
  - 3.2.7 Use in high-temperature environments
  - 3.2.8 Use in high-magnetic-field environments
  - 3.2.9 Measurements on small targets
  - 3.2.10 Measurements through semitransparent materials
- 3.3 Infrared line scanners
  - 3.3.1 Standardization of equipment
  - 3.3.2 Quantifying emissivity
  - 3.3.3 Evaluating background radiation
  - 3.3.4 Measuring (or mapping) surface radiant energy
  - 3.3.5 Measuring (or mapping) surface temperatures
  - 3.3.6 Measuring (or mapping) surface heat flows
  - 3.3.7 Use in high-temperature environments
  - 3.3.8 Use in high-magnetic-field environments
  - 3.3.9 Measurements on small targets
  - 3.3.10 Measurements through semitransparent materials
- 3.4 Thermal/infrared imaging
  - 3.4.1 Standardization of equipment
  - 3.4.2 Quantifying emissivity
  - 3.4.3 Evaluating background radiation
  - 3.4.4 Measuring (or mapping) surface radiant energy
  - 3.4.5 Measuring (or mapping) surface temperatures
  - 3.4.6 Measuring (or mapping) surface heat flows
  - 3.4.7 Use in high-temperature environments
  - 3.4.8 Use in high-magnetic-field environments
  - 3.4.9 Measurements on small targets
  - 3.4.10 Measurements through semitransparent materials
- 3.5 Heat flux indicators
  - 3.5.1 Standardization of equipment
  - 3.5.2 Measurement of heat flow
- 3.6 Exothermic or endothermic investigations. Typical examples may include, but are not limited to, the following:
  - 3.6.1 Power distribution systems
    - 3.6.1.1 Exposed electrical switchgear
    - 3.6.1.2 Enclosed electrical switchgear
    - 3.6.1.3 Exposed electrical buses
    - 3.6.1.4 Enclosed electrical buses
    - 3.6.1.5 Transformers
    - 3.6.1.6 Electric rotating equipment
    - 3.6.1.7 Overhead power lines
    - 3.6.1.8 Coils
    - 3.6.1.9 Capacitors
    - 3.6.1.10 Circuit breakers
    - 3.6.1.11 Indoor wiring
    - 3.6.1.12 Motor control center starters
    - 3.6.1.13 Lighter arrestors

3.6.2 Chemical processes
3.6.3 Foam-in-place insulation
3.6.4 Firefighting
3.6.4.1 Building investigations
3.6.4.2 Outside ground-based investigations
3.6.4.3 Outside airborne investigations
3.6.5 Moisture in airframes
3.6.6 Underground investigations
3.6.6.1 Airborne coal mine fires
3.6.6.2 Utility locating
3.6.6.3 Utility pipe leak detection
3.6.6.4 Void detection
3.6.7 Locating and mapping utilities concealed in structures
3.6.8 Mammal location and monitoring
3.6.8.1 Ground investigations
3.6.8.2 Airborne investigations
3.6.8.3 Sorting mammals according to stress levels
3.6.9 Fracture dynamics
3.6.10 Process heating or cooling
3.6.10.1 Rate
3.6.10.2 Uniformity
3.6.11 Heat tracing or channelized cooling
3.6.12 Radiant heating
3.6.13 Electronic components
3.6.13.1 Assembled circuit boards
3.6.13.2 Bare printed circuit boards
3.6.13.3 Semiconductor microcircuits
3.6.14 Welding
3.6.14.1 Welding technique parameters
3.6.14.2 Material parameters
3.6.15 Mapping of energy fields
3.6.15.1 Electromagnetic fields
3.6.15.2 Electromagnetic heating processes
3.6.15.3 Radiant heat flux distribution
3.6.15.4 Acoustic fields
3.6.16 Gaseous plumes
3.6.16.1 Monitoring
3.6.16.2 Mapping
3.6.17 Ground frost-line mapping

3.7 Friction investigations. Typical examples may include, but are not limited to, the following:
3.7.1 Bearings
3.7.2 Seals
3.7.3 Drive belts
3.7.4 Drive couplings
3.7.5 Exposed gears
3.7.6 Gearboxes
3.7.7 Machining processes
3.7.8 Aerodynamic heating

3.8 Fluid flow investigations. Typical examples may include, but are not limited to, the following:
3.8.1 Fluid piping
3.8.2 Valves
3.8.3 Heat exchangers
3.8.4 Fin fans
3.8.5 Cooling ponds
3.8.6 Cooling towers
3.8.7 Distillation towers
3.8.7.1 Packed
3.8.7.2 Trays
3.8.8 HVAC systems
3.8.9 Lake and ocean current mapping
3.8.10 Mapping civil and industrial outflows into waterways
3.8.11 Locating leaks in pressure systems
3.8.12 Filters

3.9 Thermal resistance (steady-state heat flow) investigations. Typical examples may include, but are not limited to, the following:
3.9.1 Thermal safety audits
3.9.2 Low-temperature insulating systems
3.9.3 Industrial insulation systems
3.9.4 Refractory systems
3.9.5 Semitransparent walls
3.9.6 Furnace interiors
3.9.7 Disbonds in lined process equipment

3.10 Thermal capacitance investigations. Typical examples may include, but are not limited to, the following:
3.10.1 Tank levels
3.10.2 Rigid injection molding
3.10.3 Thermal laminating processes
3.10.4 Building envelopes
3.10.5 Roof moisture
3.10.5.1 Roof-level investigations
3.10.5.2 Airborne investigations
3.10.6 Underground voids
3.10.7 Bridge-deck laminations
3.10.8 Steam traps
3.10.9 Paper manufacturing moisture profiles
3.10.10 Subsurface discontinuity detection in materials
3.10.11 Coating disbond
3.10.12 Structural materials
3.10.12.1 Subsurface discontinuity detection
3.10.12.2 Thickness variations
3.10.12.3 Disbonding

4.0 Interpretation/Evaluation

4.1 Exothermic or endothermic investigation: Typical examples may include, but are not limited to, the examples shown in paragraph 3.6

4.2 Friction investigations: Typical examples may include, but are not limited to, the examples shown in paragraph 3.7

4.3 Fluid flow investigations: Typical examples may include, but are not limited to, the examples shown in paragraph 3.8

4.4 Differences in thermal resistance (steady-state heat flow) investigations: Typical examples may include, but are not limited to, the examples shown in paragraph 3.9

4.5 Thermal capacitance investigations: Typical examples may include, but are not limited to, the examples shown in paragraph 3.10

5.0 Procedures

5.1 Existing codes and standards

5.2 Elements of IR job procedure development

6.0 Safety and Health

6.1 Safety responsibility and authority

6.2 Safety for personnel – OSHA

6.3 Safety for client and facilities

6.4 Safety for testing equipment

THERMAL/INFRARED TESTING LEVEL I, II, AND III TRAINING REFERENCES

PRIMARY BODY OF KNOWLEDGE REFERENCES

ASNT, latest edition, *ASNT Level III Study Guide: Thermal/Infrared Testing*, Columbus, OH: American Society for Nondestructive Testing Inc.*

ASNT, latest edition, *ASNT Questions & Answers Book: Thermal and Infrared Testing Method*, Columbus, OH: American Society for Nondestructive Testing Inc.*

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* Available from The American Society for Nondestructive Testing Inc.

UT
ULTRASONIC TESTING TOPICAL OUTLINES

Ultrasonic Testing Level I Topical Outline

Basic Ultrasonic Testing Course

Note: It is recommended that the trainee receive instruction in this course prior to performing work in ultrasonic testing (UT).

1.0 Introduction
- 1.1 Definition of ultrasonics
- 1.2 History of UT
- 1.3 Applications of ultrasonic energy
- 1.4 Basic math review
- 1.5 Responsibilities of levels of certification

2.0 Basic Principles of Acoustics
- 2.1 Nature of sound waves
- 2.2 Modes of sound-wave generation
- 2.3 Velocity, frequency, and wavelength of sound waves
- 2.4 Attenuation of sound waves
- 2.5 Acoustic impedance
- 2.6 Reflection
- 2.7 Refraction and mode conversion
- 2.8 Snell's law and critical angles
- 2.9 Fresnel and Fraunhofer effects

3.0 Equipment
- 3.1 Basic pulse-echo instrumentation (A-scan, B-scan, C-scan, and computerized systems)
  - 3.1.1 Electronics – time-base, pulser, receiver, and various monitor displays
  - 3.1.2 Control functions
  - 3.1.3 Standardization
    - 3.1.3.1 Basic instrument standardization
    - 3.1.3.2 Reference blocks (types and use)
- 3.2 Digital thickness instrumentation
- 3.3 Transducer operation and theory
  - 3.3.1 Piezoelectric effect
  - 3.3.2 Types of transducer elements
  - 3.3.3 Frequency (transducer elements – thickness relationships)
  - 3.3.4 Near field and far field
  - 3.3.5 Beam spread
  - 3.3.6 Construction, materials, and shapes
  - 3.3.7 Types (straight, angle, dual, etc.)
  - 3.3.8 Beam intensity characteristics
  - 3.3.9 Sensitivity, resolution, and damping
  - 3.3.10 Mechanical vibration into part
  - 3.3.11 Other type of transducers (laser UT, EMAT, etc.)
- 3.4 Couplants
  - 3.4.1 Purpose and principles
  - 3.4.2 Materials and their efficiency

4.0 Basic Testing Methods
- 4.1 Contact
- 4.2 Immersion
- 4.3 Air coupling

Ultrasonic Testing Technique Course

1.0 Testing Methods
- 1.1 Contact
  - 1.1.1 Straight beam
  - 1.1.2 Angle-beam
  - 1.1.3 Surface-wave and plate waves
  - 1.1.4 Pulse-echo transmission
  - 1.1.5 Multiple transducer
  - 1.1.6 Curved surfaces
    - 1.1.6.1 Flat entry surfaces
    - 1.1.6.2 Cylindrical and tubular shapes
- 1.2 Immersion
  - 1.2.1 Transducer in water
  - 1.2.2 Water column, wheels, etc.
  - 1.2.3 Submerged test part
  - 1.2.4 Sound beam path – transducer to part
  - 1.2.5 Focused transducers
  - 1.2.6 Curved surfaces
  - 1.2.7 Plate waves
  - 1.2.8 Pulse-echo and through-transmission
- 1.3 Comparison of contact and immersion methods

2.0 Calibration (Electronic and Functional)
- 2.1 Equipment
  - 2.1.1 Monitor displays (amplitude, sweep, etc.)
  - 2.1.2 Recorders
  - 2.1.3 Alarms
  - 2.1.4 Automatic and semiautomatic systems
  - 2.1.5 Electronic distance/amplitude correction
  - 2.1.6 Transducers

- 2.2 Standardization of equipment electronics
  - 2.2.1 Variable effects
  - 2.2.2 Transmission accuracy
  - 2.2.3 Standardization requirements
  - 2.2.4 Standardization reflectors
- 2.3 Inspection standardization
  - 2.3.1 Comparison with reference blocks
  - 2.3.2 Pulse-echo variables
  - 2.3.3 Reference for planned tests (straight beam, angle-beam, etc.)
  - 2.3.4 Transmission factors
  - 2.3.5 Transducer
  - 2.3.6 Couplants
  - 2.3.7 Materials

3.0 Straight-Beam Examination to Specific Procedures
- 3.1 Selection of parameters
- 3.2 Test standards
- 3.3 Evaluation of results
- 3.4 Test reports

4.0 Angle-Beam Examination to Specific Procedures
- 4.1 Selection of parameters
- 4.2 Test standards
- 4.3 Evaluation of results
- 4.4 Test reports

Ultrasonic Testing Level II Topical Outline

Ultrasonic Testing Evaluation Course

1.0 Review of UT Technique Course
- 1.1 Principles of ultrasonics
- 1.2 Equipment
  - 1.2.1 A-scan
  - 1.2.2 B-scan
  - 1.2.3 C-scan
  - 1.2.4 Computerized systems
- 1.3 Testing techniques
- 1.4 Standardization
  - 1.4.1 Straight beam
  - 1.4.2 Angle-beam
  - 1.4.3 Resonance
  - 1.4.4 Special applications

2.0 Evaluation of Base-Material Product Forms
- 2.1 Ingots
  - 2.1.1 Process review
  - 2.1.2 Types, origin, and typical orientation of discontinuities
  - 2.1.3 Response of discontinuities to ultrasound
  - 2.1.4 Applicable codes/standards
- 2.2 Plate and sheet
  - 2.2.1 Rolling process
  - 2.2.2 Types, origin, and typical orientation of discontinuities
  - 2.2.3 Response of discontinuities to ultrasound
  - 2.2.4 Applicable codes/standards
- 2.3 Bar and rod
  - 2.3.1 Forming process
  - 2.3.2 Types, origin, and typical orientation of discontinuities
  - 2.3.3 Response of discontinuities to ultrasound
  - 2.3.4 Applicable codes/standards
- 2.4 Pipe and tubular products
  - 2.4.1 Manufacturing process
  - 2.4.2 Types, origin, and typical orientation of discontinuities
  - 2.4.3 Response of discontinuities to ultrasound
  - 2.4.4 Applicable codes/standards
- 2.5 Forgings
  - 2.5.1 Process review
  - 2.5.2 Types, origin, and typical orientation of discontinuities
  - 2.5.3 Response of discontinuities to ultrasound
  - 2.5.4 Applicable codes/standards
- 2.6 Castings
  - 2.6.1 Process review
  - 2.6.2 Types, origin, and typical orientation of discontinuities
  - 2.6.3 Response of ultrasound to discontinuities
  - 2.6.4 Applicable codes/standards
- 2.7 Composite structures
  - 2.7.1 Process review
  - 2.7.2 Types, origin, and typical orientation of discontinuities
  - 2.7.3 Response of ultrasound to discontinuities
  - 2.7.4 Applicable codes/standards
- 2.8 Other product forms as applicable – rubber, glass, etc.

3.0 Evaluation of Weldments
- 3.1 Welding processes
- 3.2 Weld geometries
- 3.3 Welding discontinuities
- 3.4 Origin and typical orientation of discontinuities
- 3.5 Response of discontinuities to ultrasound
- 3.6 Applicable codes/standards

4.0 Evaluation of Bonded Structures
- 4.1 Manufacturing processes
- 4.2 Types of discontinuities
- 4.3 Origin and typical orientation of discontinuities
- 4.4 Response of discontinuities to ultrasound
- 4.5 Applicable codes/standards

5.0 Discontinuity Detection
- 5.1 Sensitivity to reflections
  - 5.1.1 Size, type, and location of discontinuities
  - 5.1.2 Techniques used in detection
  - 5.1.3 Wave characteristics
  - 5.1.4 Material and velocity

5.2 Resolution
- 5.2.1 Standard reference comparisons
- 5.2.2 History of part
- 5.2.3 Probability of type of discontinuity
- 5.2.4 Degrees of operator discrimination
- 5.2.5 Effects of ultrasonic frequency
- 5.2.6 Damping effects

5.3 Determination of discontinuity size
- 5.3.1 Various monitor displays and meter indications
- 5.3.2 Transducer movement versus display
- 5.3.3 Two-dimensional testing techniques
- 5.3.4 Signal patterns

5.4 Location of discontinuity
- 5.4.1 Various monitor displays
- 5.4.2 Amplitude and linear time
- 5.4.3 Search technique

6.0 Evaluation

6.1 Comparison procedures
- 6.1.1 Standards and references
- 6.1.2 Amplitude, area, and distance relationship
- 6.1.3 Application of results of other NDT methods

6.2 Object appraisal
- 6.2.1 History of part
- 6.2.2 Intended use of part
- 6.2.3 Existing and applicable code interpretation
- 6.2.4 Type of discontinuity and location

Full Matrix Capture Ultrasonic Testing Level II Topical Outline

Note: It is recommended that this course have as a minimum prerequisite an Ultrasonic Testing Level II unrestricted certification. The intent of this document is to provide "basic" knowledge on full matrix capture (FMC) ultrasonic testing consistent with other methods and to acknowledge FMC as unique enough to warrant an additional body of knowledge and qualification requirements.

1.0 Overview

1.1 Introduction

1.2 FMC terminology

1.3 History

1.4 Ultrasonic theory
- 1.4.1 Beam divergence
- 1.4.2 Wavelength

1.5 Overview of phased array ultrasonic testing (PAUT)

2.0 Basics of FMC Data Collection

3.0 Equipment

3.1 Computer-based system

3.2 Processors and throughput

3.3 Block diagram showing basic internal components

3.4 Portable versus full computer-based systems

4.0 Probe

4.1 Review of arrays
- 4.1.1 Types and configurations
- 4.1.2 Effects of pitch and element size relevant to sound transmission
- 4.1.3 Aperture size and effects

4.2 Probe selection

4.3 Dead-element check

5.0 Essential Variables

6.0 Scan Plan

6.1 Major components of a scan plan

6.2 Paths

7.0 Standardization

7.1 Single probe

7.2 Tandem probe

7.3 Reflectors versus paths

7.4 Delay and velocity

7.5 TCG

8.0 FMC Characteristics

8.1 Signal characteristics

8.2 Scale factor for FMC

8.3 FMC data size

8.4 Different FMC techniques

8.5 FMC versus other data collection

8.6 How to use FMC data

8.7 Typical FMC data explained

9.0 Total Focusing Method (TFM) Characteristics

9.1 Signal characteristics

9.2 TFM frame parameters and FMC

9.3 TFM and delay laws

9.4 Focusing capability

9.5 Coverage capability

9.6 Impact of frame parameters on amplitude

9.7 Adaptive algorithms

10.0 Examination

10.1 Types of equipment
- 10.1.1 Fully automated
- 10.1.2 Semiautomated
- 10.1.3 Manual
- 10.1.4 Advantages and disadvantages

11.0 Evaluation

11.1 Display and display settings
- 11.1.1 Imaging
- 11.1.2 3D

11.2 Flaw characterization

11.3 Flaw dimensioning

11.4 Software tools

11.5 Image artifacts and saturation

12.0 Documentation
- 12.1 Images
- 12.2 Equipment settings
- 12.3 Plotting
- 12.4 Onboard reporting, requirements

13.0 Amplitude
- 13.1 Amplitude fidelity
- 13.2 Amplitude subject to resolution
- 13.3 Amplitude and interface/dead zones

14.0 Use Cases
- 14.1 Weld examinations
  - 14.1.1 Examination volume
  - 14.1.2 Impact of geometry
  - 14.1.3 Material type
  - 14.1.4 Material thickness
  - 14.1.5 Probe considerations
  - 14.1.6 Review typical welding defects and responses
- 14.2 Corrosion examinations
  - 14.2.1 Advantages, disadvantages
  - 14.2.2 Probe considerations
- 14.3 Other examples
  - 14.3.1 Aluminum
  - 14.3.2 Composites
  - 14.3.3 Effects of probe frequency and wavelength
  - 14.3.4 Manufacturing processes and defects
  - 14.3.5 Types of welding processes
  - 14.3.6 Historical processes and defects

15.0 Procedures and Requirements
- 15.1 Codes and standards specific
- 15.2 Customized specific applications

Phased Array Ultrasonic Testing Level II Topical Outline

Note: It is recommended that this course have as a minimum prerequisite of an Ultrasonic Testing Level II unrestricted certification. The intent of this document is to provide "basic" knowledge on phased array ultrasonic testing (PAUT) consistent with other methods and to acknowledge PAUT as unique enough to warrant an additional body of knowledge and qualification requirements.

Phased Array Ultrasonic Testing Evaluation Course

1.0 Introduction
- 1.1 Terminology of PAUT
- 1.2 History of PAUT – medical ultrasound, etc.
- 1.3 Responsibilities of levels of certification

2.0 Basic Principles of PAUT
- 2.1 Review of ultrasonic wave theory – longitudinal and shear wave
- 2.2 Introduction to PAUT concepts and theory

3.0 Equipment
- 3.1 Computer-based systems
  - 3.1.1 Processors
  - 3.1.2 Control panel including input and output sockets
  - 3.1.3 Block diagram showing basic internal circuit modules
  - 3.1.4 Multielement/multichannel configurations
  - 3.1.5 Portable battery-operated versus full computer-based systems
- 3.2 Focal law generation
  - 3.2.1 Onboard focal law generator
  - 3.2.2 External focal law generator
- 3.3 Probes
  - 3.3.1 Composite materials
  - 3.3.2 Pitch, gap, and size
  - 3.3.3 Passive planes
  - 3.3.4 Active planes
  - 3.3.5 Arrays – 1D, 2D, polar, annular, special shape, etc.
  - 3.3.6 Beam and wave forming
  - 3.3.7 Grating lobes
- 3.4 Wedges
  - 3.4.1 Types of wedge designs
- 3.5 Scanners
  - 3.5.1 Mechanized
  - 3.5.2 Manual

4.0 Testing Techniques
- 4.1 Linear scans
- 4.2 Sectorial scans
- 4.3 Electronic scans

5.0 Standardization
- 5.1 Active element and probe checks
- 5.2 Wedge delay
- 5.3 Velocity
- 5.4 Exit point verifications
- 5.5 Refraction angle verifications
- 5.6 Sensitivity
- 5.7 DAC, TCG, time varied gain (TVG), and angle corrected gain (ACG) variables and parameters
- 5.8 Effects of curvature
- 5.9 Focusing effects
- 5.10 Beam steering
- 5.11 Acquisition gates

6.0 Data Collection
- 6.1 Single probes
- 6.2 Multiple probes
- 6.3 Multiple groups or multiplexing single/multiple probes
- 6.4 Nonencoded scans
  - 6.4.1 Time-based data storage
- 6.5 Encoded scans
  - 6.5.1 Line scans
  - 6.5.2 Raster scans

6.6 Zone discrimination
6.7 Scan plans and exam coverages
  6.7.1 Sectorial
  6.7.2 Linear
  6.7.3 Electronic raster scans
6.8 Probe offsets and indexing

7.0 Procedures
7.1 Specific applications
  7.1.1 Material evaluations
    7.1.1.1 Composites
    7.1.1.2 Nonmetallic materials
    7.1.1.3 Metallic materials
    7.1.1.4 Base-material scan
    7.1.1.5 Bar, rod, and rail
    7.1.1.6 Forgings
    7.1.1.7 Castings
  7.1.2 Component evaluations
    7.1.2.1 Ease with complex geometries
      7.1.2.1.1 Turbines (blades, dovetails, rotors)
      7.1.2.1.2 Shafts, keyways, etc.
      7.1.2.1.3 Nozzles
      7.1.2.1.4 Flanges
    7.1.2.2 Geometric limitations
  7.1.3 Weld inspections
    7.1.3.1 Fabrication/in-service
    7.1.3.2 Differences in material – carbon steel, stainless steel, high-temperature nickel-chromium alloy, etc.
    7.1.3.3 Review of welding discontinuities
    7.1.3.4 Responses from various discontinuities
7.2 Data presentations
  7.2.1 Standard (A-scan, B-scan, and C-scan)
  7.2.2 Other (D-scan, S-scan, etc.)
7.3 Data evaluation
  7.3.1 Codes/standards/specifications
  7.3.2 Flaw characterization
  7.3.3 Flaw dimensioning
  7.3.4 Geometry
  7.3.5 Software tools
  7.3.6 Evaluation gates
7.4 Reporting
  7.4.1 Imaging outputs
  7.4.2 Onboard reporting tools
  7.4.3 Plotting, ACAD, etc.

Time of Flight Diffraction Level II Topical Outline

Note: It is recommended that this course have as a minimum prerequisite an Ultrasonic Testing Level II unrestricted certification. The intent of this document is to provide "basic" knowledge on time of flight diffraction (TOFD) ultrasonic testing consistent with other methods and to acknowledge TOFD as unique enough to warrant an additional body of knowledge and qualification requirements.

Time of Flight Diffraction Evaluation Course

1.0 Introduction
1.1 Terminology of TOFD
1.2 History of TOFD (e.g., M.G. Silk)
1.3 Responsibilities of levels of certification

2.0 Basic Principles of TOFD
2.1 Review of ultrasonic wave theory, refracted longitudinal waves
2.2 Introduction to TOFD concepts and theory
2.3 Technique limitations

3.0 Equipment
3.1 Computer-based systems
  3.1.1 Processors
  3.1.2 Control panel including input and output sockets
  3.1.3 Block diagram showing basic internal circuit modules
  3.1.4 Portable battery-operated versus full computer-based systems
3.2 Beam profile tools
  3.2.1 Probe center separation (PCS) calculators for "flat" material/components
  3.2.2 PCS calculators for "curved" surfaces
  3.2.3 Beam-spread effects and control
  3.2.4 Multiple-zone coverage and limitations
3.3 Probes
  3.3.1 Composite materials
  3.3.2 Damping characteristics
  3.3.3 Selection of frequency and diameter
3.4 Wedges
  3.4.1 Incident- and refracted-angle selections
  3.4.2 High-temperature applications
3.5 Scanners
  3.5.1 Mechanized
  3.5.2 Manual

4.0 Testing Techniques
4.1 Line scans (single tandem-probe setups)
4.2 Line scans (multiple-probe setups)
4.3 Raster scans

5.0 Standardization
- 5.1 Material velocity calculations
- 5.2 Combined probe delay(s) calculation(s)
- 5.3 Digitization rates and sampling
- 5.4 Signal averaging
- 5.5 Pulse width control
- 5.6 PCS and angle selection
- 5.7 Sensitivity
- 5.8 Preamplifiers
- 5.9 Effects of curvature

6.0 Data Collection
- 6.1 Single probe setups
- 6.2 Multiple probe setups
- 6.3 Nonencoded scans
  - 6.3.1 Time-based data storage
- 6.4 Encoded scans
  - 6.4.1 Line scans
  - 6.4.2 Raster scans
- 6.5 Probe offsets and indexing

7.0 Procedures
- 7.1 Specific applications
  - 7.1.1 Material evaluations
    - 7.1.1.1 Base-material scans
  - 7.1.2 Weld inspections
    - 7.1.2.1 Detection and evaluation of fabrication welding flaws
    - 7.1.2.2 Detection and evaluation of in-service cracking
    - 7.1.2.3 Detection of volumetric loss such as weld root erosion and partial penetration weld dimensional verifications
    - 7.1.2.4 Geometric limitations
    - 7.1.2.5 Cladding thickness and integrity evaluations
  - 7.1.3 Complex geometries
    - 7.1.3.1 Transitions, nozzles, branch connections, tees, saddles, etc.
- 7.2 Data presentations
  - 7.2.1 Standard (A-scan, D-scan)
  - 7.2.2 Other (B-scan, C-scan)
- 7.3 Data evaluation
  - 7.3.1 Codes/standards/specifications
  - 7.3.2 Flaw characterization
  - 7.3.3 Flaw dimensioning
  - 7.3.4 Geometry
  - 7.3.5 Software tools
    - 7.3.5.1 Linearization
    - 7.3.5.2 Lateral/backwall straightening and removal
    - 7.3.5.3 Synthetic aperture focusing technique (SAFT)
    - 7.3.5.4 Spectrum processing
    - 7.3.5.5 Curved surface compensation
  - 7.3.6 Parabolic cursor(s)
- 7.4 Reporting
  - 7.4.1 Imaging outputs
  - 7.4.2 Onboard reporting tools
  - 7.4.3 Plotting, ACAD, etc.

Ultrasonic Testing Level III Topical Outline

1.0 Principles/Theory
- 1.1 General
- 1.2 Principles of acoustics
  - 1.2.1 Nature of sound waves
  - 1.2.2 Modes of sound-wave generation
  - 1.2.3 Velocity, frequency, and wavelength of sound waves
  - 1.2.4 Attenuation of sound waves
  - 1.2.5 Acoustic impedance
  - 1.2.6 Reflection
  - 1.2.7 Refraction and mode conversion
  - 1.2.8 Snell's law and critical angles
  - 1.2.9 Fresnel and Fraunhofer effects

2.0 Equipment/Materials
- 2.1 Equipment
  - 2.1.1 Pulse-echo instrumentation
    - 2.1.1.1 Controls and circuits
    - 2.1.1.2 Pulse generation (spike, square wave, and toneburst pulsers)
    - 2.1.1.3 Signal detection
    - 2.1.1.4 Display and recording methods, A-scan, B-scan, and C-scan and digital
    - 2.1.1.5 Sensitivity and resolution
    - 2.1.1.6 Gates, alarms, and attenuators
      - 2.1.1.6.1 Basic instrument standardization and calibration
      - 2.1.1.6.2 Reference blocks
  - 2.1.2 Digital thickness instrumentation
  - 2.1.3 Transducer operation and theory
    - 2.1.3.1 Piezoelectric effect
    - 2.1.3.2 Types of transducer elements
    - 2.1.3.3 Frequency (transducer elements – thickness relationships)
    - 2.1.3.4 Near field and far field
    - 2.1.3.5 Beam spread
    - 2.1.3.6 Construction, materials, and shapes
    - 2.1.3.7 Types (straight, angle, dual, etc.)
    - 2.1.3.8 Beam-intensity characteristics
    - 2.1.3.9 Sensitivity, resolution, and damping
    - 2.1.3.10 Mechanical vibration into parts
    - 2.1.3.11 Other types of transducers (laser UT, EMAT, etc.)
  - 2.1.4 Transducer operation/manipulations
    - 2.1.4.1 Tanks, bridges, manipulators, and squirters
    - 2.1.4.2 Wheels and special hand devices
    - 2.1.4.3 Transfer devices for materials
    - 2.1.4.4 Manual manipulation

- 2.1.5 Resonance testing equipment
  - 2.1.5.1 Bond testing
  - 2.1.5.2 Thickness testing
- 2.2 Materials
  - 2.2.1 Couplants
    - 2.2.1.1 Contact
      - 2.2.1.1.1 Purpose and principles
      - 2.2.1.1.2 Materials and their efficiency
    - 2.2.1.2 Immersion
      - 2.2.1.2.1 Purpose and principles
      - 2.2.1.2.2 Materials and their efficiency
    - 2.2.1.3 Air coupling
  - 2.2.2 Reference blocks
  - 2.2.3 Cables/connectors
  - 2.2.4 Test specimen
  - 2.2.5 Miscellaneous materials

3.0 Techniques/Standardization

- 3.1 Contact
  - 3.1.1 Straight beam
  - 3.1.2 Angle-beam
  - 3.1.3 Surface-wave and plate waves
  - 3.1.4 Pulse-echo transmission
  - 3.1.5 Multiple transducer
  - 3.1.6 Curved surfaces
- 3.2 Immersion
  - 3.2.1 Transducer in water
  - 3.2.2 Water column, wheels, etc.
  - 3.2.3 Submerged test part
  - 3.2.4 Sound beam path – transducer to part
  - 3.2.5 Focused transducers
  - 3.2.6 Curved surfaces
  - 3.2.7 Plate waves
  - 3.2.8 Pulse-echo and through-transmission
- 3.3 Comparison of contact and immersion methods
- 3.4 Remote monitoring
- 3.5 Standardization (electronic and functional)
  - 3.5.1 General
  - 3.5.2 Reference reflectors for standardization
    - 3.5.2.1 Balls and flat-bottom holes
    - 3.5.2.2 Area-amplitude blocks
    - 3.5.2.3 Distance-amplitude blocks
    - 3.5.2.4 Notches
    - 3.5.2.5 Side-drilled holes
    - 3.5.2.6 Special blocks – IIW and others
  - 3.5.3 Equipment
    - 3.5.3.1 Various monitor displays (amplitude, sweep, etc.)
    - 3.5.3.2 Recorders
    - 3.5.3.3 Alarms
    - 3.5.3.4 Automatic and semiautomatic systems
    - 3.5.3.5 Electronic distance amplitude correction
    - 3.5.3.6 Transducers
  - 3.5.4 Standardization of equipment electronics
    - 3.5.4.1 Variable effects
    - 3.5.4.2 Transmission accuracy
    - 3.5.4.3 Standardization and calibration requirements
    - 3.5.4.4 Standardization reflectors
  - 3.5.5 Inspection standardization
    - 3.5.5.1 Comparison with reference blocks
    - 3.5.5.2 Pulse-echo variables
    - 3.5.5.3 Reference for planned tests (straight-beam, angle-beam, etc.)
    - 3.5.5.4 Transmission factors
    - 3.5.5.5 Transducers
    - 3.5.5.6 Couplants
    - 3.5.5.7 Materials

4.0 Interpretations/Evaluations

- 4.1 Evaluation of base material product forms
  - 4.1.1 Ingots
    - 4.1.1.1 Process review
    - 4.1.1.2 Types, origin, and typical orientation of discontinuities
    - 4.1.1.3 Response of discontinuities to ultrasound
    - 4.1.1.4 Applicable codes, standards, specifications
  - 4.1.2 Plate and sheet
    - 4.1.2.1 Process review
    - 4.1.2.2 Types, origin, and typical orientation of discontinuities
    - 4.1.2.3 Response of discontinuities to ultrasound
    - 4.1.2.4 Applicable codes, standards, specifications
  - 4.1.3 Bar and rod
    - 4.1.3.1 Process review
    - 4.1.3.2 Types, origin, and typical orientation of discontinuities
    - 4.1.3.3 Response of discontinuities to ultrasound
    - 4.1.3.4 Applicable codes, standards, specifications
  - 4.1.4 Pipe and tubular products
    - 4.1.4.1 Process review
    - 4.1.4.2 Types, origin, and typical orientation of discontinuities
    - 4.1.4.3 Response of discontinuities to ultrasound
    - 4.1.4.4 Applicable codes, standards, specifications
  - 4.1.5 Forgings
    - 4.1.5.1 Process review
    - 4.1.5.2 Types, origin, and typical orientation of discontinuities
    - 4.1.5.3 Response of discontinuities to ultrasound
    - 4.1.5.4 Applicable codes, standards, specifications

- 4.1.6 Castings
  - 4.1.6.1 Process review
  - 4.1.6.2 Types, origin, and typical orientation of discontinuities
  - 4.1.6.3 Response of discontinuities to ultrasound
  - 4.1.6.4 Applicable codes, standards, specifications
- 4.1.7 Composite structures
  - 4.1.7.1 Process review
  - 4.1.7.2 Types, origin, and typical orientation of discontinuities
  - 4.1.7.3 Response of discontinuities to ultrasound
  - 4.1.7.4 Applicable codes, standards, specifications
- 4.1.8 Miscellaneous product forms as applicable (rubber, glass, etc.)
  - 4.1.8.1 Process review
  - 4.1.8.2 Types, origin, and typical orientation of discontinuities
  - 4.1.8.3 Response of discontinuities to ultrasound
  - 4.1.8.4 Applicable codes, standards, specifications

4.2 Evaluation of weldments
- 4.2.1 Process review
- 4.2.2 Weld geometries
- 4.2.3 Types, origin, and typical orientation of discontinuities
- 4.2.4 Response of discontinuities to ultrasound
- 4.2.5 Applicable codes, standards, specifications

4.3 Evaluation of bonded structures
- 4.3.1 Manufacturing process
- 4.3.2 Types, origin, and typical orientation of discontinuities
- 4.3.3 Response of discontinuities to ultrasound
- 4.3.4 Applicable codes, standards, specifications

4.4 Variables affecting test results
- 4.4.1 Instrument performance variations
- 4.4.2 Transducer performance variations
- 4.4.3 Test specimen variations
  - 4.4.3.1 Surface condition
  - 4.4.3.2 Part geometry
  - 4.4.3.3 Material structure
- 4.4.4 Discontinuity variations
  - 4.4.4.1 Size and geometry
  - 4.4.4.2 Relation to entry surface
  - 4.4.4.3 Type of discontinuity
- 4.4.5 Procedure variations
  - 4.4.5.1 Recording criteria
  - 4.4.5.2 Acceptance criteria
- 4.4.6 Personnel variations
  - 4.4.6.1 Skill level in interpretation of results
  - 4.4.6.2 Knowledge level in interpretation of results

4.5 Evaluation (general)
- 4.5.1 Comparison procedures
  - 4.5.1.1 Standards and references
  - 4.5.1.2 Amplitude, area, distance relationship
  - 4.5.1.3 Application of results of other NDT methods
- 4.5.2 Object appraisal
  - 4.5.2.1 History of part
  - 4.5.2.2 Intended use of part
  - 4.5.2.3 Existing and applicable code interpretation
  - 4.5.2.4 Type of discontinuity and location

5.0 Procedures

5.1 Specific applications
- 5.1.1 General
- 5.1.2 Flaw detection
- 5.1.3 Thickness measurement
- 5.1.4 Bond evaluation
- 5.1.5 Fluid flow measurement
- 5.1.6 Material properties measurements
- 5.1.7 Computer control and defect analysis
- 5.1.8 Liquid level sensing
- 5.1.9 Process control
- 5.1.10 Field inspection

5.2 Codes, standards, specifications

Phased Array Ultrasonic Testing

1.0 Introduction

- 1.1 Terminology of PAUT
- 1.2 History of PAUT – medical ultrasound, etc.
- 1.3 Responsibilities of levels of certification

2.0 Basic Principles of PAUT

- 2.1 Review of ultrasonic wave theory: longitudinal and shear wave
- 2.2 Introduction to PAUT concepts and theory
  - 2.2.1 Phasing
  - 2.2.2 Beam scanning patterns
  - 2.2.3 Delay laws or focal laws
  - 2.2.4 Imaging
  - 2.2.5 Dynamic depth focusing

3.0 Equipment

3.1 Computer-based systems
- 3.1.1 Processors
- 3.1.2 Control panel including input and output sockets
- 3.1.3 Block diagram showing basic internal circuit modules
- 3.1.4 Multielement/multichannel configurations
- 3.1.5 Portable battery-operated versus full computer-based systems

3.2 Focal law generation
- 3.2.1 Onboard focal law generator
- 3.2.2 External focal law generator

- 3.3 Probes
  - 3.3.1 Composite materials
  - 3.3.2 Passive planes
  - 3.3.3 Active planes
  - 3.3.4 Arrays – 1D, 2D, polar, annular, special shape, etc.
    - 3.3.4.1 Linear arrays
      - 3.3.4.1.1 Aperture (active, effective, minimum, passive)
      - 3.3.4.1.2 Element pitch, gap, width, and size
  - 3.3.5 Beam and wave forming
    - 3.3.5.1 Sweep range
    - 3.3.5.2 Steering focus power
    - 3.3.5.3 Compensation gain
    - 3.3.5.4 Beam (length and width)
    - 3.3.5.5 Focal depth, depth of field, and focal range
    - 3.3.5.6 Resolution
      - 3.3.5.6.1 Near-surface resolution
      - 3.3.5.6.2 Far-surface resolution
      - 3.3.5.6.3 Lateral and axial resolution
      - 3.3.5.6.4 Angular-surface resolution
  - 3.3.6 Lobes
    - 3.3.6.1 Main lobes
    - 3.3.6.2 Side lobes
    - 3.3.6.3 Grating lobes
    - 3.3.6.4 Grating lobe amplitude
  - 3.3.7 Beam apodization
- 3.4 Wedges
  - 3.4.1 Types of wedge designs
- 3.5 Scanners
  - 3.5.1 Mechanized
  - 3.5.2 Manual

4.0 Testing Techniques

- 4.1 Linear scans
- 4.2 Sectorial scans
- 4.3 Electronic scans

5.0 Standardization

- 5.1 Active element and probe checks
- 5.2 Wedge delay
- 5.3 Velocity
- 5.4 Exit point verifications
- 5.5 Refraction angle verifications
- 5.6 Sensitivity
- 5.7 DAC, TCG, TVG, and ACG variables and parameters
- 5.8 Effects of curvature
- 5.9 Focusing effects
- 5.10 Beam steering
- 5.11 Acquisition gates

6.0 Data Collection

- 6.1 Single probes
- 6.2 Multiple probes
- 6.3 Multiple groups or multiplexing single/multiple probes
- 6.4 Nonencoded scans
  - 6.4.1 Time-based data storage
- 6.5 Encoded scans
  - 6.5.1 Line scans
  - 6.5.2 Raster scans
- 6.6 Zone discrimination
- 6.7 Scan plans and exam coverages
  - 6.7.1 Sectorial
  - 6.7.2 Linear
  - 6.7.3 Electronic raster scans
- 6.8 Probe offsets and indexing

7.0 Procedures

- 7.1 Specific applications
  - 7.1.1 Material evaluations
    - 7.1.1.1 Composites
    - 7.1.1.2 Nonmetallic materials
    - 7.1.1.3 Metallic materials
    - 7.1.1.4 Base-material scan
    - 7.1.1.5 Bar, rod, and rail
    - 7.1.1.6 Forgings
    - 7.1.1.7 Castings
  - 7.1.2 Component evaluations
    - 7.1.2.1 Ease with complex geometries
      - 7.1.2.1.1 Turbines (blades, dovetails, rotors)
      - 7.1.2.1.2 Shafts, keyways, etc.
      - 7.1.2.1.3 Nozzles
      - 7.1.2.1.4 Flanges
    - 7.1.2.2 Geometric limitations
  - 7.1.3 Weld inspections
    - 7.1.3.1 Fabrication/in-service
    - 7.1.3.2 Differences in material – carbon steel, stainless steel, high-temperature nickel-chromium alloy, etc.
    - 7.1.3.3 Review of welding discontinuities
    - 7.1.3.4 Responses from various discontinuities
- 7.2 Data presentations
  - 7.2.1 Standard (A-scan, B-scan, and C-scan)
  - 7.2.2 Other (D-scan, S-scan, etc.)
- 7.3 Data evaluation
  - 7.3.1 Codes, standards, specifications
  - 7.3.2 Flaw characterization
  - 7.3.3 Flaw dimensioning
  - 7.3.4 Geometry
  - 7.3.5 Software tools
  - 7.3.6 Evaluation gates
- 7.4 Reporting
  - 7.4.1 Imaging outputs
  - 7.4.2 Onboard reporting tools
  - 7.4.3 Plotting, ACAD, etc.

Time of Flight Diffraction (TOFD)

1.0 Introduction
- 1.1 Terminology of TOFD
- 1.2 History of TOFD (e.g., M.G. Silk)
- 1.3 Responsibilities of levels of certification

2.0 Basic Principles of TOFD
- 2.1 Review of ultrasonic wave theory, refracted longitudinal waves
- 2.2 Introduction to TOFD concepts and theory
- 2.3 Technique limitations

3.0 Equipment
- 3.1 Computer-based systems
  - 3.1.1 Processors
  - 3.1.2 Control panel including input and output sockets
  - 3.1.3 Block diagram showing basic internal circuit modules
  - 3.1.4 Portable battery-operated versus full computer-based systems
- 3.2 Beam profile tools
  - 3.2.1 PCS calculators for "flat" material/components
  - 3.2.2 PCS calculators for "curved" surfaces
  - 3.2.3 Beam-spread effects and control
  - 3.2.4 Multiple-zone coverage and limitations
- 3.3 Probes
  - 3.3.1 Composite materials
  - 3.3.2 Damping characteristics
  - 3.3.3 Selection of frequency and diameter
- 3.4 Wedges
  - 3.4.1 Incident and refracted-angle selections
  - 3.4.2 High-temperature applications
- 3.5 Scanners
  - 3.5.1 Mechanized
  - 3.5.2 Manual

4.0 Testing Techniques
- 4.1 Line scans (single tandem probe setups)
- 4.2 Line scans (multiple-probe setups)
- 4.3 Raster scans

5.0 Standardization
- 5.1 Material velocity calculations
- 5.2 Combined probe delay(s) calculation(s)
- 5.3 Digitization rates and sampling
- 5.4 Signal averaging
- 5.5 Pulse width control
- 5.6 PCS and angle selection
- 5.7 Sensitivity
- 5.8 Preamplifiers
- 5.9 Effects of curvature

6.0 Data Collection
- 6.1 Single-probe setups
- 6.2 Multiple-probe setups
- 6.3 Nonencoded scans
  - 6.3.1 Time-based data storage
- 6.4 Encoded scans
  - 6.4.1 Line scans
  - 6.4.2 Raster scans
- 6.5 Probe offsets and indexing

7.0 Procedures
- 7.1 Specific applications
  - 7.1.1 Material evaluations
    - 7.1.1.1 Base-material scans
  - 7.1.2 Weld inspections
    - 7.1.2.1 Detection and evaluation of fabrication welding flaws
    - 7.1.2.2 Detection and evaluation of in-service cracking
    - 7.1.2.3 Detection of volumetric loss such as weld root erosion and partial penetration weld dimensional verifications
    - 7.1.2.4 Geometric limitations
    - 7.1.2.5 Cladding thickness and integrity evaluations
  - 7.1.3 Complex geometries
    - 7.1.3.1 Transitions, nozzles, branch connections, tees, saddles, etc.
- 7.2 Data presentations
  - 7.2.1 Standard (A-scan, D-scan)
  - 7.2.2 Other (B-scan, C-scan)
- 7.3 Data evaluation
  - 7.3.1 Codes/standards/specifications
  - 7.3.2 Flaw characterization
  - 7.3.3 Flaw dimensioning
  - 7.3.4 Geometry
  - 7.3.5 Software tools
    - 7.3.5.1 Linearization
    - 7.3.5.2 Lateral/backwall straightening and removal
    - 7.3.5.3 Synthetic aperture focusing technique (SAFT)
    - 7.3.5.4 Spectrum processing
    - 7.3.5.5 Curved surface compensation
  - 7.3.6 Parabolic cursor(s)
- 7.4 Reporting
  - 7.4.1 Imaging outputs
  - 7.4.2 Onboard reporting tools
  - 7.4.3 Plotting, ACAD, etc.

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* Available from The American Society for Nondestructive Testing Inc., Columbus, OH.

Limited Certification for Ultrasonic Digital Thickness Measurement Topical Outline

1.0 Principles/Theory
- 1.1 General
- 1.2 Principles of acoustics
  - 1.2.1 Nature of sound waves
  - 1.2.2 Modes of sound-wave generation
  - 1.2.3 Velocity, frequency, and wavelength of sound waves
  - 1.2.4 Attenuation/scattering of sound waves

2.0 Equipment/Materials
- 2.1 Equipment
  - 2.1.1 Pulse-echo instrumentation
    - 2.1.1.1 Pulse generation
    - 2.1.1.2 Signal detection
    - 2.1.1.3 Display and recording methods, A-scan, B-scan, C-scan, and digital
    - 2.1.1.4 Sensitivity and resolution
  - 2.1.2 Digital thickness instrumentation
  - 2.1.3 Transducer operation and theory
    - 2.1.3.1 Piezoelectric effect
    - 2.1.3.2 Frequency (crystal-thickness relationships)
    - 2.1.3.3 Types (straight, angle, single, dual, etc.)
- 2.2 Materials
  - 2.2.1 Couplants
    - 2.2.1.1 Purpose and principles
    - 2.2.1.2 Material and their efficiency
  - 2.2.2 Reference blocks
  - 2.2.3 Cables/connectors
  - 2.2.4 Test specimen

3.0 Techniques/Standardization – Contact Straight Beam

4.0 Variables Affecting Test Results
- 4.1 Instrument performance variations
- 4.2 Transducer performance variations
- 4.3 Test specimen variations
  - 4.3.1 Surface condition
  - 4.3.2 Part geometry
  - 4.3.3 Material structure

5.0 Procedure/Specification Applications/Thickness Measurement

Limited Certification for Ultrasonic A-scan Thickness Measurement Topical Outline

1.0 Principles/Theory
- 1.1 General
- 1.2 Principles of acoustics
  - 1.2.1 Nature of sound waves
  - 1.2.2 Modes of sound-wave generation
  - 1.2.3 Velocity, frequency, and wavelength of sound waves
  - 1.2.4 Attenuation of sound waves
  - 1.2.5 Acoustic impedance
  - 1.2.6 Reflection

2.0 Equipment/Materials
- 2.1 Equipment
  - 2.1.1 Pulse-echo instrumentation
    - 2.1.1.1 Controls and circuits
    - 2.1.1.2 Pulse generation
    - 2.1.1.3 Signal detection
    - 2.1.1.4 Display and recording methods – A-scan, B-scan, C-scan, and digital
    - 2.1.1.5 Sensitivity and resolution
    - 2.1.1.6 Gates, alarms, and attenuators
    - 2.1.1.7 Basic instrument standardization
    - 2.1.1.8 Reference blocks
  - 2.1.2 Digital thickness instrumentation
  - 2.1.3 Transducer operation and theory
    - 2.1.3.1 Piezoelectric effect
    - 2.1.3.2 Types of crystals
    - 2.1.3.3 Frequency (crystal-thickness relationships)
    - 2.1.3.4 Types (straight, angle, single, dual, etc.)
  - 2.1.4 Resonance testing equipment
    - 2.1.4.1 Thickness testing

2.2 Materials
- 2.2.1 Couplants
  - 2.2.1.1 Purpose and principles
  - 2.2.1.2 Material and their efficiency
- 2.2.2 Reference blocks
- 2.2.3 Cables/connectors
- 2.2.4 Test specimen
- 2.2.5 Miscellaneous materials

3.0 Techniques/Standardization – Contact Straight Beam

3.1 Contact
- 3.1.1 Straight beam
- 3.1.2 Pulse-echo transmission

4.0 Variables Affecting Test Results

4.1 Instrument performance variations

4.2 Transducer performance variations

4.3 Test specimen variations
- 4.3.1 Surface condition
- 4.3.2 Part geometry
- 4.3.3 Material structure

4.4 Personnel variations
- 4.4.1 Skill level in interpretation of results
- 4.4.2 Knowledge level in interpretation of results

5.0 Procedures

5.1 Thickness measurement

VA

VIBRATION ANALYSIS TOPICAL OUTLINES

Vibration Analysis Level I Topical Outline

Basic Vibration Analysis Physics Course

1.0 Introduction
- 1.1 Brief history of NDT/predictive maintenance (PdM) and vibration analysis (VA)
- 1.2 The purpose of VA (condition monitoring)
  - 1.2.1 Industrial Internet of Things (IIoT)
  - 1.2.2 Permanent versus portable monitoring
- 1.3 Basic principles of VA
- 1.4 Basic terminology of VA to include:
  - 1.4.1 Measurement units
  - 1.4.2 Measurement orientation
  - 1.4.3 Hardware
  - 1.4.4 Software and signal processing
  - 1.4.5 Machine components
  - 1.4.6 Data presentation

2.0 Transducers
- 2.1 Types (wired and wireless)
  - 2.1.1 Displacement – proximity
  - 2.1.2 Velocity –seismic
  - 2.1.3 Acceleration – piezo-electro
- 2.2 Applications
- 2.3 Effect on signal with various mounting methods
- 2.4 Benefits and limitations

3.0 Instrumentation
- 3.1 Types
- 3.2 Applications
- 3.3 Limitations

Basic Vibration Analysis Operating Course

1.0 Machinery Basics
- 1.1 Machine types to include:
  - 1.1.1 Prime movers (fixed speed and variable speed)
    - 1.1.1.1 Motors
    - 1.1.1.2 Turbines (steam, water, wind, etc.)
  - 1.1.2 Pumps (positive displacement, centrifugal, multistage, etc.)
  - 1.1.3 Gearboxes (spur, helical, worm, bevel, planetary, etc.)
  - 1.1.4 Air handlers
  - 1.1.5 Compressors
  - 1.1.6 Rolls
- 1.2 Machine components to include:
  - 1.2.1 Bearings
  - 1.2.2 Couplings
  - 1.2.3 Rotors
  - 1.2.4 Gears
  - 1.2.5 Impellers
  - 1.2.6 Belts/chains
- 1.3 Machine orientations

2.0 Data Collection
- 2.1 Sensor location
- 2.2 Check (instrumentation) standardization
- 2.3 Turning speed identification
- 2.4 Data acquisition
  - 2.4.1 Recognize valid data
  - 2.4.2 Recognize abnormal vibration signature
  - 2.4.3 Recognize abnormal conditions
- 2.5 Field diagnostics

3.0 Signal Processing
- 3.1 F_{max} and Nyquist frequency
- 3.2 Spectral resolution
  - 3.2.1 Effect on data collection time
  - 3.2.2 Impact on analysis confidence
- 3.3 Sampling rate
- 3.4 Time block duration
- 3.5 Overlap
- 3.6 Averaging methods
- 3.7 Demodulation techniques
- 3.8 Integration modes (analog versus digital)
- 3.9 Overall modes (analog versus digital)

4.0 Analysis
- 4.1 Time domain characteristics and effect on frequency domain
  - 4.1.1 Sinusoidal
  - 4.1.2 Truncation
  - 4.1.3 Modulation
  - 4.1.4 Bias offset
  - 4.1.5 Settling
  - 4.1.6 Impacting (periodic and random)
  - 4.1.7 Beat
  - 4.1.8 Transient events

- 4.2 Spectral characteristics and sources in time domain
  - 4.2.1 Harmonics
  - 4.2.2 Sidebands
  - 4.2.3 Broadband noise
  - 4.2.4 Ski-slope
  - 4.2.5 Aliasing
- 4.3 Rule-based analysis for common problems
  - 4.3.1 Subsynchronous (belts, flow related, resonance, etc.)
  - 4.3.2 Synchronous (balance, alignment, looseness, coupling, blade pass frequency, etc.)
  - 4.3.3 Nonsynchronous (anti-friction bearings, electrical, resonance, etc.)

5.0 Corrective actions
- 5.1 Balancing (single and multiple plane)
- 5.2 Alignment (hot and cold)

6.0 Safety and Health
- 6.1 When working in close proximity to operating equipment containing a great deal of energy, special care must be taken to avoid injury in addition to using specific personal protective equipment (PPE).
  - 6.1.1 Mechanical
  - 6.1.2 Electrical
  - 6.1.3 Chemical
  - 6.1.4 Environmental

Vibration Analysis Level II Topical Outline

Intermediate Vibration Analysis Physics Course

1.0 Review
- 1.1 Basic principles
- 1.2 Maintenance practices
- 1.3 Reliability concepts
- 1.4 Instrumentation
  - 1.4.1 Handheld
  - 1.4.2 Permanent
  - 1.4.3 Portable multichannel recorders
  - 1.4.4 Cloud based

2.0 Additional Terminology
- 2.1 IIoT
  - 2.1.1 Machine learning
  - 2.1.2 Artificial intelligence
- 2.2 Data requisition
  - 2.2.1 Process data acquisition
  - 2.2.2 Transient capture
  - 2.2.3 Surveillance, continuous, intermittent
  - 2.2.4 Local (e.g., route based)
- 2.3 Signal processing
  - 2.3.1 Dynamic range
  - 2.3.2 Frequency response function
  - 2.3.3 Coherence
  - 2.3.4 Leakage

3.0 Diagnostic Tools
- 3.1 Phase
  - 3.1.1 Cross channel
  - 3.1.2 Peak and phase
- 3.2 Fast Fourier transform (FFT)
- 3.3 Time waveform
- 3.4 Orbit analysis
- 3.5 Bode/Nyquist
- 3.6 Trend analysis
- 3.7 Transient capture/data recording
- 3.8 Modal/operating deflection shape (ODS)

Intermediate Vibration Analysis Techniques Course

1.0 Transducers
- 1.1 Transducer selection
- 1.2 Special transducers
- 1.3 Transducer limitations

2.0 Signal Processing and Data Acquisition
- 2.1 Data acquisition
  - 2.1.1 Aliasing and its prevention
  - 2.1.2 Sampling rate
  - 2.1.3 Lines of resolution (e.g., bins)
  - 2.1.4 Filters
    - 2.1.4.1 High pass
    - 2.1.4.2 Low pass
    - 2.1.4.3 Bandpass
  - 2.1.5 Demodulation
- 2.2 F_{max} considerations
  - 2.2.1 Rotational speed
  - 2.2.2 Shaft components (e.g., bearing type, vanes, rotor bars, etc.)
  - 2.2.3 Electrostatic discharge
  - 2.2.4 Effective frequency range of sensor and mounting method
  - 2.2.5 Correct amplitude units for expected frequency
  - 2.2.6 Check (instrument) standardization
- 2.3 Averaging
  - 2.3.1 Benefit
  - 2.3.2 Effect on collection time
  - 2.3.3 Application of:
    - 2.3.3.1 Normal (linear) averaging
    - 2.3.3.2 Order tracking
    - 2.3.3.3 Synchronous time averaging
    - 2.3.3.4 Peak-hold averaging
    - 2.3.3.5 Exponential averaging
  - 2.3.4 Overlap

- 2.4 Windows and weighting
  - 2.4.1 Leakage
  - 2.4.2 Uniform, hanning, flat-top, force-exponential
  - 2.4.3 Selection of proper window
- 2.5 Analog/digital integration
  - 2.5.1 Time domain versus frequency domain
  - 2.5.3 Overall versus FFT frequency range
- 2.6 Alarming
  - 2.6.1 Codes, standards, and specifications
  - 2.6.2 Overall and crest factor
  - 2.6.3 Narrow-band parameters
  - 2.6.4 Enveloped parameters
  - 2.6.5 Statistical
- 2.7 Monitoring interval
  - 2.7.1 Route based
  - 2.7.2 Online systems

3.0 Data Presentation

- 3.1 Scope and limitations of different testing methods
- 3.2 Waterfall/cascades
- 3.3 Linear versus logarithmic
- 3.4 Trend analysis
- 3.5 Conversion of amplitude units
- 3.6 True zoom versus expansion
- 3.7 Absolute frequency versus order-based frequency

4.0 Problem Identification

- 4.1 Unbalance
- 4.2 Misalignment
- 4.3 Resonance
  - 4.3.1 Single-channel identification
  - 4.3.2 Two-channel identification
- 4.4 Bearing defects
- 4.5 Looseness
- 4.6 Bent shafts
- 4.7 Gear defects
- 4.8 Electrical defects
- 4.9 Hydraulic/flow dynamics
- 4.10 Rubs
- 4.11 Belts and couplings
- 4.12 Eccentricity

5.0 Reporting Methodology

- 5.1 Technical reports
- 5.2 Management-oriented reports
- 5.3 Oral reports

6.0 Safety and Health

When working in close proximity to operating equipment containing a great deal of energy, special care must be taken to avoid injury in addition to using specific personal protective equipment.

- 6.1 Mechanical
- 6.2 Electrical
- 6.3 Chemical
- 6.4 Environmental

Vibration Analysis Level III Topical Outline

The principles and theory section, or any other section, is not intended to be covered as a completely separate section. This category just means that somewhere in the material for training it is necessary to cover the basic theory and principles on those topics.

1.0 Principles/Theory

The vibration data provides detailed information about the condition of a machine and its components. Data can be processed and presented in different ways to help the analyst in diagnosing specific problems. The section on principles and theory provides the concepts of VA.

- 1.1 Physical concepts
  - 1.1.1 Sources of vibration
  - 1.1.2 Stiffness
  - 1.1.3 Mass
  - 1.1.4 Damping
  - 1.1.5 Phase
  - 1.1.6 Modes of vibration
  - 1.1.7 Resonance
- 1.2 Data presentation
  - 1.2.1 Spectrum (amplitude and frequency units)
  - 1.2.2 Waveform (amplitude and time units)
  - 1.2.3 Phase, coherence, and Nyquist
- 1.3 Sources of vibration
  - 1.3.1 Reciprocating machinery analysis
  - 1.3.2 Specialty machine concepts
    - 1.3.2.1 Nonlinear behavior
- 1.4 Correction methods
  - 1.4.1 Absorbers
  - 1.4.2 Damping treatments
  - 1.4.3 Changing mass
  - 1.4.4 Changing stiffness
  - 1.4.5 Changing operating speed

2.0 Equipment

- 2.1 Sensors
  - 2.1.1 Attachments (i.e., brackets, connectors, sensor mounting)
  - 2.1.2 Cabling
- 2.2 Signal conditioning
  - 2.2.1 Averaging methods
  - 2.2.2 Windows and weighting
  - 2.2.3 Triggering
  - 2.2.4 Spectral and time-domain resolution
- 2.3 Instruments
  - 2.3.1 Portable, route-based data collector
  - 2.3.2 Online surveillance data collector
  - 2.3.3 Unfiltered meter (e.g., vibration pen)
  - 2.3.4 Multichannel transient data recorder
- 2.4 Equipment response to environments (performance based)
  - 2.4.1 Temperature gradients
  - 2.4.2 Moisture

3.0 Techniques/Standardization

3.1 Standardization
- 3.1.1 Point sensor standardization
- 3.1.2 Instrument standardization
- 3.1.3 Test instrument standardization

3.2 Measurement and techniques
- 3.2.1 Order tracking
- 3.2.2 Time synchronous averaging
- 3.2.3 Cross channel measurements
- 3.2.4 Transient analysis
- 3.2.5 Modal analysis fundamentals
- 3.2.6 Operating deflection shape analysis
  - 3.2.6.1 Frequency domain
  - 3.2.6.2 Time domain
- 3.2.7 Natural frequency tests
- 3.2.8 Torsional vibration techniques
- 3.2.9 Specialized VA techniques (high-frequency detection, demodulated spectrum, spike-energy spectrum, negative averaging, shock pulse, etc.)

3.3 Vibration correction techniques
- 3.3.1 Change mass, stiffness, and/or damping
- 3.3.2 Alignment
- 3.3.3 Clearances on journal bearings
- 3.3.4 Correct beats
- 3.3.5 Static/dynamic balancing
- 3.3.6 Dynamic absorber
- 3.3.7 Eliminate looseness
- 3.3.8 Isolation treatments
- 3.3.9 Speed change

4.0 Analysis/Evaluation

Ability to analyze test data, perform an evaluation, and recommend remedial action

4.1 Equipment considerations
- 4.1.1 Low-speed (<600 rpm)
- 4.1.2 High-speed (>4000 rpm)
- 4.1.3 Variable-speed drive (electrical)
- 4.1.4 Variable-speed drive (i.e., gas or steam turbine)
- 4.1.5 Reciprocating machinery
- 4.1.6 Gear type (e.g., spur, helical, worm, epicyclic, etc.)
- 4.1.7 Pump type (e.g., centrifugal, positive displacement, etc.)
- 4.1.8 Bearing type (e.g., fluid film, anti-friction, etc.)
- 4.1.9 Process variables

4.2 Data analysis
- 4.2.1 Operational effects
- 4.2.2 Correlation of test data
- 4.2.3 Transient analysis
- 4.2.4 In-depth time waveform analysis
- 4.2.5 Cross-channel analysis
- 4.2.6 Multichannel analysis
- 4.2.7 Machinery-specific analysis

4.3 Data evaluation
- 4.3.1 Evaluation of data to standards/codes
- 4.3.2 Specifications or acceptance criteria
- 4.3.3 Failure mode and effects analysis
- 4.3.4 Root cause analysis
- 4.3.5 Cost justification or return on investment analysis

4.4 Corrective actions
- 4.4.1 Engineering study [e.g., finite element analysis (FEA)] to determine design changes
- 4.4.2 Proactive repair to address root cause failure analysis (RCFA)
- 4.4.3 Balancing
- 4.4.4 Alignment
- 4.4.5 Component replacement/adjustment
- 4.4.6 Variable frequency drive tuning
- 4.4.7 Process changes (flow, temperature, speed, etc.)

5.0 Procedures

- 5.1 Develop procedures for performing the various types of testing techniques needed to determine equipment condition
- 5.2 Create and maintain databases used to collect appropriate CM data

6.0 Safety and Health

When working in close proximity to operating equipment containing a great deal of energy, special care must be taken to avoid injury in addition to using specific personal protective equipment.
- 6.1 Mechanical
- 6.2 Electrical
- 6.3 Chemical
- 6.4 Environmental/chemical

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VT
VISUAL TESTING TOPICAL OUTLINES

Visual Testing Level I Topical Outline

Note: The guidelines listed below should be implemented using equipment and procedures relevant to the employer's industry. No times are given for a specific subject; this should be specified in the employer's written practice. Based upon the employer's product, not all of the referenced subcategories need apply.

1.0 Introduction
- 1.1 Definition of visual testing (VT)
- 1.2 History of VT
- 1.3 Overview of VT applications

2.0 Definitions

Standard terms and their meanings in the employer's industry

3.0 Fundamentals
- 3.1 Vision
- 3.2 Lighting
- 3.3 Material attributes
- 3.4 Environmental factors
- 3.5 Visual perception
- 3.6 Direct and indirect methods

4.0 Equipment (as applicable)
- 4.1 Mirrors
- 4.2 Magnifiers
- 4.3 Borescopes
- 4.4 Fiberscopes
- 4.5 Video borescopes
- 4.6 Remote visual inspection systems
- 4.7 Light sources and special lighting
- 4.8 Gauges (welding, go/no-go, etc.) templates, scales, micrometers, calipers, special tools, etc.
- 4.9 Automated systems
- 4.10 Computer-enhanced systems

5.0 Employer-Defined Applications

(Includes a description of inherent, processing, and service-induced discontinuities.)
- 5.1 Mineral-based material
- 5.2 Metallic materials, including welds
- 5.3 Organic-based materials
- 5.4 Other materials (employer defined)

6.0 VT to Specific Procedures
- 6.1 Selection of parameters
  - 6.1.1 Inspection objectives
  - 6.1.2 Inspection checkpoints
  - 6.1.3 Sampling plans
  - 6.1.4 Inspection patterns
  - 6.1.5 Documented procedures
- 6.2 Test standards/standardization
- 6.3 Classification of indications per acceptance criteria
- 6.4 Reports and documentation

Visual Testing Level II Topical Outline

The guidelines listed below should be implemented using equipment and procedures relevant to the employer's industry. The employer should tailor the program to the company's particular application area. Discontinuity cause, appearance, and how to best visually detect and identify these discontinuities should be emphasized. No times are given for a specific subject; this should be specified in the employer's written practice. Depending upon the employer's product, not all the referenced subcategories need apply.

A Level I class should precede every Level II class. When a Level II certification is sought in one step, the Level I topics shall be blended into this Level II topical outline. The combined outline must be reviewed and accepted by the responsible Level III.

1.0 Purpose and Scope of VT
- 1.1 Scope
  - 1.1.1 Internal quality control
  - 1.1.2 Quality control by the customer
  - 1.1.3 Quality control by authorities
- 1.2 VT looks for:
  - 1.2.1 Discontinuities
  - 1.2.2 Shape and geometry deviations
  - 1.2.3 Surface finish
- 1.3 Time of application
  - 1.3.1 During manufacture
  - 1.3.2 In service

2.0 Elements of Vision
- 2.1 Mechanics of vision
- 2.2 Adaptation and accommodation
- 2.3 Vision limitations
  - 2.3.1 Perception and environmental conditions
  - 2.3.2 Orientation, visual angle, and distance
  - 2.3.3 Ophthalmic disorders

- 2.3.4 Mental attitude and fatigue
- 2.3.5 Physiology and health
- 2.4 Vision acuity examination and charts
  - 2.4.1 Vision acuity tests
  - 2.4.2 Color and gray shade differentiation

3.0 Elements of Lighting
- 3.1 Fundamentals of light
- 3.2 Light sources
  - 3.2.1 Incandescent radiators
  - 3.2.2 Luminescent radiators (fluorescent light, high-intensity discharge light, light-emitting diodes, lasers)
- 3.3 Adequate light levels
- 3.4 Glare and fatigue
- 3.5 General lighting requirements

4.0 Contrast and Resolution
- 4.1 Reflection at smooth and rough/textured surfaces
- 4.2 Law of illumination
- 4.3 Reflectivity and luminance
- 4.4 Luminous contrast
- 4.5 Influence of cleanliness on contrast
- 4.6 Dark-field contrast
- 4.7 Colors and contribution of colors to contrast
- 4.8 Surface geometry and contrast

5.0 Optics
- 5.1 Transmission of light through solid and liquid media
- 5.2 Refraction of light
- 5.3 Refractive indexes of glasses
- 5.4 How prisms change the direction of light
- 5.5 How lenses focus and disperse light
- 5.6 Lens optics and lens trains
- 5.7 Fiber optics and fiber bundles
- 5.8 Digitization and digital technology

6.0 VT Equipment
- 6.1 Generic tools such as magnifiers and mirrors
- 6.2 Rigid borescopes
- 6.3 Fiberscopes
- 6.4 Video borescopes
- 6.5 Video borescopes measurement techniques
  - 6.5.1 Comparison technique
  - 6.5.2 Shadow technique
  - 6.5.3 Stereo technique
  - 6.5.4 Laser-based measurements
- 6.6 Specialized inspection systems
  - 6.6.1 Push-tube cameras
  - 6.6.2 Pipe crawler camera systems
  - 6.6.3 Subsea remote camera systems

7.0 Visual Appearance of Discontinuities
- 7.1 Primary manufacturing discontinuities (e.g., castings)
- 7.2 Secondary manufacturing discontinuities (e.g., forgings)
- 7.3 Service-induced discontinuities
  - 7.3.1 Caused by mechanical loads (e.g., fatigue)
  - 7.3.2 Caused by thermal loads (e.g., thermal shock)
  - 7.3.3 Caused by corrosion (e.g., pitting)
  - 7.3.4 Caused by abrasive wear (e.g., erosion)
- 7.4 Inherent discontinuities

8.0 Evaluation and Reporting
- 8.1 General evaluation scheme (ASTM E1316)
- 8.2 Evaluation criteria
  - 8.2.1 Verbal descriptions
  - 8.2.2 Comparison standards or catalogs (e.g., photos/replica)
  - 8.2.3 Size-based criteria (measures)
  - 8.2.4 Mixtures of the above
- 8.3 Evaluation techniques
  - 8.3.1 Visual-tactile recognition
  - 8.3.2 Grading by comparison with a standard
  - 8.3.3 Measurement
- 8.4 Reporting and documentation
  - 8.4.1 Technique reports
  - 8.4.2 Data reports
  - 8.4.3 Image reports (sketches, hard-copy photo, or digital photo)
- 8.5 Completion of testing confirmed with a checklist

9.0 Codes, Standards, and Specifications
- 9.1 VT as an engineering task
- 9.2 VT as a technician task
- 9.3 US standards (e.g., ASME BPVC, AWS D1.1)
- 9.4 European standards (based on PED)
- 9.5 ISO standards

10.0 Employer-specific Topics
- 10.1 Applications and techniques
- 10.2 Specifications
- 10.3 Lighting techniques
- 10.4 Materials tested
- 10.5 Special evaluation criteria
- 10.6 Safety rules

11.0 Glossary
- 11.1 Refer to Chapter 13 of ASNT, 2010, *Nondestructive Testing Handbook, Vol. 9: Visual Testing*, third edition
- 11.2 Topics may be deleted if the VT is only required to perform direct visual inspection

Visual Testing Level III Topical Outline

1.0 Principles/Theory
- 1.1 Vision and light
  - 1.1.1 Physiology of sight
  - 1.1.2 Visual acuity
  - 1.1.3 Visual angle and distance
  - 1.1.4 Color vision
  - 1.1.5 Physics and measurement of light
- 1.2 Environmental factors
  - 1.2.1 Lighting
  - 1.2.2 Cleanliness
  - 1.2.3 Distance
  - 1.2.4 Air contamination
- 1.3 Test object characteristics
  - 1.3.1 Texture
  - 1.3.2 Color
  - 1.3.3 Cleanliness
  - 1.3.4 Geometry

2.0 Equipment Accessories
- 2.1 Magnifiers
- 2.2 Mirrors
- 2.3 Dimensional
  - 2.3.1 Linear measurement
  - 2.3.2 Micrometers/calipers
  - 2.3.3 Optical comparators
  - 2.3.4 Dial indicators
  - 2.3.5 Gauges
- 2.4 Borescopes
  - 2.4.1 Rigid
  - 2.4.2 Fiber-optic
  - 2.4.3 Special purpose
- 2.5 Video systems (robotics)
  - 2.5.1 Photoelectric devices
  - 2.5.2 Microscopy
  - 2.5.3 Video borescopes
  - 2.5.4 Video imaging/resolution/image processing (enhancement)
  - 2.5.5 Charge-coupled devices (CCDs)
- 2.6 Automated systems
  - 2.6.1 Lighting techniques
  - 2.6.2 Optical filtering
  - 2.6.3 Image sensors
  - 2.6.4 Signal processing
- 2.7 Video technologies
- 2.8 Machine vision
- 2.9 Replication
- 2.10 Surface comparators
- 2.11 Chemical aids
- 2.12 Photography
- 2.13 Eye

3.0 Techniques/Standardization
- 3.1 Diagrams and drawings
- 3.2 Raw materials
  - 3.2.1 Ingots
  - 3.2.2 Blooms/billets/slabs
- 3.3 Primary process materials
  - 3.3.1 Plates/sheets
  - 3.3.2 Forgings
  - 3.3.3 Castings
  - 3.3.4 Bars
  - 3.3.5 Tubing
  - 3.3.6 Extrusions
  - 3.3.7 Wire
- 3.4 Joining processes
  - 3.4.1 Joint configuration
  - 3.4.2 Welding
  - 3.4.3 Brazing
  - 3.4.4 Soldering
  - 3.4.5 Bonding
- 3.5 Fabricated components
  - 3.5.1 Pressure vessels
  - 3.5.2 Pumps
  - 3.5.3 Valves
  - 3.5.4 Fasteners
  - 3.5.5 Piping systems
- 3.6 In-service materials
  - 3.6.1 Wear
  - 3.6.2 Corrosion/erosion
  - 3.6.3 Microscopy
- 3.7 Coatings
  - 3.7.1 Paint
  - 3.7.2 Insulation
  - 3.7.3 Cathodic protection (conversion coatings)
  - 3.7.4 Anodizing
- 3.8 Other applications
  - 3.8.1 Ceramics
  - 3.8.2 Composites
  - 3.8.3 Glasses
  - 3.8.4 Plastics
  - 3.8.5 Bearings
- 3.9 Requirements
  - 3.9.1 Codes
  - 3.9.2 Standards
  - 3.9.3 Specifications
  - 3.9.4 Techniques (direct, indirect, video, etc.)
  - 3.9.5 Personnel qualification and certification

4.0 Interpretation/Evaluation

4.1 Equipment variables affecting test results including type and intensity of light
4.2 Material variables affecting test results including the variations of surface finish
4.3 Discontinuity variables affecting test results
4.4 Determination of dimensions (e.g., depth, width, length, etc.)
4.5 Sampling/scanning procedure variables affecting test results
4.6 Process for reporting visual discontinuities
4.7 Personnel (human factors) variables affecting test results
4.8 Detection
    4.8.1 Interpretation
    4.8.2 Evaluation

5.0 Procedures and Documentation

5.1 Hard copy – general applications
    5.1.1 Mineral-based materials
    5.1.2 Organic-based materials
    5.1.3 Composite materials
    5.1.4 Metallic materials
5.2 Photography – specific applications
    5.2.1 Metal joining processes
    5.2.2 Pressure vessels
    5.2.3 Pumps
    5.2.4 Valves
    5.2.5 Bolting
    5.2.6 Castings
    5.2.7 Forgings
    5.2.8 Extrusions
    5.2.9 Microcircuits
5.3 Audio/video – requirements
    5.3.1 Codes (AWS, ASME, etc.)
    5.3.2 Standards (ASTM, NAVSEA, MIL-STD, etc.)
    5.3.3 Specifications
    5.3.4 Procedures (Level III exam specific)
5.4 Electronic and magnetic media
5.5 Personnel qualification and certification

6.0 Safety and Health

6.1 Electrical shock
6.2 Mechanical hazards
6.3 Lighting hazards physiological deleterious effects of light
6.4 Chemicals contamination
6.5 Radioactive materials
6.6 Explosive environments

VISUAL TESTING LEVEL I, II, AND III TRAINING REFERENCES

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BASIC

BASIC EXAMINATION TOPICAL OUTLINES

Basic Examination Level III Topical Outline

The Basic examination will cover three main topical areas:

1.0 Personnel Qualification and Certification Programs Covering Recommended Practice No. SNT-TC-1A, ANSI/ASNT-CP-189, and the ASNT NDT Level III Program
2.0 General Familiarity with Other NDT Methods, Covering the 11 NDT Test Methods Listed in this Section, and;
3.0 General Knowledge of Materials, Fabrication, and Product Technology

The above topics are further subdivided into topical outlines below, followed by the reference materials used in the development of these outlines and sample questions typical of those in the examinations.

1.0 Personnel Qualification and Certification Programs
- 1.1 Recommended Practice No. SNT-TC-1A
  - 1.1.1 Scope
  - 1.1.2 Definitions
  - 1.1.3 Nondestructive testing methods
  - 1.1.4 Levels of qualification
  - 1.1.5 Written practice
  - 1.1.6 Education, training, and experience for initial qualification
  - 1.1.7 Training programs
  - 1.1.8 Examinations
  - 1.1.9 Certification
  - 1.1.10 Technical performance evaluation
  - 1.1.11 Interrupted service
  - 1.1.12 Recertification
  - 1.1.13 Termination
  - 1.1.14 Reinstatement
  - 1.1.15 Referenced publications
- 1.2 ASNT Standard ANSI/ASNT-CP-189
  - 1.2.1 Scope
  - 1.2.2 Definitions
  - 1.2.3 Levels of qualification
  - 1.2.4 Qualification requirements
  - 1.2.5 Qualification and certification
  - 1.2.6 Examinations
  - 1.2.7 Expiration, suspension, revocation, and reinstatement of employer certification
  - 1.2.8 Employer recertification
  - 1.2.9 Records
  - 1.2.10 Referenced publications
- 1.3 NDT Level III certification program
  - 1.3.1 Scope
  - 1.3.2 Definitions
  - 1.3.3 Certification outcome
  - 1.3.4 Eligibility for examination
  - 1.3.5 Qualification examinations
  - 1.3.6 Examinations results
  - 1.3.7 Certification
  - 1.3.8 Validity
  - 1.3.9 Recertification

2.0 General Familiarity with Other NDT Methods
- 2.1 Acoustic emission testing (AE)
  - 2.1.1 Fundamentals
    - 2.1.1.1 Principles/theory of AE
    - 2.1.1.2 Sources of acoustic emissions
    - 2.1.1.3 Equipment and material
  - 2.1.2 Proper selection of acoustic emission technique
    - 2.1.2.1 Instrumentation and signal processing
    - 2.1.2.2 Cables (types)
    - 2.1.2.3 Signal conditioning
    - 2.1.2.4 Signal detection
    - 2.1.2.5 Noise discrimination
    - 2.1.2.6 Electronic technique
    - 2.1.2.7 Attenuation materials
    - 2.1.2.8 Data-filtering techniques
  - 2.1.3 Interpretation and evaluation of test results
- 2.2 Electromagnetic testing (ET)
  - 2.2.1 Sensors
  - 2.2.2 Basic types of equipment, types of readout
  - 2.2.3 Reference standards
  - 2.2.4 Applications and test result interpretation
    - 2.2.4.1 Flaw detection
    - 2.2.4.2 Conductivity and permeability sorting
    - 2.2.4.3 Thickness gauging
    - 2.2.4.4 Process control
- 2.3 Leak testing (LT)
  - 2.3.1 Fundamentals
    - 2.3.1.1 Bubble leak testing
    - 2.3.1.2 Pressure leak testing
    - 2.3.1.3 Halogen detector leak testing
    - 2.3.1.4 Mass spectrometer leak testing

2.3.2 LT procedures and techniques
- 2.3.2.1 System factors
- 2.3.2.2 Relative sensitivity
- 2.3.2.3 Evacuated systems
- 2.3.2.4 Pressurized systems – ambient fluids, tracer fluids
- 2.3.2.5 Locating leaks
- 2.3.2.6 Standardization

2.3.3 Test result interpretation
2.3.4 Essentials of safety
2.3.5 Test equipment
2.3.6 Applications
- 2.3.6.1 Piping and pressure vessels
- 2.3.6.2 Evacuated systems
- 2.3.6.3 Low-pressure fluid containment vessels, pipes, and tubing
- 2.3.6.4 Hermetic seals
- 2.3.6.5 Electrical and electronic components

2.4 Liquid penetrant testing (PT)
2.4.1 Fundamentals
- 2.4.1.1 Interaction of penetrants and discontinuity openings
- 2.4.1.2 Fluorescence and contrast

2.4.2 PT
- 2.4.2.1 Penetrant processes
- 2.4.2.2 Test equipment and systems factors
- 2.4.2.3 Test result interpretation, discontinuity indications
- 2.4.2.4 Applications
  - 2.4.2.4.1 Castings
  - 2.4.2.4.2 Welds
  - 2.4.2.4.3 Wrought metals
  - 2.4.2.4.4 Machined parts
  - 2.4.2.4.5 Leaks
  - 2.4.2.4.6 Field inspections

2.5 Magnetic particle testing (MT)
2.5.1 Fundamentals
- 2.5.1.1 Magnetic field principles
- 2.5.1.2 Magnetization by means of electric current
- 2.5.1.3 Demagnetization

2.5.2 MT
- 2.5.2.1 Basic types of equipment and inspection materials
- 2.5.2.2 Test results interpretation, discontinuity indications
- 2.5.2.3 Applications
  - 2.5.2.3.1 Welds
  - 2.5.2.3.2 Castings
  - 2.5.2.3.3 Wrought metals
  - 2.5.2.3.4 Machined parts
  - 2.5.2.3.5 Field applications

2.6 Neutron radiographic testing (NR)
2.6.1 Fundamentals
- 2.6.1.1 Sources
  - 2.6.1.1.1 Isotopic
  - 2.6.1.1.2 Neutron
- 2.6.1.2 Detectors
  - 2.6.1.2.1 Imaging
  - 2.6.1.2.2 Nonimaging
- 2.6.1.3 Nature of penetrating radiation and interactions with matter
- 2.6.1.4 Essentials of safety

2.6.2 NR
- 2.6.2.1 Basic imaging considerations
- 2.6.2.2 Test result interpretation, discontinuity indications
- 2.6.2.3 Systems factors (source/test object/detector interactions)
- 2.6.2.4 Applications
  - 2.6.2.4.1 Explosives and pyrotechnic devices
  - 2.6.2.4.2 Assembled components
  - 2.6.2.4.3 Bonded components
  - 2.6.2.4.4 Corrosion detection
  - 2.6.2.4.5 Nonmetallic materials

2.7 Radiographic testing (RT)
2.7.1 Fundamentals
- 2.7.1.1 Sources
- 2.7.1.2 Detectors
  - 2.7.1.2.1 Imaging
  - 2.7.1.2.2 Nonimaging
- 2.7.1.3 Nature of penetrating radiation and interactions with matter
- 2.7.1.4 Essentials of safety

2.7.2 RT
- 2.7.2.1 Basic imaging considerations
- 2.7.2.2 Test result interpretation, discontinuity indications
- 2.7.2.3 Systems factors (source/test object/detector interactions)
- 2.7.2.4 Applications
  - 2.7.2.4.1 Castings
  - 2.7.2.4.2 Welds
  - 2.7.2.4.3 Assemblies
  - 2.7.2.4.4 Electronic components
  - 2.7.2.4.5 Field inspections

2.8 Thermal/infrared testing (IR)
2.8.1 Fundamentals
- 2.8.1.1 Principles and theory of IR
- 2.8.1.2 Temperature measurement principles
- 2.8.1.3 Proper selection of IR technique

2.8.2 Equipment/materials
- 2.8.2.1 Temperature measurement equipment
- 2.8.2.2 Heat flux indicators
- 2.8.2.3 Noncontact devices
- 2.8.2.4 Contact temperature indicators

2.8.2.5 Noncontact pyrometers
2.8.2.6 Line scanners
2.8.2.7 Thermal imaging
2.8.3 Applications
2.8.3.1 Exothermic or endothermic investigations
2.8.3.2 Friction investigations
2.8.3.3 Fluid flow investigations
2.8.3.4 Thermal resistance investigations
2.8.3.5 Thermal capacitance investigations
2.8.4 Interpretation and evaluation
2.9 Ultrasonic testing (UT)
2.9.1 Fundamentals
2.9.1.1 Wave propagation
2.9.1.1.1 Sound fields
2.9.1.1.2 Wave travel modes
2.9.1.1.3 Refraction, reflection, scattering, and attenuation
2.9.1.2 Transducers and sound beam coupling
2.9.2 UT
2.9.2.1 Basic types of equipment
2.9.2.2 Reference standards
2.9.2.3 Test result interpretation, discontinuity indications
2.9.2.4 System factors
2.9.2.5 Applications
2.9.2.5.1 Flaw detection and evaluation
2.9.2.5.2 Thickness measurement
2.9.2.5.3 Bond evaluation
2.9.2.5.4 Process control
2.9.2.5.5 Castings
2.9.2.5.6 Weldments
2.10 Visual testing (VT)
2.10.1 Fundamentals
2.10.1.1 Principles and theory of VT
2.10.1.2 Selection of correct visual technique
2.10.1.3 Equipment and materials
2.10.2 Specific applications
2.10.2.1 Metal joining processes
2.10.2.2 Pressure vessels
2.10.2.3 Pumps
2.10.2.4 Valves
2.10.2.5 Bolting
2.10.2.6 Castings
2.10.2.7 Forgings
2.10.2.8 Extrusions
2.10.2.9 Microcircuits
2.10.3 Interpretation and evaluation
2.10.3.1 Codes and standards
2.10.3.2 Environmental factors
2.11 Magnetic flux leakage testing (MFL)
2.11.1 Fundamentals
2.11.1.1 Magnetic field principles
2.11.1.2 Magnetization by means of electric current
2.11.1.3 Flux leakage
2.11.2 MFL
2.11.2.1 Basic types of equipment and inspection materials
2.11.2.2 Types of discontinuities found by MFL
2.11.2.3 Sensors used in MFL
2.11.3 Applications
2.11.3.1 Wire rope inspection
2.11.3.2 Pipe body inspection
2.11.3.3 Tank floor/steel plate inspection

3.0 Basic Materials, Fabrication, and Product Technology
3.1 Fundamentals of material technology
3.1.1 Properties of materials
3.1.1.1 Strength and elastic properties
3.1.1.2 Physical properties
3.1.1.3 Material properties testing
3.1.2 Origin of discontinuities and failure modes
3.1.2.1 Inherent discontinuities
3.1.2.2 Process-induced discontinuities
3.1.2.3 Service-induced discontinuities
3.1.2.4 Failures in metallic materials
3.1.2.5 Failures in nonmetallic materials
3.1.3 Statistical nature of detecting and characterizing discontinuities
3.2 Fundamentals of fabrication and product technology
3.2.1 Raw materials processing
3.2.2 Metals processing
3.2.2.1 Primary metals
3.2.2.1.1 Metal ingot production
3.2.2.1.2 Wrought primary metals
3.2.2.2 Castings
3.2.2.2.1 Green sand molded
3.2.2.2.2 Metal molded
3.2.2.2.3 Investment molded
3.2.2.3 Welding
3.2.2.3.1 Common processes
3.2.2.3.2 Hard-surfacing
3.2.2.3.3 Solid-state
3.2.2.4 Brazing
3.2.2.5 Soldering
3.2.2.6 Machining and material removal
3.2.2.6.1 Turning, boring, and drilling
3.2.2.6.2 Milling
3.2.2.6.3 Grinding
3.2.2.6.4 Electrochemical
3.2.2.6.5 Chemical
3.2.2.7 Forming
3.2.2.7.1 Cold-working processes
3.2.2.7.2 Hot-working processes

3.2.2.8 Powdered metal processes
3.2.2.9 Heat treatment
3.2.2.10 Surface finishing and corrosion protection
3.2.2.10.1 Shot peening and grit blasting
3.2.2.10.2 Painting
3.2.2.10.3 Plating
3.2.2.10.4 Chemical conversion coatings
3.2.2.11 Adhesive joining
3.2.3 Nonmetals and composite materials processing
3.2.3.1 Basic materials processing and process control
3.2.3.2 Nonmetals and composites fabrication
3.2.3.3 Adhesive joining
3.2.4 Dimensional metrology
3.2.4.1 Fundamental units and standards
3.2.4.2 Gauging
3.2.4.3 Interferometry

BASIC EXAMINATION TRAINING REFERENCES

PERSONNEL QUALIFICATION AND CERTIFICATION PROGRAMS

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APPENDIX A
RADIOGRAPHIC SAFETY

Appendix A: Radiographic Safety Operations and Emergency Instructions Course

Note: This outline provides radiation safety subject matter that applies to multiple types of penetrating radiation. Instructors should consider the applicable type of radiation source and delivery system to be covered and tailor their use of these subjects accordingly.

1.0 Personnel Safety and Radiation Protection
- 1.1 Hazards of excessive exposure
  - 1.1.1 General: alpha-, beta-, gamma, neutron, and X-radiation
    - 1.1.1.1 Alpha particles
    - 1.1.1.2 Beta particles
    - 1.1.1.3 X-radiation
    - 1.1.1.4 Gamma radiation
  - 1.1.2 Specific neutron hazards
    - 1.1.2.1 Relative biological effectiveness
    - 1.1.2.2 Neutron activation
- 1.2 Methods of controlling radiation dose
  - 1.2.1 Time
  - 1.2.2 Distance
  - 1.2.3 Shielding
    - 1.2.3.1 Half-value layers
    - 1.2.3.2 Tenth-value layers
  - 1.2.4 Exposure shields and/or exposure rooms
    - 1.2.4.1 Operation
    - 1.2.4.2 Alarms
- 1.3 Personnel monitoring
  - 1.3.1 Difference between dose and dose rate
    - 1.3.1.1 Coulomb per kilogram (C/kg)
    - 1.3.1.2 Gray (Gy)
    - 1.3.1.3 Sievert (Sv)
  - 1.3.2 Wearing of monitoring badges
    - 1.3.2.1 Pocket dosimeters
      - 1.3.2.1.1 Neutron monitoring dosimeters
      - 1.3.2.1.2 Gamma and X-ray dosimeters
    - 1.3.2.2 Film badges
    - 1.3.2.3 Thermoluminescent detectors (TLDs)
  - 1.3.3 Reading of pocket dosimeters
  - 1.3.4 Recording of daily dosimeter readings
  - 1.3.5 "Off-scale" dosimeter – required activity
  - 1.3.6 Permissible exposure limits
  - 1.3.7 As low as reasonably achievable (ALARA) concept

2.0 Radiation Survey Instruments
- 2.1 Types of radiation instruments
  - 2.1.1 Geiger-Müller tube
  - 2.1.2 Ionization chambers
  - 2.1.3 Scintillation chambers, counters
- 2.2 Neutron radiation survey equipment
- 2.3 Reading and interpreting meter indications
- 2.4 Calibration frequency
- 2.5 Calibration expiration – action to be taken
- 2.6 Battery check – importance

3.0 Radiation-Area Surveys
- 3.1 Type and quantity of radiation
- 3.2 Establishment of restricted areas
- 3.3 Posting and surveillance of restricted areas
  - 3.3.1 Radiation areas
  - 3.3.2 High radiation areas
- 3.4 Use of time, distance, and shielding to reduce personnel radiation exposure
- 3.5 Applicable regulatory requirements for surveys, posting, control of radiation, and high radiation areas
- 3.6 Establishment of time limits

4.0 Radiation Survey Reports
- 4.1 Requirements for completion
- 4.2 Description of report format

5.0 Neutron Radiographic (NR) Work Practices
- 5.1 Radioactive contamination
  - 5.1.1 Clothing requirements
  - 5.1.2 Contamination control
  - 5.1.3 Contamination cleanup
- 5.2 Operation and emergency procedures
- 5.3 Specific procedures

6.0 NR Explosive-Device Safety[**]
- 6.1 Static electricity
- 6.2 Grounding devices
- 6.3 Clothing requirements
- 6.4 Handling and storage requirements and procedures
- 6.5 Shipping and receiving procedures
- 6.6 State and federal explosive-licensing requirements

7.0 Safety and Health

- 7.1 Radiation hazards
  - 7.1.1 Exposure hazards
  - 7.1.2 Methods of controlling radiation exposure
  - 7.1.3 Operation and emergency procedures

8.0 Biological Effects of Radiation

- 8.1 "Natural" background radiation
- 8.2 Unit of radiation dose – sievert (Sv)
- 8.3 Difference between radiation and contamination
- 8.4 Radiation damage – repair concept
- 8.5 Symptoms of radiation injury
- 8.6 Acute radiation exposure and somatic injury
- 8.7 Personnel monitoring for tracking exposure
- 8.8 Organ radiosensitivity

9.0 Exposure Devices

- 9.1 Daily inspection and maintenance
- 9.2* Radiation exposure limits for gamma ray exposure devices
- 9.3 Labeling
- 9.4 Use
- 9.5 Use of collimators to reduce personnel exposure
- 9.6* Use of source changers for gamma ray sources

10.0 Emergency Procedures

- 10.1* Vehicle accidents with radioactive sealed sources
- 10.2* Fire involving sealed sources
- 10.3* "Source out" – failure to return to safe shielded conditions
- 10.4* Emergency call list

11.0 Storage and Shipment of Exposed Devices and Sources

- 11.1* Vehicle storage
- 11.2* Storage vault – permanent
- 11.3* Shipping instructions – sources
- 11.4* Receiving instructions – radioactive material

12.0 State and Federal Regulations

- 12.1 NRC and Agreement States – authority
- 12.2 License reciprocity
- 12.3* Radioactive materials license requirements for industrial radiography
- 12.4 Occupational Safety and Health Administration (OSHA)
- 12.5 Qualification requirements for radiography personnel
- 12.6 Regulations for the control of radiation (state or NRC as applicable)
- 12.7* Department of Transportation regulations for radiographic source shipment
- 12.8 Regulatory requirements for X-ray machines (state and federal as applicable)

* Topics may be deleted if the radiography is limited to X-ray exposure devices.
** Required only by those personnel who will be involved in neutron radiography of explosive devices.

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NRC, Code of Federal Regulations, Title 10, Energy, Part 34, *Licenses for Industrial Radiography and Radiation Safety Requirements for Industrial Radiographic Operation*s, Sub-Part 43, Training, US Nuclear Regulatory Commission.

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